

Example Problem 10-10

Is the wormgear set described in Example Problem 8-4 satisfactory with regard to strength and wear when operating under the conditions of Example Problem 10-9? The wormgear has a face width of 1.25 in.

Solution From previous problems and solutions,

$$\begin{aligned} W_{tG} &= 962 \text{ lb} & VR &= m_G = 17.33 \\ v_{tG} &= 229 \text{ ft/min} & v_s &= 944 \text{ ft/min} \\ D_G &= 8.667 \text{ in} & D_W &= 2.000 \text{ in} \end{aligned}$$

Assume 58 HRC minimum for the steel worm. Assume that the bronze gear is sand cast.

Stress:

$$\begin{aligned} K_V &= 1200/(1200 + v_{tG}) = 1200/(1200 + 229) = 0.84 \\ W_d &= W_{tG}/K_V = 962/0.84 = 1145 \text{ lb} \\ F &= 1.25 \text{ in} \\ y &= 0.125 \text{ (from Table 10-5)} \\ p_n &= p \cos \lambda = (0.5236) \cos 14.04^\circ = 0.508 \text{ in} \end{aligned}$$

Then,

$$\sigma = \frac{W_d}{y F p_n} = \frac{1145}{(0.125)(1.25)(0.508)} = 14\,430 \text{ psi}$$

The guidelines in Section 10-11 indicate that this stress level would be adequate for either manganese or phosphor gear bronze.

Surface Durability: Use Equation (10-42):

$$W_{tR} = C_s D_G^{0.8} F_e C_m C_v \quad (10-42)$$

C Factors: The values for the *C* factors can be found from Figures 10-27 to 10-29. We find

$$\begin{aligned} C_s &= 740 \text{ for sand-cast bronze and } D_G = 8.667 \text{ in} \\ C_m &= 0.814 \text{ for } m_G = 17.33 \\ C_v &= 0.265 \text{ for } v_s = 944 \text{ ft/min} \end{aligned}$$

We can use $F_e = F = 1.25$ in if this value is not greater than 0.67 times the worm diameter. For $D_W = 2.000$ in,

$$0.67 D_W = (0.67)(2.00 \text{ in}) = 1.333 \text{ in}$$

Therefore, use $F_e = 1.25$ in. Then the rated tangential load is

$$W_{tR} = (740)(8.667)^{0.8}(1.25)(0.814)(0.265) = 1123 \text{ lb}$$

Because this value is greater than the actual tangential load of 962 lb, the design should be satisfactory, provided that the conditions defined for the application of Equation (10-42) are met.

10-13 EMERGING TECHNOLOGY AND SOFTWARE FOR GEAR DESIGN

Comprehending the fundamentals of gear design presented in Chapters 8-10 in this book should provide a foundation on which further growth in capabilities to perform reasonable design decisions for gear-type power transmissions can be built. The gear design field is continually adding new technologies and improving upon the methods presented here.

Literature in the field often promotes the advantages of gaining additional experience to enable more refined designs and to explore emerging technologies. This section discusses a few emerging technologies and topics beyond the scope of this book. While some technologies mentioned are not truly new and emerging, they are becoming more readily adopted by gear drive design practitioners. Also presented is a brief overview of software available to assist in the gear design process.

Emerging Gearing Technologies

It is strongly recommended that gear design professionals remain aware of the continual upgrading and updating of gear design technology as represented in the extensive suite of standards published by the American Gear Manufacturers Association (AGMA), the International Organization for Standardization (ISO), and other such organizations around the world. With experience, design professionals should also become involved in the standards development process as most of the standards are produced by volunteers representing prominent companies who produce gear drive products and the machinery to make them, or who employ such drives in their products and systems. Industry publications such as *Power Transmission Engineering* and *Gear Technology* provide timely, ongoing commentary and reporting on the state of the art in gear drives and their manufacture. (See Internet sites 10 and 14 in Chapter 9.)

Non-standard Gearing and Gear Tooth Forms. To provide a foundation for further study, this book has focused on standard gear tooth forms of the full-depth involute type with standard fillet geometry, addenda, dedenda, center distances, and other features, using methods that were discovered in the late eighteenth century. Major equipment manufacturers provide general-purpose machines to produce standard designs and suppliers of gears and gear drives benefit from standardization in terms of replacement components and are able to specify products from a variety of vendors.

Experience has shown that modifications to some standard features can be beneficial in special applications and in fields where extensive testing can be performed. Examples are automotive, ship propulsion, industrial machinery, construction equipment, and aerospace propulsion. Crowning of teeth and tip relief to promote smooth engagement, geometry modifications that accommodate deformations of teeth under heavy loads, and modified fillets to optimize bending stresses are a few areas of constant exploration. Materials technologies are also areas where fruitful developments are made for steels, other metallic materials, and the widely diverse array of plastic materials. Others pursue radical changes in gear tooth form to enhance strength, stiffness, noise reduction, or pitting resistance.

Internet site 1 describes proprietary designs called Megagear[®] and Unimegear[®] for which increased power density, efficiency, and durability are obtained. The gear tooth profiles are optimized resulting in greater contact area between gear teeth that reduces compressive stress, increases fatigue life, and makes load distribution more uniform.

Internet site 2 describes Direct Gear Design[®], an alternative method of analysis and design of involute gears, which separates gear geometry definition from tool selection to achieve the best possible performance for a particular product and application. The result is gear

geometry with asymmetric teeth and optimized fillets at the root area. Two involutes of two different base circles are defined that simultaneously increase the contact ratio and operating pressure angle beyond conventional gear limits. Specially designed gear cutting tools are used to produce the gears by hobbing or shaping. Plastic gear molding, die casting, gear forging, and powder metal processes can also be used. The application of Direct Gear Design must be justified by a significant improvement in gear performance. See also Reference 23.

Internet site 3 describes the patented MGT Frictionless Drive System[®] that uses the repulsive forces of magnets to provide a link between a drive shaft and the driven one while allowing both shafts to rotate completely independently of each other. Consequently there is no friction generated between them and the efficiency and life of the equipment are improved. The system can be applied either as gearing or couplings. Safety is improved because if the system is overloaded by a certain amount, the driver slips with respect to the driven member. Coupler faces do not touch and the gap between them is typically a few millimeters.

Internet site 4 and Reference 24 describe the field of non-circular gears and their manufacture. Forms can be elliptical, square, or designed to a wide variety of special functions. Sometimes used as cam substitutes in applications where torque or radial loads must be handled. Applications include speed matching on assembly lines, variable cutoff cycles for cutting products from continuous webs, linear motion with quick return, positive and negative rotation mechanisms, flowmeters, bicycle sprockets, variable-speed windshield wipers, and stop and dwell motion devices.

Peening. We have emphasized that successful gear design must demonstrate that the bending stress and contact stress in gears must be controlled to limits of the fatigue strength and pitting resistance of the material from which the gear is made. It stands to reason, therefore, that enhancing the fatigue strength and pitting resistance of the material can lead to safer, lighter, and longer lasting gears. One method of accomplishing that is called *peening*, a process of bombarding the high-stress surfaces of gears with hard materials, called *shot*, at high velocity in a controlled manner. Each impact of a piece of shot permanently dents the target material producing the final condition of compression. Shot media come in a variety of shapes such as follows:

- Cast spheres (steel, stainless steel, cast iron, aluminum oxide, zinc, titanium, and others)
- Cut wire (steel, aluminum, zinc, copper, and others)
- Beads (ceramic, glass, aluminosilicate, and plastic)
- Crushed glass, coal slag, hard rock, and garnet

The process leaves the surface with a desirable residual compressive stress and a high hardness that extends the load-carrying ability and life of the treated

component. One major objective for using shot peening is to selectively treat areas of a part that have inherently high tensile stress due to operating conditions or those for which tensile stresses at the surface are created by previous processing steps such as grinding, welding, and aggressive machining. Peening counteracts the residual tensile stresses resulting in lower net final stresses under load. Applications are found in numerous types of products from aerospace (landing gear; turbine components), automotive (gears, engine components, and structural elements), engines (crankshafts, camshafts, connecting rods, shafts, leaf springs, and coil springs), construction equipment (high-wear surfaces, actuator arms, and shafts), medical devices (replacement knees), energy production and transmission, and recreation equipment.

Internet sites 5 and 6 describe the types of equipment used for shot peening and the materials and forms of shot available. Two common types are as follows:

- Wheel blasting—Shot is introduced at the center of a rapidly spinning wheel and centrifugally accelerated as it passes to the periphery and is then flung at high velocity toward the target area.
- Air blasting—Pressurized air passes through a nozzle into which the shot is injected and then blown at high velocity toward the target.

Related products commonly called sand blasters are used to clean surfaces of scale and to produce a textured surface finish without necessarily producing the residual compressive stress in the surface. Vibratory finishing employs hard media of many materials and shapes into which parts are discharged to remove scale and deburr machined edges and surfaces, by repetitive rubbing of the media against the parts.

Software for Gear Design

As may be obvious from working through the topics and design methodologies covered in this chapter and Chapter 9, the subject of gear design is computationally demanding. This book includes a few focused calculation aids in the form of spreadsheets that give a hint to the advantages of using such computer-aided engineering approaches. Developers of several commercially available software packages with far greater capabilities for gear design covering a wide range of gear types are listed as Internet sites 7–13. Some of the developers also offer consulting services to aid in planning and implementing complex gear drives.

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9. Universal Technical Systems, Inc. Developer of the versatile UTS technical analysis software that includes the Integrated Gear Software (IGS) for gear design.
10. Drive System Technology, Inc. Mechanical power transmission consultants and developer of the PowerGear gear design software. Company founded by Raymond Drago, Chief Engineer of Drive Systems Technology, Inc. (see Reference 17).
11. Gleason Plastic Gears. Engineering consultants for the design and production of plastic gears, utilizing their own custom plastic gear design software. The website's *Engineering and Design* page includes an animated, dynamic, full-color FEA image of gear teeth in contact that displays bending and contact stresses during power transmission.
12. KISSsoft, U.S.A., LLC. Developer of the KISSsoft gear design software for several common gear styles along with related machine components and complete power transmissions.
13. AGMA Software Products. Developer of a set of software tied directly to the AGMA standards. Included are Bevel Gears Rating Suite, Gear Rating Suite (for spur and helical gears), AGMA 2015-1-A01 Calculator (Accuracy Classification System—Tangential Measurements for Cylindrical Gears). From the home page, select *Software Products*.

INTERNET SITES RELATED TO HELICAL GEARS, BEVEL GEARS, AND WORMGEARING

Refer to the list of Internet sites at the end of Chapter 9, Spur Gears. Virtually all sites listed there are also relevant to the design of helical gears, bevel gears, and wormgearing.

1. **Power Engineering and Manufacturing, Ltd.** Manufacturer of custom speed reducers and creator of the Megagear® gear tooth form.
2. **AKGears, LLC.** Creator of the Direct Gear Design® gear tooth form and developer of its application to a variety of products and mechanical systems.
3. **Magnetic Gearing and Turbine Corporation.** Developer and producer of the MGT Frictionless Drive System® that uses the repulsive forces of magnets to provide torque and power transmission without physical contact between the mating parts.
4. **Cunningham Industries, Inc.** Manufacturer of elliptical and other non-circular gears.
5. **The Shot Peener.** A site dedicated to providers of shot peening services or machinery to perform shot peening. The site contains valuable information about the technology and its applications.
6. **Wheelabrator, Inc.** Manufacturer of equipment and media for shot peening including wheelblast and airblast processes along with vibratory finishing equipment.

PROBLEMS

Helical Gearing

1. A helical gear has a transverse diametral pitch of 8, a transverse pressure angle of $14\frac{1}{2}^\circ$, 45 teeth, a face width of 2.00 in, and a helix angle of 30° .
 - (a) If the gear transmits 5.0 hp at a speed of 1250 rpm, compute the tangential force, the axial force, and the radial force.
 - (b) If the gear operates with a pinion having 15 teeth, compute the bending stress in the pinion teeth. The power comes from an electric motor, and the drive is to a reciprocating pump. Specify a quality number for the teeth.
 - (c) Specify a suitable material for the pinion and the gear considering both strength and pitting resistance.
2. A helical gear has a normal diametral pitch of 12, a normal pressure angle of 20° , 48 teeth, a face width of 1.50 in, and a helix angle of 45° .
 - (a) If the gear transmits 2.50 hp at a speed of 1750 rpm, compute the tangential force, the axial force, and the radial force.
 - (b) If the gear operates with a pinion having 16 teeth, compute the bending stress in the pinion teeth. The power comes from an electric motor, and the drive

is to a centrifugal blower. Specify a quality number for the teeth.

- (c) Specify a suitable material for the pinion and the gear considering both strength and pitting resistance.
3. A helical gear has a transverse diametral pitch of 6, a transverse pressure angle of $14\frac{1}{2}^\circ$, 36 teeth, a face width of 1.00 in, and a helix angle of 45° .
 - (a) If the gear transmits 15.0 hp at a speed of 2200 rpm, compute the tangential force, the axial force, and the radial force.
 - (b) If the gear operates with a pinion having 12 teeth, compute the bending stress in the pinion teeth. The power comes from a six-cylinder gasoline engine, and the drive is to a concrete mixer. Specify a quality number for the teeth.
 - (c) Specify a suitable material for the pinion and the gear considering both strength and pitting resistance.
4. A helical gear has a normal diametral pitch of 24, a normal pressure angle of $14\frac{1}{2}^\circ$, 72 teeth, a face width of 0.25 in, and a helix angle of 45° .
 - (a) If the gear transmits 0.50 hp at a speed of 3450 rpm, compute the tangential force, the axial force, and the radial force.
 - (b) If the gear operates with a pinion having 16 teeth, compute the bending stress in the pinion teeth. The power comes from an electric motor, and the drive is to a winch that will experience moderate shock. Specify a quality number for the teeth.
 - (c) Specify a suitable material for the pinion and the gear considering both strength and pitting resistance.

For Problem 5–11, complete the design of a pair of helical gears to operate under the stated conditions. Specify the geometry of the gears and the material and its heat treatment. Assume that the drive is from an electric motor unless otherwise specified. Consider both strength and pitting resistance.

5. A pair of helical gears is to be designed to transmit 5.0 hp while the pinion rotates at 1200 rpm. The gear drives a reciprocating compressor and must rotate between 385 and 390 rpm.
6. A helical gear pair is to be a part of the drive for a milling machine requiring 20.0 hp with the pinion speed at 550 rpm and the gear speed to be between 180 and 190 rpm.
7. A helical gear drive for a punch press requires 50.0 hp with the pinion rotating at 900 rpm and the gear speed at 225 to 230 rpm.
8. A single-cylinder gasoline engine has the pinion of a helical gear pair on its output shaft. The gear is attached to the shaft of a small cement mixer. The mixer requires 2.5 hp while rotating at approximately 75 rpm. The engine is governed to run at approximately 900 rpm.
9. A four-cylinder industrial engine runs at 2200 rpm and delivers 75 hp to the input gear of a helical gear drive for a large wood chipper used to prepare pulpwood chips for paper making. The output gear must run between 4500 and 4600 rpm.
10. A small commercial tractor is being designed for chores such as lawn mowing and snow removal. The wheel drive

system is to be through a helical gear pair in which the pinion runs at 450 rpm while the gear, mounted on the hub of the wheel, runs at 75 to 80 rpm. The wheel has an 18-in diameter. The two-cylinder gasoline engine delivers 20.0 hp to the wheels.

11. A water turbine transmits 15.0 hp to a pair of helical gears at 4500 rpm. The output of the gear pair must drive an electric power generator at 3600 rpm. The center distance for the gear pair must not exceed 4.00 in.
12. Determine the power-transmitting capacity of a pair of helical gears having a normal pressure angle of 20° , a helix angle of 15° , a normal diametral pitch of 10, 20 teeth in the pinion, 75 teeth in the gear, and a face width of 2.50 in, if they are made from SAE 4140 OQT 1000 steel. They are of typical commercial quality. The pinion will rotate at 1725 rpm on the shaft of an electric motor. The gear will drive a centrifugal pump.
13. Repeat Problem 12 with the gears made from SAE 4620 DOQT 300 carburized, case-hardened steel. Then compute the axial and radial forces on the gears.

Bevel Gears

14. A straight bevel gear pair has the following data: $N_p = 15$; $N_G = 45$; $P_d = 6$; and 20° pressure angle. If the gear pair is transmitting 3.0 hp, compute the forces on both the pinion and the gear. The pinion speed is 300 rpm. The face width is 1.25 in. Compute the bending stress and the contact stress for the teeth, and specify a suitable material and heat treatment. The gears are driven by a gasoline engine, and the load is a concrete mixer providing moderate shock. Assume that neither gear is straddle mounted.
15. A straight bevel gear pair has the following data: $N_p = 25$; $N_G = 50$; $P_d = 10$; and 20° pressure angle. If the gear pair is transmitting 3.5 hp, compute the forces on both the pinion and the gear. The pinion speed is 1250 rpm. The face width is 0.70 in. Compute the bending stress and the contact stress for the teeth, and specify a suitable material and heat treatment. The gears are driven by a gasoline engine, and the load is a conveyor providing moderate shock. Assume that neither gear is straddle mounted.
16. Design a pair of straight bevel gears to transmit 5.0 hp at a pinion speed of 850 rpm. The gear speed should be approximately 300 rpm. Consider both strength and pitting resistance. The driver is a gasoline engine, and the driven machine is a heavy-duty conveyor.
17. Design a pair of straight bevel gears to transmit 0.75 hp at a pinion speed of 1800 rpm. The gear speed should be approximately 475 rpm. Consider both strength and pitting resistance. The driver is an electric motor, and the driven machine is a reciprocating saw.

Wormgearing

18. A wormgear set has a single-thread worm with a pitch diameter of 1.250 in, a diametral pitch of 10, and a normal pressure angle of 14.5° . If the worm meshes with a wormgear having 40 teeth and a face width of 0.625 in, compute the gear pitch diameter, the center distance, and the velocity ratio. If the wormgear set is transmitting 924 lb · in of torque at its output shaft, which is rotating

at 30 rpm, compute forces on the gear, efficiency, input speed, input power, and stress on the gear teeth. If the worm is hardened steel and the gear is chilled bronze, evaluate the rated load, and determine whether the design is satisfactory for pitting resistance.

19. Three designs are being considered for a wormgear set to produce a velocity ratio of 20 when the worm gear rotates at 90 rpm. All three have a diametral pitch of 12, a worm pitch diameter of 1.000 in, a gear face width of 0.500 in, and a normal pressure angle of 14.5° . One has a single-thread worm and 20 wormgear teeth; the second has a double-thread worm and 40 wormgear teeth; the third has a four-thread worm and 80 wormgear teeth. For each design, compute the rated output torque, considering both strength and pitting resistance. The worms are hardened steel, and the wormgears are chilled bronze.

20. For each of the three designs proposed in Problem 19, compute the efficiency.

The data for Problems 21–23 are given in Table 10–6. Design a wormgear set to produce the desired velocity ratio when transmitting the given torque at the output shaft for the given output rotational speed.

24. Compare the two designs described in Table 10–7 when each is transmitting 1200 lb·in of torque at its output shaft, which rotates at 20 rpm. Compute the forces on the worm and the wormgear, the efficiency, and the input power required.

TABLE 10–6 Data for Problems 21–23

Problem	VR	Torque (lb·in)	Output speed (rpm)
21.	7.5	984	80
22.	3	52.5	600
23.	40	4200	45

TABLE 10–7 Designs for Problem 24

Design	P_d	N_t	N_G	D_w	F_G	Pressure angle
A	6	1	30	2.000	1.000	14.5°
B	10	2	60	1.250	0.625	14.5°

KEYS, COUPLINGS, AND SEALS

The Big Picture

You Are the Designer

- 11-1 Objectives of This Chapter
- 11-2 Keys
- 11-3 Materials for Keys
- 11-4 Stress Analysis to Determine Key Length
- 11-5 Splines
- 11-6 Other Methods of Fastening Elements to Shafts
- 11-7 Couplings
- 11-8 Universal Joints
- 11-9 Retaining Rings and Other Means of Axial Location
- 11-10 Types of Seals
- 11-11 Seal Materials

THE BIG PICTURE

Keys, Couplings, and Seals

Discussion Map

- Keys and couplings connect functional parts of mechanisms and machines, allowing moving parts to transmit power or to locate parts relative to each other.
- Retaining rings hold assemblies together or hold parts on shafts, such as keeping a sprocket in position or holding a wheel on an axle.
- Seals protect critical components by excluding contaminants or by retaining fluids inside the housing of a machine.

This chapter will help you understand the functions and design requirements of such devices. In addition, you will learn to recognize commercially available designs and apply them properly.

Think of how two or more parts of a machine can be connected for the purpose of locating one part with respect to another. Now think about how that connection must be designed if the parts are moving and if power must be transmitted between them.

This chapter presents information on commercially available products that accomplish these functions. The generic categories of keys, couplings, and seals actually encompass numerous different designs.

A *key* is used to connect a drive member such as a belt pulley, chain sprocket, or gear to the shaft that carries it. (See Figure 11-1.) Torque and power are transmitted across the key to or from the shaft. But how does

Discover

Look around you and identify several examples of the use of keys, couplings, retaining rings, and seals in automobiles, trucks, home appliances, shop tools, gardening equipment, or bicycles.

the power get into or out of the shaft? One way might be that the output from the shaft of a motor or engine is connected to the input shaft of a transmission through a *flexible coupling* that reliably transmits power but allows for some misalignment between the shafts during operation because of the flexing of frame members or through progressive misalignment due to wear.

Seals may be difficult to see because they are typically encased in a housing or are covered in some way. Their function is to protect critical elements of a machine from contamination due to dust, dirt, or water or other fluids while allowing rotating or translating machine elements to move to accomplish their desired

functions. Seals exclude undesirable materials from the inside of a mechanism, or they hold critical lubrication or cooling fluids inside the housing of the mechanism.

Look at machines that you interact with each day, and identify parts that fit the descriptions given here. Look in the engine compartment of a car or a truck. How are drive pulleys, linkages, latches, the hinges for the hood, the fan, the windshield wipers, and any other moving part connected to something else—the frame of the car, a rotating shaft, or some other moving part? If you are familiar with the inner workings of the engine and transmission, describe how those parts are connected. Look at the steering and suspension systems, the water pump, the fuel pump, the brake fluid reservoir, and the suspension struts or shock absorbers. Try to see where seals are used. Can you see the universal joints, sometimes called the *constant-velocity* (CV) joints, in the drive train? They should be connecting the output shaft from the transmission to the final parts of the drive train as the power is delivered to the wheels.

Find a small lawn or garden tractor at home or at a local home store. Typically their mechanisms are

accessible, although protected from casual contact for safety reasons. Trace how power is transmitted from the engine, through a transmission, through a drive chain or belt, and all the way to the wheels or to the blade of a mower. How are functional parts connected together?

Look at home appliances, power tools in a home shop, and gardening equipment. Can you see parts that are held in place with *retaining rings*? These are typically thin, flat rings that are pressed onto shafts or inserted into grooves to hold a wheel on a shaft, to hold a gear or a pulley in position along the length of a shaft, or to simply hold some part of the device in place.

How are the keys, couplings, seals, retaining rings, and other connecting devices made? What materials are used? How are they installed? Can they be removed? What kinds of forces must they resist? How does their special geometry accomplish the desired function? How could they fail?

This chapter will help you become familiar with such mechanical components, with some of the manufacturers that offer commercial versions, and with correct application methods.

YOU ARE THE DESIGNER

In the first part of Chapter 8, you were the designer of a gear-type speed reducer whose conceptual design is shown in Figure 1-12. It has four gears, three shafts, and six bearings, all contained within a housing. How are the gears attached to the shafts? One way is to use keys at the interface between the hub of the gears and the shaft. You should be able to design the keys. How is the input shaft connected to the motor or engine that delivers the power? How is the output shaft connected to the driven machine? One way is to use flexible couplings. You should be able to specify commercially available couplings and apply them properly, considering the amount of torque they must transmit and how much misalignment they should permit.

How are the gears located axially along the shafts? Part of this function may be provided by shoulders machined on the shaft. But

that works only on one side. On the other side, one type of locating means is a retaining ring that is installed into a groove in the shaft after the gear is in place. Rings or spacers may be used at the left of gears *A* and *B* and at the right of gears *C* and *D*. Notice that the input and output shafts extend outside the housing. How can you keep contaminants from the outside from getting inside? How can you keep lubricating oil inside? Shaft seals can accomplish that function. Seals may also be provided on the bearings to retain lubricant inside and in full contact with the rotating balls or rollers of the bearing.

You should be familiar with the kinds of materials used for seals and with their special geometries. These concepts are discussed in this chapter.

11-1 OBJECTIVES OF THIS CHAPTER

After completing this chapter, you will be able to:

1. Describe several kinds of *keys*.
2. Specify a suitable size key for a given size shaft.
3. Specify suitable materials for keys.
4. Complete the design of keys and the corresponding keyways and keyseats, giving their complete geometries.
5. Describe *splines* and determine their torque capacity.
6. Describe several alternate methods of fastening machine elements to shafts.
7. Describe *rigid couplings* and *flexible couplings*.

8. Describe several types of flexible couplings.

9. Describe *universal joints*.

10. Describe *retaining rings* and other means of locating elements on shafts.

11. Specify suitable seals for shafts and other types of machine elements.

11-2 KEYS

A *key* is a machinery component placed at the interface between a shaft and the hub of a power-transmitting element for the purpose of transmitting torque [see Figure 11-1(a)]. The key is demountable to facilitate assembly and disassembly of the shaft system. It is installed in an axial groove machined into the shaft, called a *keyseat*.

A similar groove in the hub of the power-transmitting element is usually called a *keyway*, but it is more properly also a keyseat. The key is typically installed into the shaft keyseat first; then the hub keyseat is aligned with the key, and the hub is slid into position.

Square and Rectangular Parallel Keys

The most common type of key for shafts up to $6\frac{1}{2}$ in in diameter is the square key, as illustrated in Figure 11-1(b). The rectangular key, Figure 11-1(c), is recommended for larger shafts and is used for smaller shafts where the shorter height can be tolerated. Both the square and the rectangular keys are referred to as *parallel keys* because the top and bottom and the sides of the key are parallel. (See Internet sites 1 and 20.)

Table 11-1 gives the preferred dimensions for parallel keys as a function of shaft diameter for both U.S. sizes and SI Metric sizes. The width is approximately one-quarter of the diameter of the shaft. See References 7 and 9 for more detailed dimensions and tolerances.

The keyseats in the shaft and the hub are designed so that exactly one-half of the height of the key is bearing on the side of the shaft keyseat and the other half on the side of the hub keyseat. Figure 11-2 shows the resulting geometry. The distance Y is the radial distance from the theoretical top of the shaft, before the keyseat is machined, to the top edge of the finished keyseat to produce a keyseat depth of exactly $H/2$. To assist in machining and inspecting the shaft or the hub, the dimensions S and T can be computed and shown on the part drawings. The equations are given in Figure 11-2. Tabulated values of Y , S , and T are available in References 7 and 9.

As discussed later in Chapter 12, keyseats in shafts are usually machined with either an end mill or a circular milling cutter, producing the profile or sled runner

keyseat, respectively (refer Figure 12-7). In general practice, the keyseats and keys are left with essentially square ends and edges. But radiused keyseats and chamfered keys can be used to reduce the stress concentrations. Table 11-2 shows suggested values from ANSI Standard B17.1.

As alternates to the use of parallel keys, taper keys, gib head keys, pin keys, and Woodruff keys can be used to provide special features of installation or operation. Figure 11-3 shows the general geometry of these types of keys.

Taper Keys and Gib Head Keys

Taper keys [Figure 11-3(a) and (b)] are designed to be inserted from the end of the shaft after the hub is in position rather than installing the key first and then sliding the hub over the key as with parallel keys. The taper extends over at least the length of the hub, and the height, H , measured at the end of the hub, is the same as for the parallel key. The taper is typically 1/8 in per foot. Note that this design gives a smaller bearing area on the sides of the key, and the bearing stress must be checked.

The *gib head key* [Figure 11-3(c)] has a tapered geometry inside the hub that is the same as that of the plain taper key. But the extended head provides the means of extracting the key from the same end at which it was installed. This is very desirable if the opposite end is not accessible to drive the key out.

Pin Keys

The *pin key*, shown in Figure 11-3(d), is a cylindrical pin placed in a cylindrical groove in the shaft and hub. Lower stress concentration factors result from this design as compared with parallel or taper keys. A close

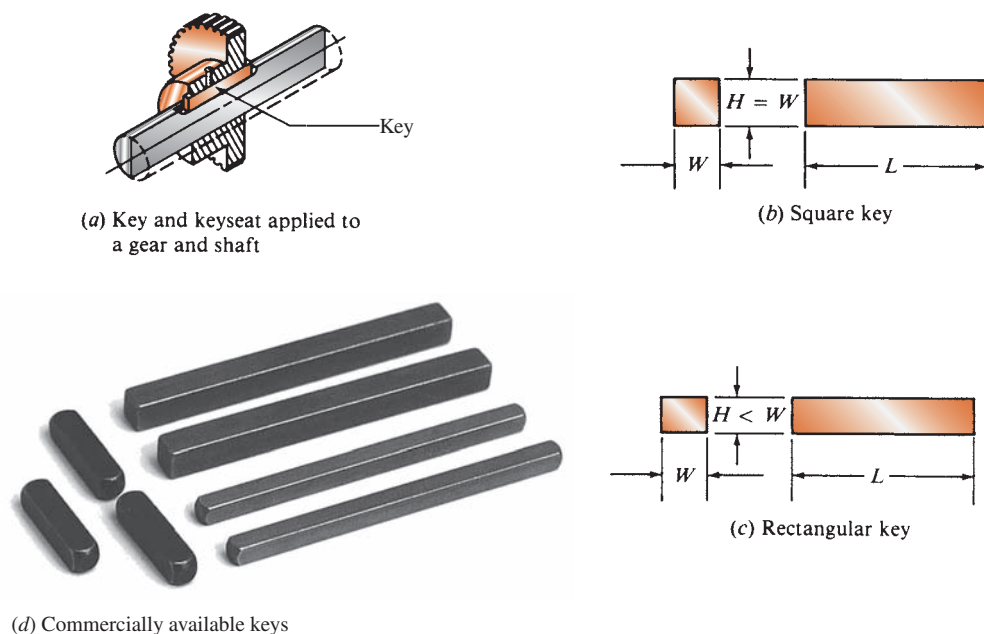


FIGURE 11-1 Parallel keys

TABLE 11-1 Key Size vs. Shaft Diameter

U.S. inch sizes				SI metric sizes			
Nominal shaft diameter		Key dimensions		Nominal shaft diameter		Key dimensions	
Over (in)	to-including (in)	Width, W (in)	Height, H (in)	Over (mm)	to-including (mm)	Width, W (mm)	Height, H (mm)
0.3125	0.4375	0.09375	0.09375	6	8	2	2
0.4375	0.5625	0.1250	0.1250	8	10	3	3
0.5625	0.875	0.1875	0.1875	10	12	4	4
0.875	1.250	0.2500	0.2500	12	17	5	5
1.250	1.375	0.3125	0.3125	17	22	6	6
1.375	1.75	0.375	0.375	22	30	8	7
1.75	2.25	0.500	0.500	30	38	10	8
2.25	2.75	0.625	0.625	38	44	12	8
2.75	3.25	0.750	0.750	44	50	14	9
3.25	3.75	0.875	0.875	50	58	16	10
3.75	4.50	1.00	1.00	58	65	18	11
4.50	5.50	1.25	1.25	65	75	20	12
5.50	6.50	1.50	1.50	75	85	22	14
6.50	7.50	1.75	1.50	85	95	25	14
7.50	9.00	2.00	1.50	95	110	28	16
9.00	11.00	2.50	1.75	110	130	32	18
11.00	13.00	3.00	2.00	130	150	36	20
13.00	15.00	3.50	2.50	150	170	40	22
15.00	18.00	4.00	3.00	170	200	45	25
18.00	22.00	5.00	3.50	200	230	50	28
22.00	26.00	6.00	4.00	230	260	56	32
26.00	30.00	7.00	5.00	260	290	63	32
				290	330	70	36
				330	380	80	40
				380	440	90	45
				440	500	100	50

Note: Key sizes above the horizontal line are square; others are rectangular.

fit between the pin and the groove is required to ensure that the pin does not move and that the bearing is uniform along the length of the pin.

Woodruff Keys

Where light loading and relatively easy assembly and disassembly are desired, the *Woodruff key* should be considered. Figure 11-3(e) shows the standard configuration. The circular groove in the shaft holds the key in position while the mating part is slid over the key. The stress analysis for this type of key proceeds in the manner discussed for the parallel key, taking into

consideration the particular geometry of the Woodruff key. ANSI Standard B17.2-1967 lists the dimensions for a large number of standard Woodruff keys and their mating keyseats. (See Reference 8.) Table 11-3 provides a sampling. Note that the *key number* indicates the nominal key dimensions. The last two digits give the nominal diameter, B , in eighths of an inch, and the digits preceding the last two give the nominal width, W , in thirty-seconds of an inch. For example, key number 1210 has a diameter of 10/8 in ($1\frac{1}{4}$ in), and a width of 12/32 in ($3/8$ in). The actual size of the key is slightly smaller than half of the full circle, as shown in dimensions C and F in Table 11-3.

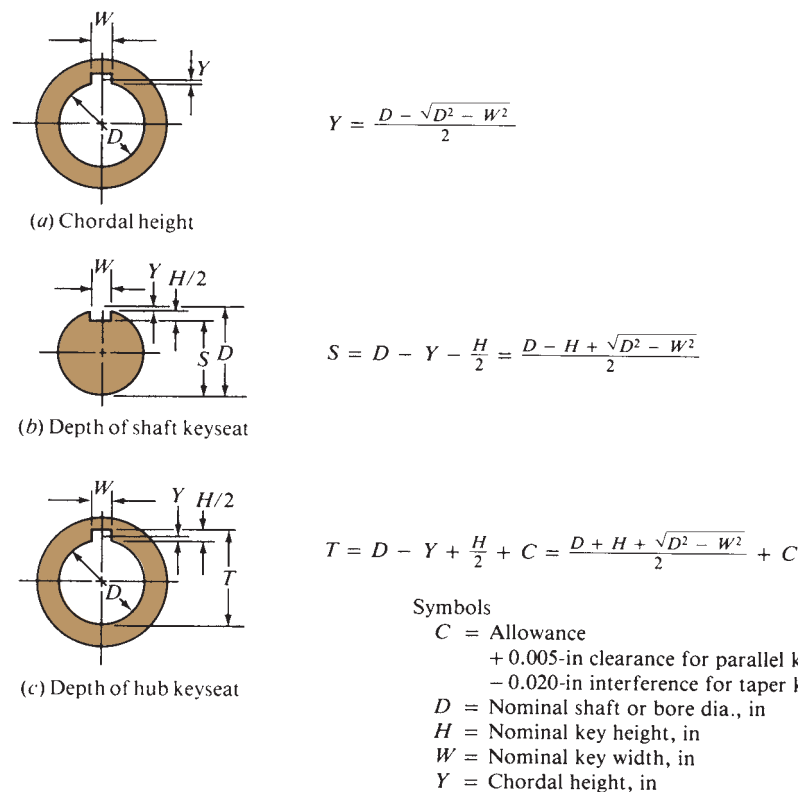


FIGURE 11-2 Dimensions for parallel keyseats

TABLE 11-2 Suggested Fillet Radii and Key Chamfers

$H/2$, keyseat depth		Fillet radius	45° chamfer
Over	To (incl.)		
1/8	1/4	1/32	3/64
1/4	1/2	1/16	5/64
1/2	7/8	1/8	5/32
7/8	$1\frac{1}{4}$	3/16	7/32
$1\frac{1}{4}$	$1\frac{3}{4}$	1/4	9/32
$1\frac{3}{4}$	$2\frac{1}{2}$	3/8	13/32

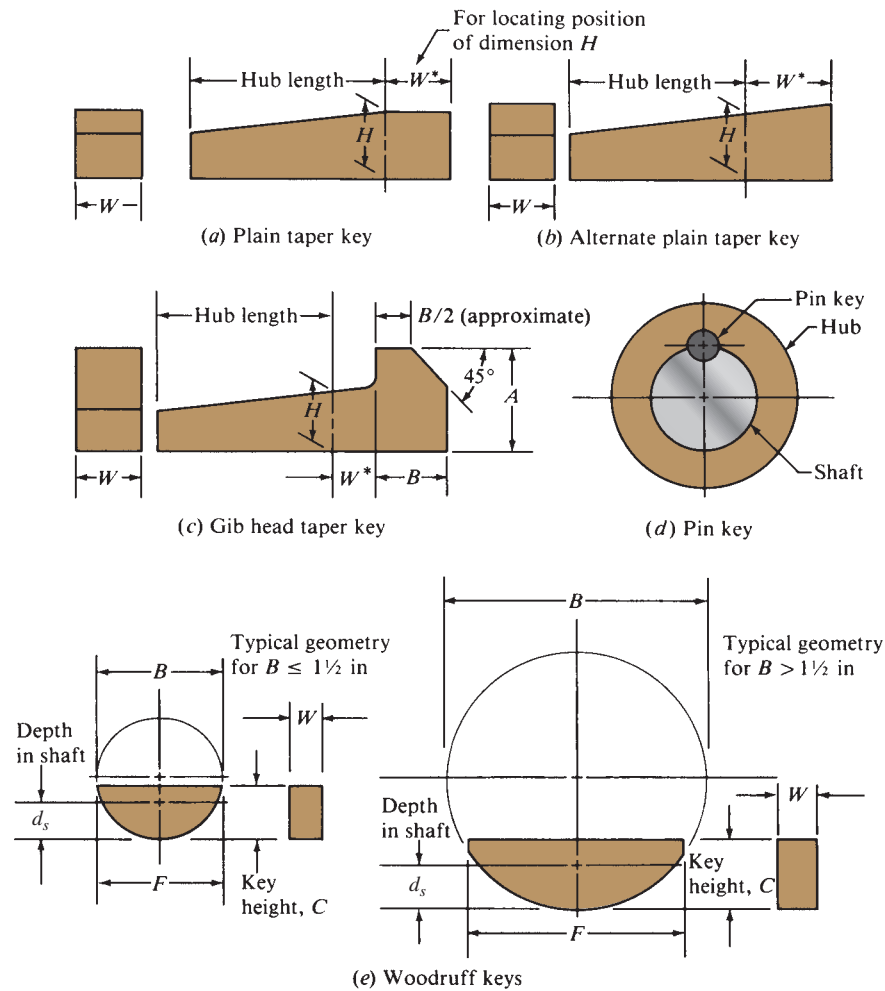
Note: All dimensions are given in inches.

Selection and Installation of Keys and Keyseats

The key and the keyseat for a particular application are usually designed after the shaft diameter is specified by the methods of Chapter 12. Then, with the shaft diameter as a guide, the size of the key is selected from Table 11-1. The only remaining variables are the length of the key and its material. One of these can be specified, and the requirements for the other can then be computed.

Typically the length of a key is specified to be a substantial portion of the hub length of the element in which it is installed to provide for good alignment and stable operation. But if the keyseat in the shaft is to be in the vicinity of other geometric changes, such as shoulder fillets and ring grooves, it is important to provide some axial clearance between them so that the effects of the stress concentrations are not compounded.

The key can be cut off square at its ends or provided with a radius at each end when installed in a profile



*Note: Plain and gib head taper keys have a 1/8-in taper in 12 in.

FIGURE 11-3 Key types

TABLE 11-3 Woodruff Key Dimensions

Key number	Nominal key size, $W \times B$	Actual length, F	Height of key, C	Shaft keyseat depth	Hub keyseat depth
202	$1/16 \times 1/4$	0.248	0.104	0.0728	0.0372
204	$1/16 \times 1/2$	0.491	0.200	0.1668	0.0372
406	$1/8 \times 3/4$	0.740	0.310	0.2455	0.0685
608	$3/16 \times 1$	0.992	0.435	0.3393	0.0997
810	$1/4 \times 1\frac{1}{4}$	1.240	0.544	0.4170	0.1310
1210	$3/8 \times 1\frac{1}{4}$	1.240	0.544	0.3545	0.1935
1628	$1/2 \times 3\frac{1}{2}$	2.880	0.935	0.6830	0.2560
2428	$3/4 \times 3\frac{1}{2}$	2.880	0.935	0.5580	0.3810

Note: All dimensions are given in inches.

keyseat to improve location. Square-cut keys are usually used with the sled-runner-type keyseat.

The key is sometimes held in position with a set screw in the hub over the key. However, the reliability of this

approach is questionable because of the possibility of the set screw's backing out with vibration of the assembly. Axial location of the assembly should be provided by more positive means, such as shoulders, retaining rings, or spacers.

11-3 MATERIALS FOR KEYS

Keys are made from plain carbon steel, alloy steels, stainless steels, and some nonferrous metals. Even plastics are used for small devices under low loads. This book will focus primarily on steels for general industrial applications. Table 11-4 lists a variety of steels and one aluminum alloy to illustrate the types of materials available. (See Internet site 20.)

For problem solutions in this book, it is recommended that the low-carbon steel SAE 1018 be considered for most applications. It is a low-cost, readily available material and its strength is generally adequate.

If higher strength is necessary to produce a design with a reasonable length, medium-carbon SAE 1035 or 1045, or alloy steels SAE 4140 or 8630 are recommended. The high-carbon steel SAE 1095 may be used but it may have low ductility.

Where corrosion resistance is necessary, the stainless steels listed in Table 11-4 may be considered. Aluminum 6061 is less frequently used for keys but may be desirable for material compatibility reasons.

11-4 STRESS ANALYSIS TO DETERMINE KEY LENGTH

There are two basic modes of potential failure for keys transmitting power: (1) shear across the shaft/hub interface and (2) compression failure due to the

bearing action between the sides of the key and the shaft or hub material. The analysis for either failure mode requires an understanding of the forces that act on the key. Figure 11-4 shows the idealized case in which the torque on the shaft creates a force on the left side of the key. The key in turn exerts a force on the right side of the hub keyseat. The reaction force of the hub back on the key then produces a set of opposing forces that place the key in direct shear over its cross section, $W \times L$. The magnitude of the shearing force can be found from

$$F = T/(D/2)$$

The shearing stress is then

$$\tau = \frac{F}{A_s} = \frac{T}{(D/2)(WL)} = \frac{2T}{DWL} \quad (11-1)$$

In design, we can set the shearing stress equal to a design stress in shear for the maximum shear stress theory of failure:

$$\tau_d = 0.5s_y/N$$

Then the required length of the key is

➤ **Minimum Required Key Length for Shear**

$$L_{\min} = \frac{2T}{\tau_d DW} \quad (11-2)$$

TABLE 11-4 Examples of Materials Used for Keys

Material designation	Tensile strength s_u		Yield strength s_y	
	(ksi)	(MPa)	(ksi)	(MPa)
Carbon steels (SAE)				
1018	64	441	54	372
1035	72	496	39.5	272
1045	91	627	77	531
1095	140	965	83	572
Alloy steels (SAE)				
4140	102	703	90	621
8630	100	690	95	655
Stainless steels (SAE)				
303	90	621	35	241
304	85	586	35	241
316	85	586	35	241
416	75	517	40	276
Aluminum				
6061	18	124	12	83

Source: Adapted from Internet site 20.

Note: Strength properties typical, not guaranteed.

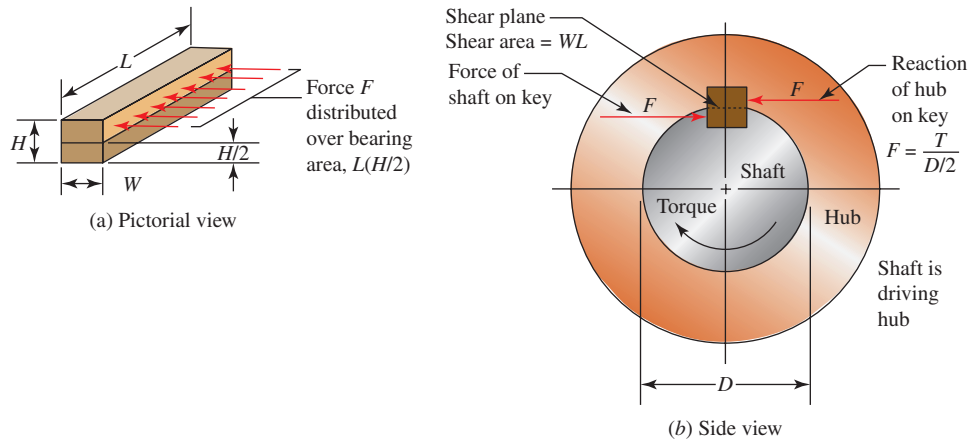


FIGURE 11-4 Forces on a key

The failure in bearing is related to the compressive stress on the side of the key, the side of the shaft keyseat, or the side of the hub keyseat. The area in compression is the same for either of these zones, $L \times (H/2)$. Thus, the failure occurs on the surface with the lowest compressive yield strength. Let's define a *design stress for compression* as

$$\sigma_d = s_y/N$$

Then the compressive stress is

$$\sigma = \frac{F}{A_c} = \frac{T}{(D/2)(L)(H/2)} = \frac{4T}{DLH} \quad (11-3)$$

Letting this stress equal the design compressive stress allows the computation of the required length of the key for this mode of failure:

⇒ Minimum Required Key Length for Compression

$$L_{\min} = \frac{4T}{\sigma_d DH} \quad (11-4)$$

For the design of a square key in which the strength of the key material is lower than that of the shaft or the hub, Equations (11-2) and (11-4) produce the same result. Substituting the design stress into either equation would give

⇒ Minimum Required Key Length if Key Material Is Weakest

$$L_{\min} = \frac{4TN}{DWs_y} \quad (11-5)$$

But be sure to evaluate the length from Equation (11-4) if either the shaft or the hub has a lower yield strength than the key.

DESIGN PROCEDURE FOR PARALLEL KEYS

1. Complete the design of the shaft into which the key will be installed, and specify the actual diameter at the location of the keyseat.
2. Select the size of the key from Table 11-1.
3. Specify a suitable design factor, N . In typical industrial applications, $N = 3$ is adequate to accommodate accidental overloads and shock.
4. Specify the material for the key, usually SAE 1018 steel. A higher-strength material can be used.
5. Determine the yield strength of the materials for the key, the shaft, and the hub.
6. If a square key is used and the key material has the lowest strength, use Equation (11-5) to compute the minimum required length of the key. This length will be satisfactory for both shear and bearing stress.
7. If a rectangular key is used, or if either the shaft or the hub has a lower strength than the key, use Equation (11-4) to compute the minimum required length of the key based on bearing stress. Also, use Equation (11-2) or (11-5) to compute the minimum required length based on shear of the key. The larger of the two computed lengths governs the design. Check to be sure that the computed length is shorter than the hub length. If not, a higher-strength material must be selected and the design process repeated. Alternatively, two keys or a spline can be used instead of a single key.
8. Specify the actual length of the key to be equal to or longer than the computed minimum length. A convenient standard size should be specified using the preferred basic sizes shown in Appendix A2-1. **The key should extend over all or a substantial part of the length of the hub. But the keyseat should not run into other stress raisers such as shoulders or grooves.**
9. Complete the design of the keyseat in the shaft and the keyway in the hub using the equations in Figure 11-2. ANSI Standard B17.1 should be consulted for standard tolerances on dimensions for the key and the keyseats.
10. See also Chapter 15 for additional details concerning tolerancing and chamfering.

Example Problem 11-1

A portion of a shaft where a gear is to be mounted has a diameter of 2.00 in. The gear transmits 2965 lb·in of torque. The shaft is to be made of SAE 1040 cold-drawn steel. The gear is made from SAE 8650 OQT 1000 steel. The width of the hub of the gear mounted at this location is 1.75 in. Design the key.

Solution From Table 11-1, the standard key dimension for a 2.00-in-diameter shaft would be 0.500-in square. See Figure 11-5 for the proposed design.

Material selection is a design decision. Let's choose SAE 1018 steel with $s_y = 54\,000$ psi as listed in Table 11-4.

A check of the yield strengths of the three materials in the key, the shaft, and the hub indicates that the key is the weakest material. Then Equation (11-5) can be used to compute the minimum required length of the key:

$$L = \frac{4TN}{DWs_y} = \frac{4(2965 \text{ lb} \cdot \text{in})(3)}{(2.00 \text{ in})(0.500 \text{ in})(54\,000 \text{ lb/in}^2)} = 0.659 \text{ in}$$

This length is well below the width of the hub of the gear. Notice that the design of the shaft includes retaining rings on both sides of the gear. It is desirable to keep the keyseat well clear of the ring grooves. Therefore, let's specify the length of the key to be 1.50 in.

Summary In summary, the key has the following characteristics:

Material: SAE 1018 steel keystock

Width: 0.500 in

Height: 0.500 in

Length: 1.50 in

Figure 11-5 shows some details of the completed design. A profile keyseat in the shaft is shown.

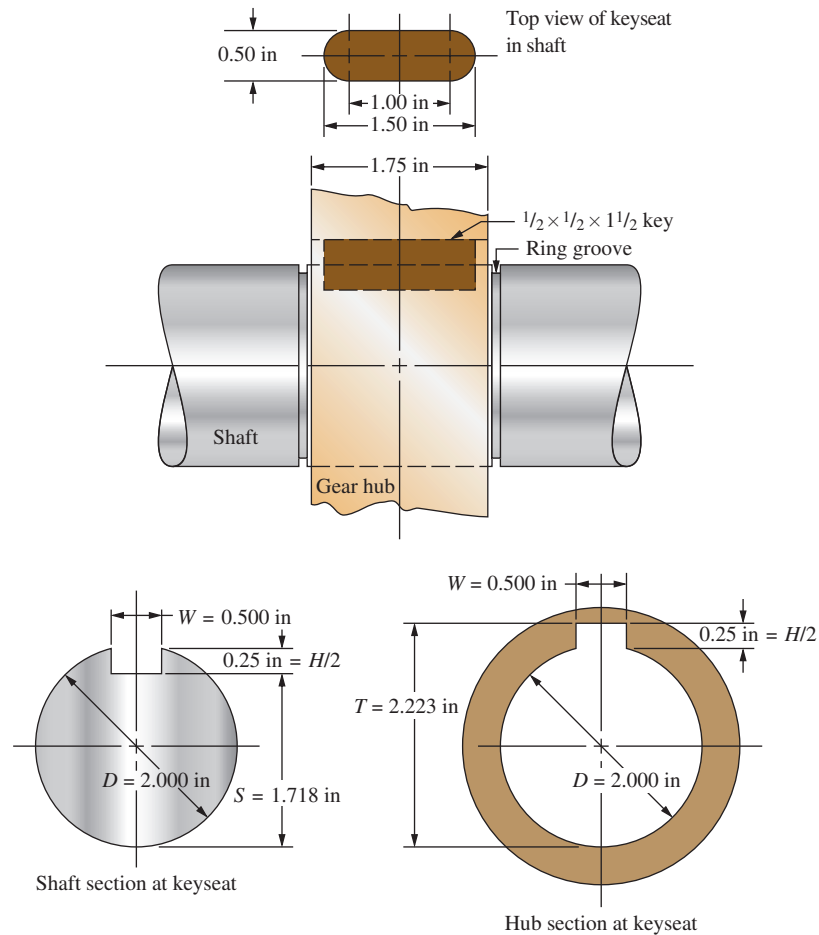


FIGURE 11-5 Details for proposed design of key and keyseats

Shear and Bearing Areas for Woodruff Keys

The geometry of Woodruff keys makes it more difficult to determine the shear area and the bearing area for use in stress analyses. Figure 11-3(e) shows that the bearing area on the side of the key in the keyseat is a segment of a circle. The shear area is the product of the chord of that segment times the thickness of the key. The following equations describe the geometry:

Given

- B = nominal diameter of the cylinder of which the key is a part
 W = width (thickness) of the key
 C = full height of the key
 d_s = depth of the keyseat in the shaft

Results

$$\text{Shear area} = A_s = 2W\sqrt{d_s(B - d_s)} \quad (11-6)$$

To define the equations for the bearing areas on the side of the key in the shaft and in the hub, we first define three geometric variables, G , L , and J , as follows:

$$G = (\pi/180)B \cos^{-1} \{ 2[(B/2) - d_s]/B \}$$

$$L = 2\sqrt{d_s(B - d_s)}$$

$$J = (\pi/180)B \cos^{-1} \{ 2[(B/2) - C]/B \}$$

Bearing Area in Shaft

$$A_{c \text{ shaft}} = 0.5 \{ G(B/2) - L[(B/2) - d_s] \} \quad (11-7)$$

$$A_{c \text{ hub}} = 0.5 \{ J(B/2) - F[(B/2) - C] \} - A_{c \text{ shaft}} \quad (11-8)$$

11-5 SPLINES

A *spline* can be described as a series of axial keys machined into a shaft, with corresponding grooves machined into the bore of the mating part (gear, sheave, sprocket, and so on; see Figure 11-6). The splines perform the same function as a key in transmitting torque from the shaft to the mating element. The advantages of splines over keys are many. Because usually four or more splines are used, as compared with one or two keys, a more uniform transfer of the torque and a lower loading on a given part of the shaft/hub interface result. The splines are integral with the shaft, so no relative motion can occur as between a key and the shaft. Splines are accurately machined to provide a controlled fit between the mating internal and external splines. The surface of the spline is often hardened to resist wear and to facilitate its use in applications in which axial motion of the mating element is desired. Sliding motion between a standard parallel key and the mating element should not be permitted. Because of the multiple splines on the shaft, the mating element can be indexed to various positions.

Splines can be either straight-sided or involute. The involute form is preferred because it provides for self-centering of the mating element and because it can be machined with standard hobs used to cut gear teeth.

Straight-Sided Splines

Straight splines are made according to the specifications of the Society of Automotive Engineers (SAE) and usually contain 4, 6, 10, or 16 splines. Figure 11-6 shows three styles. The six-spline version shows the basic design

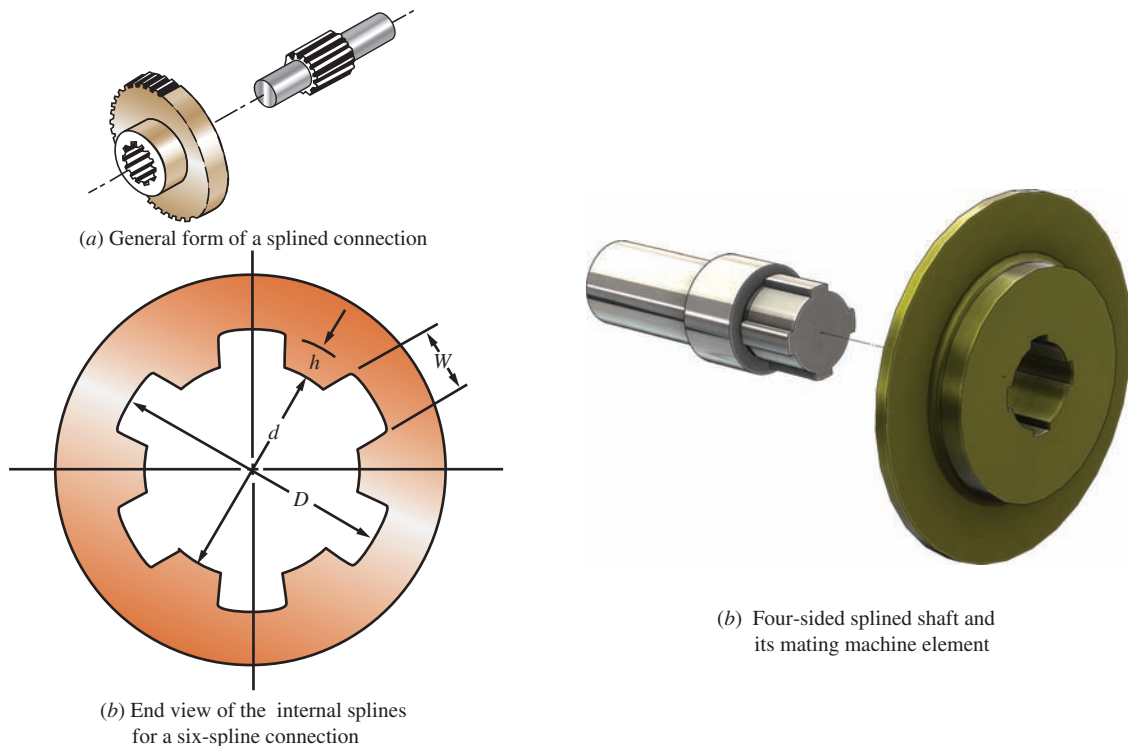


FIGURE 11-6 Straight-sided splines with different numbers of splines

TABLE 11-5 Formulas for SAE Straight Splines

Number of splines	W , for all fits	A: Permanent fit		B: To slide without load		C: To slide under load	
		h	d	h	d	h	d
Four	$0.241D$	$0.075D$	$0.850D$	$0.125D$	$0.750D$		
Six	$0.250D$	$0.050D$	$0.900D$	$0.075D$	$0.850D$	$0.100D$	$0.800D$
Ten	$0.156D$	$0.045D$	$0.910D$	$0.070D$	$0.860D$	$0.095D$	$0.810D$
Sixteen	$0.098D$	$0.045D$	$0.910D$	$0.070D$	$0.860D$	$0.095D$	$0.810D$

Note: These formulas give the maximum dimensions for W , h , and d .

parameters of D (major diameter), d (minor diameter), W (spline width), and h (spline depth). The dimensions for d , W , and h are related to the nominal major diameter D by the formulas given in Table 11-5. Note that the values of h and d differ according to the use of the spline. The permanent fit, A , is used when the mating part is not to be moved after installation. The B fit is used if the mating part will be moved along the shaft without a torque load. When the mating part must be moved under load, the C fit is used.

The torque capacity for SAE splines is based on the limit of 1000-psi bearing stress on the sides of the splines, from which the following formula is derived:

✦ Torque Capacity for a Spline

$$T = 1000NRh \quad (11-9)$$

where N = number of splines

R = mean radius of the splines

h = depth of the splines (from Table 11-5)

The torque capacity is per inch of length of the spline. But note that

$$R = \frac{1}{2} \left[\frac{D}{2} + \frac{d}{2} \right] = \frac{D + d}{4}$$

$$h = \frac{1}{2} (D - d)$$

Then

$$T = 1000N \frac{(D + d)}{4} \frac{(D - d)}{2} = 1000N \frac{(D^2 - d^2)}{8} \quad (11-10)$$

This equation can be further refined for each of the types of splines in Table 11-5 by substitution of the appropriate relationships for N and d . For example, for the six-spline version and the B fit, $N = 6$, $d = 0.850D$, and $d^2 = 0.7225D^2$.

Then

$$T = 1000(6) \frac{[D^2 - 0.7225D^2]}{8} = 208D^2$$

Thus, the required diameter to transmit a given torque would be

✦ Required Spline Diameter for a given Torque

$$D = \sqrt{T/208}$$

In these formulas, dimensions are in inches and the torque is in pound-inches. We use this same approach to find the torque capacities and required diameters for the other versions of straight splines (Table 11-6).

The graphs in Figure 11-7 enable you to choose an acceptable diameter for a straight-sided spline to carry a given torque, depending on the desired fit, A , B , or C . The data were taken from Table 11-6.

Involute Splines

Involute splines are typically made with pressure angles of 30° , 37.5° , or 45° . The 30° form is illustrated in Figure 11-8, showing the two types of fit that can be

TABLE 11-6 Torque Capacity for Straight Splines per Inch of Spline Length

Number of splines	Fit	Torque capacity	Required diameter
4	A	$139D^2$	$\sqrt{T/139}$
4	B	$219D^2$	$\sqrt{T/219}$
6	A	$143D^2$	$\sqrt{T/143}$
6	B	$208D^2$	$\sqrt{T/208}$
6	C	$270D^2$	$\sqrt{T/270}$
10	A	$215D^2$	$\sqrt{T/215}$
10	B	$326D^2$	$\sqrt{T/326}$
10	C	$430D^2$	$\sqrt{T/430}$
16	A	$344D^2$	$\sqrt{T/344}$
16	B	$521D^2$	$\sqrt{T/521}$
16	C	$688D^2$	$\sqrt{T/688}$

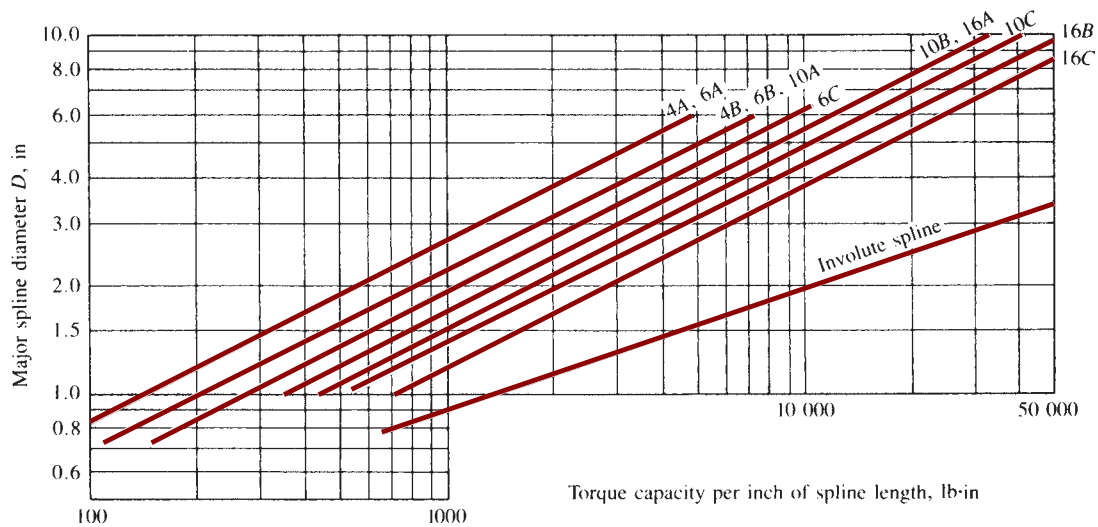
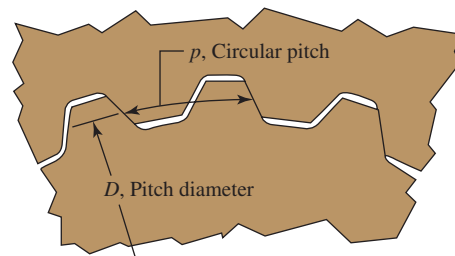
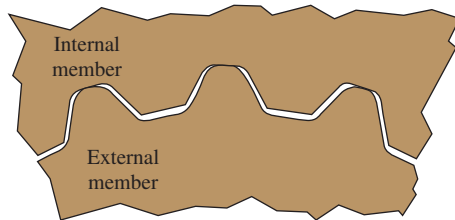


FIGURE 11-7 Torque capacity per inch of spline length, lb · in for straight-sided splines



(a) Side-fit spline



Note chamfer on tips of
external spline teeth

(b) Major diameter fit spline

N = Number of spline teeth
 P = Diametral pitch
 $D = N/P$ = Pitch diameter
 $p = \pi/P$ = Circular pitch

Minor diameter:

$$\text{Internal: } \frac{N-1}{P}$$

$$\text{External: } \frac{N-1.35}{P}$$

Major dia.:

$$\text{Internal: } \frac{N+1.35}{P} \quad \text{side fit}$$

$$\frac{N+1}{P} \quad \text{major diameter fit}$$

$$\text{External: } \frac{N+1}{P}$$

FIGURE 11-8 30° involute spline

specified. The *major diameter fit* produces accurate concentricity between the shaft and the mating element. In the *side fit*, contact occurs only on the sides of the teeth, but the involute form tends to center the shaft in the mating splined hub.

Figure 11-8 also gives some of the basic formulas for key features of involute splines in the U.S. Customary Unit System with dimensions in inches. (See Reference 5.) The terms are similar to those for involute spur gears, which are discussed more completely in Chapter 8. The basic spline size is governed by its *diametral pitch*, P :

$$P = N/D \quad (11-11)$$

where N = number of spline teeth

D = pitch diameter

The diametral pitch, then, is *the number of teeth per inch of pitch diameter*. Only even numbers of teeth from 6 to 60 are typically used. Up to 100 teeth are used on some 45° splines. Note that the pitch diameter lies *within* the tooth and is related to the major and minor diameters by the relationships shown in Figure 11-8.

The *circular pitch*, p , is the distance from one point on a tooth to the corresponding point on the next adjacent tooth, measured along the pitch circle. To find the

nominal value of p , divide the circumference of the pitch circle by the number of spline teeth. That is,

$$p = \pi D/N \quad (11-12)$$

But because $P = N/D$, we can also say

$$p = \pi/P \quad (11-13)$$

The *tooth thickness*, t , is the thickness of the tooth measured along the pitch circle. Then the theoretical value is

$$t = p/2 = \pi/2P$$

The nominal value of the width of the tooth space is equal to t .

Standard Diametral Pitches. The following are the 17 standard diametral pitches in common use:

2.5	3	4	5	6	8	10	12	16
20	24	32	40	48	64	80	128	

The common designation for an involute spline is given as a fraction, P/P_s , where P_s is called the *stub pitch* and is always equal to $2P$. Thus, if a spline had a diametral pitch of 4, it would be called a 4/8 pitch spline. For convenience, we will use only the diametral pitch.

Length of Splines. Common designs use spline lengths from $0.75D$ to $1.25D$, where D is the pitch diameter of the spline. If these standards are used, the shear strength of the splines will exceed that of the shaft on which they are machined.

Metric Module Splines. The dimensions of splines made to metric standards are related to the *module*, m , where

$$m = D/N \quad (11-14)$$

and both D and m are in millimeters. (See Reference 6.) Note that the symbol Z is used in place of N for the number of teeth in standards describing metric splines. Other features of metric splines can be found from the following formulas:

$$\text{Pitch diameter} = D = mN \quad (11-15)$$

$$\text{Circular pitch} = p = \pi m \quad (11-16)$$

$$\text{Basic tooth thickness} = t = \pi m/2 \quad (11-17)$$

Standard Modules. There are 15 standard modules:

0.25	0.50	0.75	1.00	1.25	1.50	1.75	2.00
2.50	3	4	5	6	8	10	

Design Aids

Refer References 9–11 for additional design guidelines. Reference 9 includes extensive tables of data for splines and offers application and design information. Reference 10 gives information on the application, operation, dimensioning, and manufacture of involute splines as applied to automotive applications. Data are included on allowable shear stress, allowable compressive stress, wear life factor, spline overload factor, and fatigue life factor. Reference 11 focuses on metric module splines.

11-6 OTHER METHODS OF FASTENING ELEMENTS TO SHAFTS

The following discussion will acquaint you with some of the ways in which power-transmitting elements can be attached to shafts without keys or splines. In most cases the designs have not been standardized, and analysis of individual cases, considering the forces exerted on the elements and the manner of loading of the fastening means, is necessary. In several of the designs, the analysis of shear and bearing will follow a procedure similar to that shown for keys. If a satisfactory analysis is not possible, testing of the assembly is recommended.

Pinning

With the element in position on the shaft, a hole can be drilled through both the hub and the shaft, and a pin can be inserted in the hole. Figure 11-9 shows three examples of this approach. The straight, solid, cylindrical pin is subjected to shear over two cross sections. If there is a

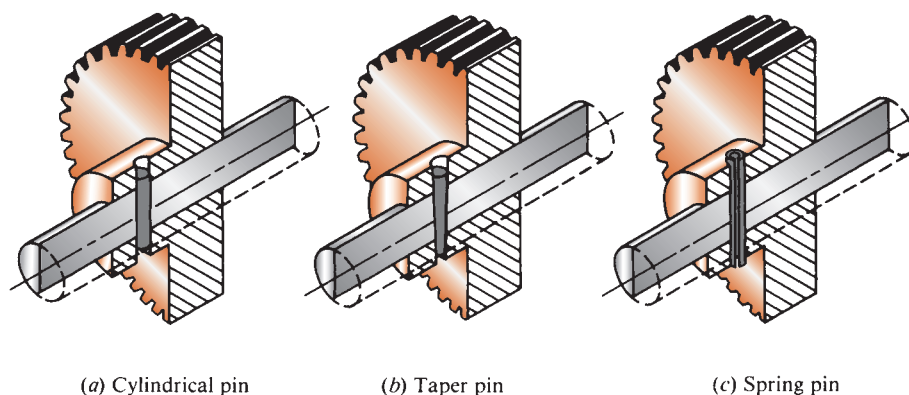


FIGURE 11-9 Pinning

force, F , on each end of the pin at the shaft/hub interface, and if the shaft diameter is D , then

$$T = 2F(D/2) = FD$$

or $F = T/D$. With the symbol d representing the pin diameter, the shear stress in the pin is

$$\tau = \frac{F}{A_s} = \frac{T}{D(\pi d^2/4)} = \frac{4T}{D(\pi d^2)} \quad (11-18)$$

Letting the shear stress equal the design stress in shear as before, solving for d gives the required pin diameter:

➤ Required Diameter for a Pin

$$d = \sqrt{\frac{4T}{D(\pi)(\tau_d)}} \quad (11-19)$$

Sometimes the diameter of the pin is purposely made small to ensure that the pin will break if a moderate overload is encountered, in order to protect critical parts of a mechanism. Such a pin is called a *shear pin*.

One problem with a cylindrical pin is that fitting it adequately to provide precise location of the hub and to prevent the pin from falling out is difficult. The *taper pin* overcomes some of these problems, as does the *split spring pin* shown in Figure 11-9(c). For the split spring pin, the hole is made slightly smaller than the pin diameter so that a light force is required to assemble the pin in the hole. The spring force retains the pin in the hole and holds the assembly in position. But, of course, the presence of any of the pin-type connections produces stress concentrations in the shaft. (See Internet site 1.)

Keyless Hub to Shaft Connections

Using a steel ring compressed tightly around a smooth shaft allows torque to be transmitted between the hub of a power-transmitting element and a shaft without having a key between the two elements. Figure 11-10 shows a commercially available product called a *Locking Assembly*™ from Ringfeder® Corporation that employs this principle.

The Locking Assembly™ employs steel rings with opposing mating tapers held together with a series of fasteners. With the Locking Assembly™ placed completely within a counter bore of the hub of virtually any kind of power-transmitting element such as a gear, sprocket, fan wheel, cam, coupling, or turbine rotor, the fasteners can then be tightened. Initially there is a small clearance between the inside diameter of the locking device and the shaft as well as the hub bore. This clearance facilitates easy assembly and positioning of the hub. After the hub is positioned in the desired location on the shaft, the fasteners are tightened to a specified torque in a specific sequence. As the bolts are tightened, they draw the opposing tapered rings together, generating a radial movement of the inner ring toward the shaft and a simultaneous outward movement of the outer ring toward the



(a) A variety of styles



(b) Locking assembly applied to a gear

FIGURE 11-10 Ringfeder® Locking Assemblies (Used by Ringfeder Power Transmission USA Corp.)

ID of the hub. Once the initial clearances are eliminated, further tightening of the bolts results in a high pressure against the shaft and the hub. When the bolts are properly torqued using a torque wrench, the final contact pressure combined with friction allows for the transmission of a predetermined and predictable amount of torque between the hub and the shaft.

The connection can transmit axial forces in the form of thrust loads as well as torque as with helical gears, for example. Table 11-7 lists examples of torque and axial force capacities for selected sizes of one model of the Ringfeder Locking Assemblies®. All the metric data are for catalog metric sizes with capacities given for torque in $\text{kN} \cdot \text{m}$ and axial force in kN . Data in the upper nonshaded part for the inch-size models are from catalog data with capacities given for torque in $\text{lb} \cdot \text{ft}$ and axial force in lb . The shaft sizes listed are fairly close to the metric sizes in the left part of the table to illustrate comparisons. The lower shaded part shows data in U.S. units converted from the metric data for the larger sizes because the maximum standard inch-size model is for a shaft size of 7.875 in.

Unlike a thermal or pressure fit connection, the locking device can be easily removed since it is a *mechanical shrink fit*. The pressures generated within the locking device itself allow the stresses to remain within the elastic limits of the materials. Removal is simply done by carefully loosening the screws, thus allowing the rings to slide apart and return the Locking Assembly™ to its original relaxed condition. The element can then be repositioned or removed at any time. Some locking devices have self-releasing taper angles, which allow them to self-release when the fasteners are loosened. Other locking devices have self-locking tapers that require the gentle pressing

TABLE 11-7 Capacities of Selected Sizes of Ringfeder Locking Assemblies

Metric sizes			Inch sizes		
Shaft size (mm)	Transmissible torque (kN · m)	Transmissible axial force (kN)	Shaft size (in)	Transmissible torque (lb · ft)	Transmissible axial force (lb)
25	0.460	30.0	1.000	337	8088
50	2.07	80.0	2.000	1808	21 696
75	5.81	160	3.000	4332	34 656
100	11.1	220	4.000	8489	50 934
150	28.0	380	6.000	22 762	91 048
200	63.5	640	7.875	46 707	142 345
300	183	1220	11.8	134 987	274 281
400	384	1920	15.7	283 252	431 655
600	896	2989	23.6	660 921	671 987
800	1550	3876	31.5	1 143 334	871 402
1000	2375	4749	39.4	1 751 145	1 067 670
			Converted from metric data		

Note: All unshaded data are from catalog listings.

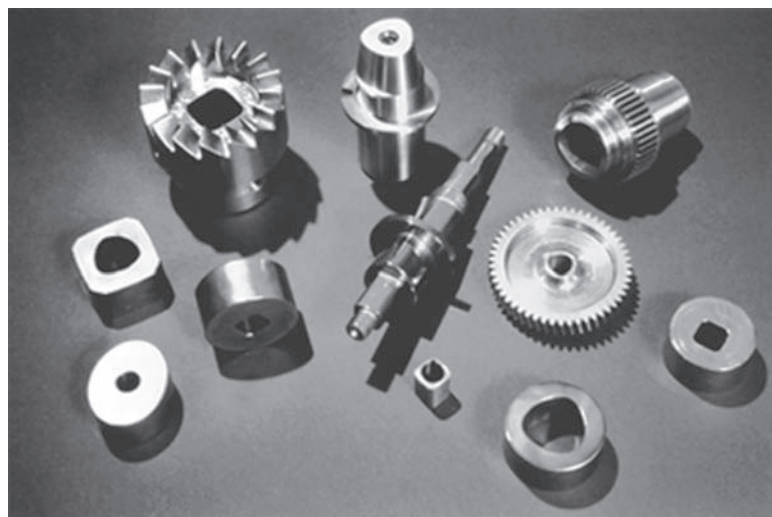
apart of the locking device parts. Different applications dictate which type of device is better suited.

Advantages of the keyless connection are the elimination of keys, keyways, or splines, and the cost of machining them; tight fit of the driving element around the shaft; the ability to transmit reversing or dynamically changing loads; and easy assembly, disassembly, and adjustment of the elements. General engineering information is provided on Internet site 2 for Ringfeder® Corporation for installation dimensions, tolerances, lubrication, and surface finishes required. The rated torque values are for use on solid shafts; hollow shafts require additional analysis.

Special considerations are necessary for the hub design to ensure that stresses remain below the yield strength of the hub material.

Polygon Hub to Shaft Connection

Figure 11-11 shows a shaft to hub connection that employs special mating polygon shapes to transmit torque without keys or splines. See Internet site 18 for available sizes and application information. German standards DIN 32711 and 32712 describe the forms. They can be produced on shaft sizes from 0.188 in



Polygon profile produced in a variety of products

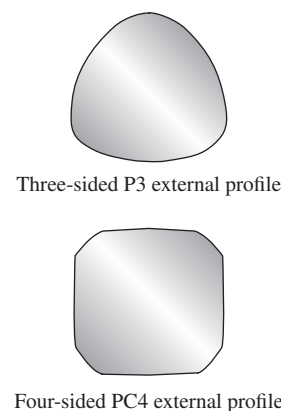


FIGURE 11-11 Polygon hub to shaft connections (Used by General Polygon Systems, Inc.)

(4.76 mm) to 8.00 in (203 mm). The three-sided configuration is called the P3 profile, and the four-sided design is called the PC4 profile. CNC turning and grinding can be used to produce the external form while broaching typically produces the internal form. Torque is transmitted by distributing the load on each side of the polygon, eliminating the shearing action inherent with keys or splines. Dimensions can be controlled with tight or press fits for precision, backlash-free location or with a sliding fit for ease of assembly.

Split Taper Bushing

A *split taper bushing* (see Figure 11–12) uses a key to transmit torque. Axial location on the shaft is provided by the clamping action of a split bushing having a small taper on its outer surface. When the bushing is pulled into a mating hub with a set of capscrews, the bushing is brought into tight contact with the shaft to hold the assembly in the proper axial position. The small taper locks the assembly together. Removal of the bushing is accomplished by removing the capscrews and using them in push-off holes to force the hub off the taper. The assembly can then be easily disassembled.

Set Screws

A *set screw* is a threaded fastener driven radially through a hub to bear on the outer surface of a shaft (see Figure 11–13). The point of the set screw is flat, oval, cone-shaped, cupped, or any of several proprietary forms. The point bears on the shaft or digs slightly into its surface. Thus, the set screw transmits torque by the friction between the point and the shaft or by the resistance of the material in shear. The capacity for torque transmission is somewhat variable, depending on the hardness of the shaft material and the clamping force created when the screw is installed. Furthermore, the screw may loosen during operation because of vibration. For these reasons, set screws should be used with care. Some manufacturers provide set screws with plastic inserts in the side among the threads. When the set screw is screwed into a tapped hole, the plastic is deformed by the threads and holds the

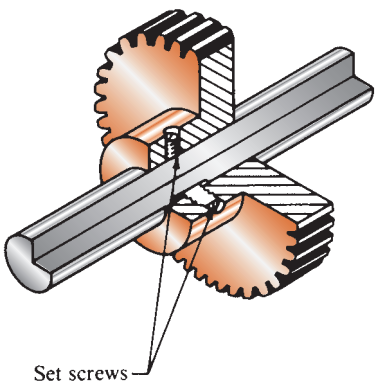


FIGURE 11–13 Set screws

screw securely, resisting vibration. Using a liquid adhesive also helps resist loosening. (See Internet site 21.)

Another problem with using set screws is that the shaft surface is damaged by the point; this damage may make disassembly difficult. Machining a flat on the surface of the shaft may help reduce the problem and also produce a more consistent assembly.

When set screws are properly assembled on typical industrial shafting, their force capability is approximately as follows (see Reference 9):

Screw Diameter (in)	Holding Force (lb)
1/4	100
3/8	250
1/2	500
3/4	1300
1	2500

Taper and Screw

The power-transmitting element (gear, sheave, sprocket, or other) that is to be mounted at the end of a shaft can be secured with a screw and a washer in the manner shown in Figure 11–14(a). The taper provides good concentricity and moderate torque transmission capacity. Because of the machining required, the connection is fairly costly. A modified form uses the tapered shaft with

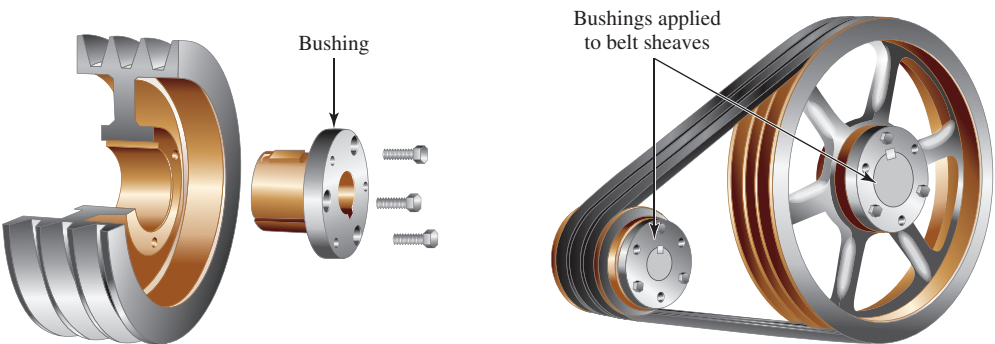


FIGURE 11–12 V-belt sheaves having three grooves with a split taper bushing for shaft mounting

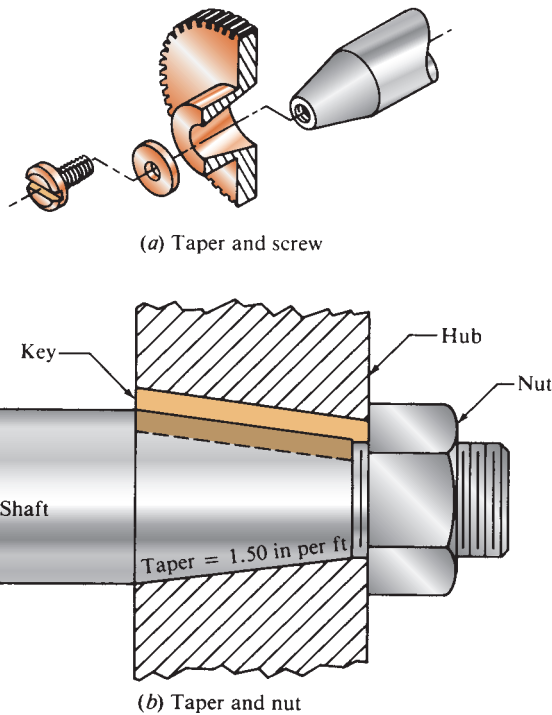


FIGURE 11-14 Tapered shaft for fastening machine elements to shafts

a threaded end for the application of a nut, as shown in Figure 11-14(b). The inclusion of a key lying in a keyseat machined parallel to the taper increases the torque-transmitting capacity greatly and ensures positive alignment.

Press Fit

Making the diameter of the shaft greater than the bore diameter of the mating element results in an interference fit. The resulting pressure between the shaft and the hub permits the transmission of torque at fairly high levels, depending on the degree of interference. This is discussed in more detail in Chapter 13. Sometimes the press fit is combined with a key, with the key providing the positive drive and the press fit ensuring concentricity and holding the part in position axially.

Molding

Plastic and die cast gears can be molded directly to their shafts. Often the gear is applied to a location that is knurled to improve the ability to transmit torque. A modification of this procedure is to take a separate gear blank with a prepared hub, locate it over the proper position on a shaft, and then cast zinc into the space between the shaft and the hub to lock them together.

11-7 COUPLINGS

The term *coupling* refers to a device used to connect two shafts together at their ends for the purpose of transmitting power. There are two general types of couplings: rigid and flexible.

Rigid Couplings

Rigid couplings are designed to draw two shafts together tightly so that no relative motion can occur between them. This design is desirable for certain kinds of equipment in which precise alignment of two shafts is required and can be provided. In such cases, the coupling must be designed to be capable of transmitting the torque in the shafts.

Figure 11-15 shows three styles of rigid couplings. (See Internet site 3.) For the style in part (a), the two flanged parts are installed on the shafts to be coupled and the bolts in the flange are drawn together by a series of bolts. The load path is then from the driving shaft to its flange, through the bolts, into the mating flange, and out to the driven shaft. The torque places the bolts in shear. The total shear force on the bolts depends on the radius of the bolt circle, $D_{bc}/2$, and the torque, T . That is,

$$F = T/(D_{bc}/2) = 2T/D_{bc}$$

Letting n be the number of bolts, the shear stress in each bolt is

$$\tau = \frac{F}{A_s} = \frac{F}{n(\pi d^2/4)} = \frac{2T}{D_{bc}n(\pi d^2/4)} \quad (11-20)$$

Letting the stress equal the design stress in shear and solving for the bolt diameter, we have

Required Bolt Diameter for Rigid Coupling

$$d = \sqrt{\frac{8T}{D_{bc}n\pi\tau_d}} \quad (11-21)$$

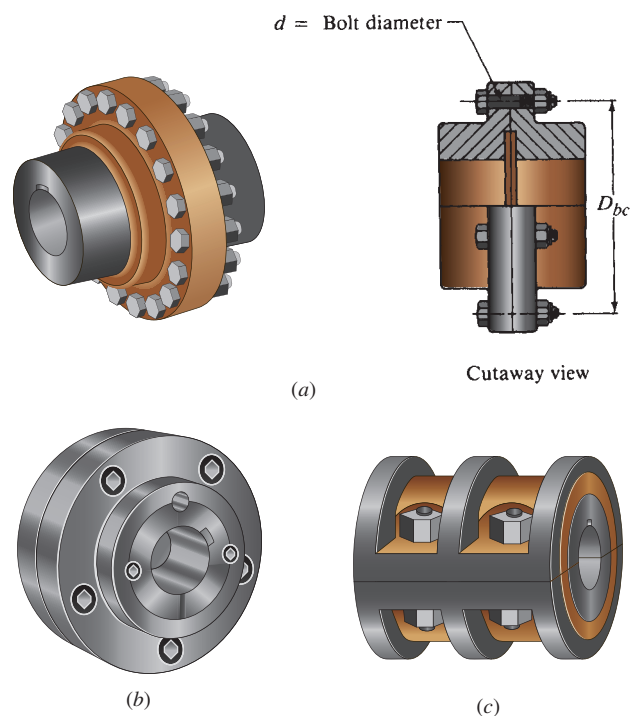


FIGURE 11-15 Rigid coupling

Notice that this analysis is similar to that for pinned connections in Section 11–6. The analysis assumes that the bolts are the weakest part of the coupling.

The style in Figure 11–15(b) is called a *taper-lock rigid coupling*, available in eight sizes capable of transmitting torques from 5050 lb·in to 254 500 lb·in (655 N·m to 33 000 N·m) and accommodating shaft sizes from 0.50 in to 6.00 in (12.7 mm to 152 mm). Figure 11–15(c) is called a *ribbed rigid coupling* available in bore sizes up to 7.00 in (178 mm) and can transmit up to 254 400 lb·in (33 000 N·m) of torque. (See Internet site 3.)

Rigid couplings should be used only when the alignment of the two shafts can be maintained very accurately, not only at the time of installation but also during operation of the machines. If significant angular, radial, or axial misalignment occurs, stresses that are difficult to predict and that may lead to early failure due to fatigue will be induced in the shafts. These difficulties can be overcome by the use of flexible couplings.

Flexible Couplings

Flexible couplings are designed to transmit torque smoothly while permitting some axial, radial, and angular misalignment. The flexibility is such that when misalignment does occur, parts of the coupling move with little or no resistance. Thus, no significant axial or bending stresses are developed in the shaft.

Many types of flexible couplings are available commercially, as shown in Figures 11–16 through 11–25. Each is designed to transmit a given limiting torque. The manufacturer's catalog lists the design data from which you can choose a suitable coupling. Remember that torque equals power divided by rotational speed. So for a given size of coupling, as the speed of rotation increases, the amount of power that the coupling can transmit also increases, although not always in direct proportion. Of course, centrifugal effects determine the upper limit of speed.

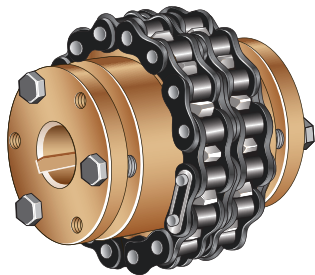


FIGURE 11–16 Chain coupling. Torque is transmitted through a double roller chain. Clearances between the chain and the sprocket teeth on the two coupling halves accommodate misalignment

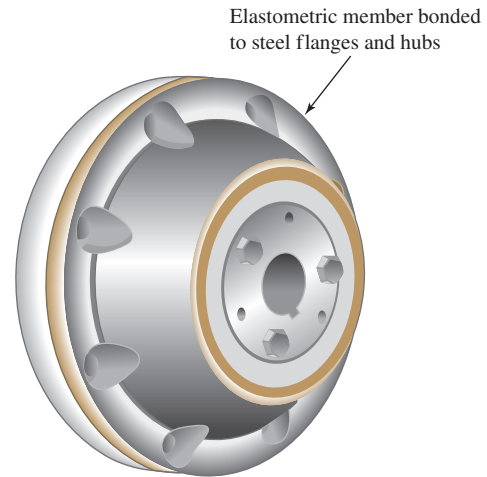


FIGURE 11–17 Elastomeric coupling. The features of this coupling are that it (1) generally minimizes torsional vibration; (2) cushions shock loads; (3) compensates for parallel misalignment up to 1/32 in; (4) accommodates angular misalignment of $\pm 3^\circ$; and (5) provides adequate end float, $\pm 1/32$ in

The degree of misalignment that can be accommodated by a given coupling should be obtained from the manufacturer's catalog data, with values varying with the size and design of the coupling. Small couplings may be limited to parallel misalignment of 0.005 in, although larger couplings may allow 0.030 in or more. Typical allowable angular misalignment is $\pm 3^\circ$. Axial movement allowed, sometimes called *end float*, is up

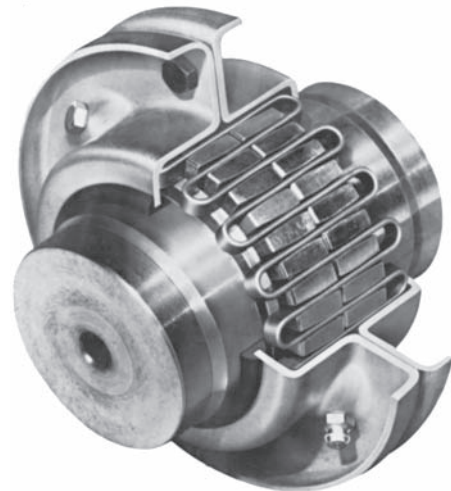


FIGURE 11–18 Flexible grid coupling. Torque is transmitted through a flexible spring steel grid. Flexing of the grid permits misalignment and makes it torsionally resilient to resist shock loads

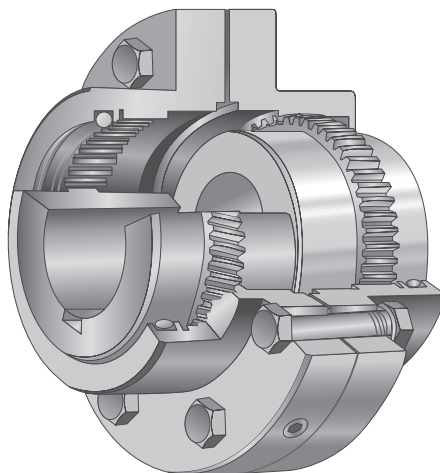


FIGURE 11-19 Gear coupling. Torque is transmitted between crown-hobbed teeth from the coupling half to the sleeve. The crown shape on the gear teeth permits misalignment



FIGURE 11-20 Bellows coupling. The inherent flexibility of the bellows accommodates the misalignment (Stock Drive Products, New Hyde Park, NY)

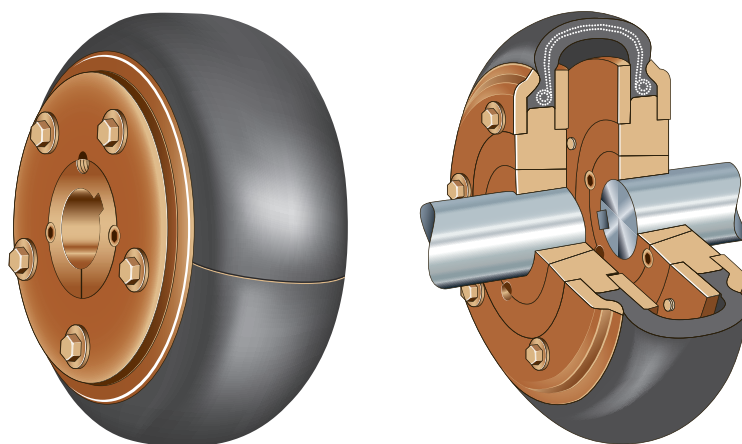


FIGURE 11-21 PARA-FLEX® coupling. Using an elastomeric element permits misalignment and cushions shocks

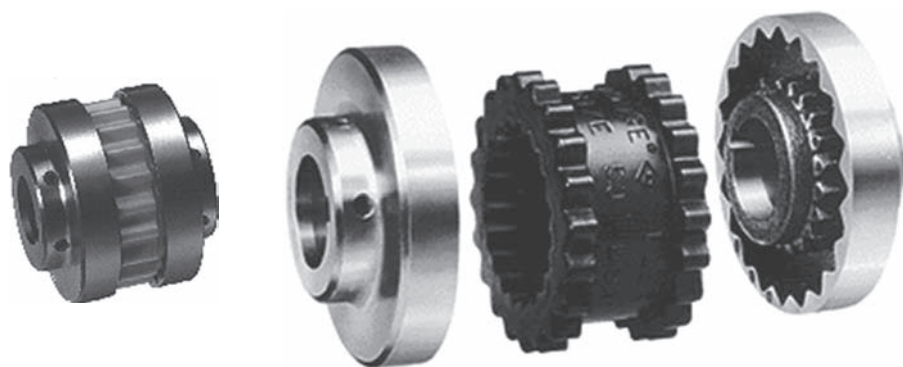


FIGURE 11-22 D-Flex coupling used mainly for connecting pumps and motors (Baldor/Dodge, Greenville, SC)

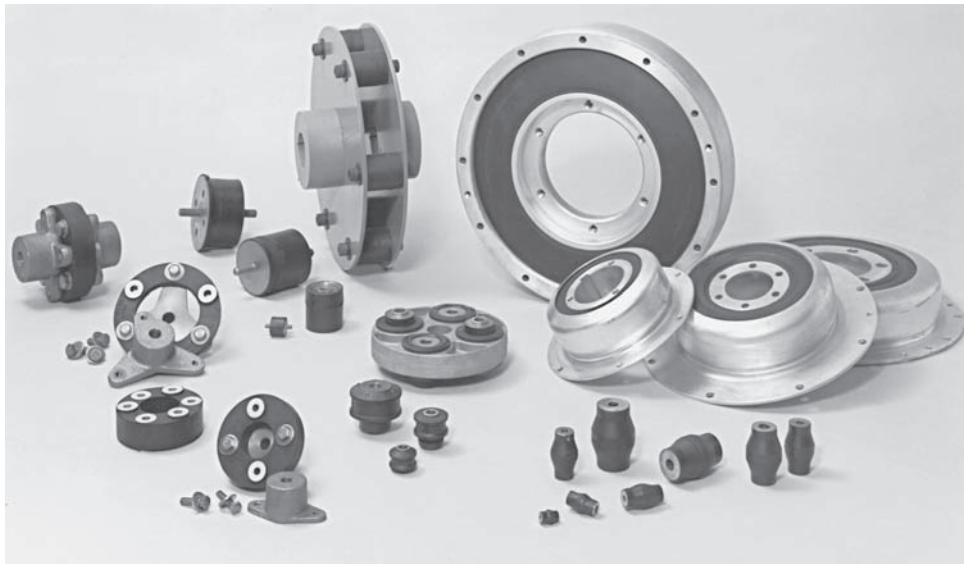
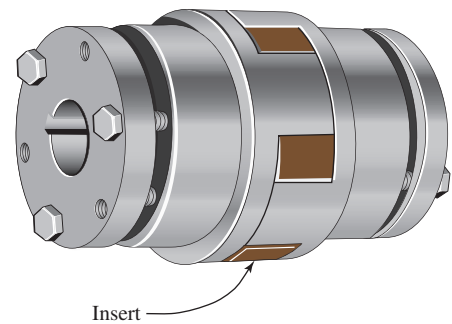
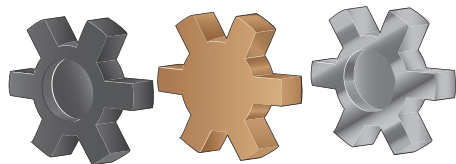


FIGURE 11-23 Dynaflex® coupling. Torque is transmitted through elastomeric material that flexes to permit misalignment and to attenuate shock loads (Courtesy of LORD Corporation)



(a) Assembled coupling



Neoprene
(normal-duty
applications)

Bronze,
oil-impregnated
(low-speed,
high-torque
applications)

Polyurethane
(extra capacity
at medium to
high speed)

(b) Types of inserts

FIGURE 11-24 Jaw-type coupling

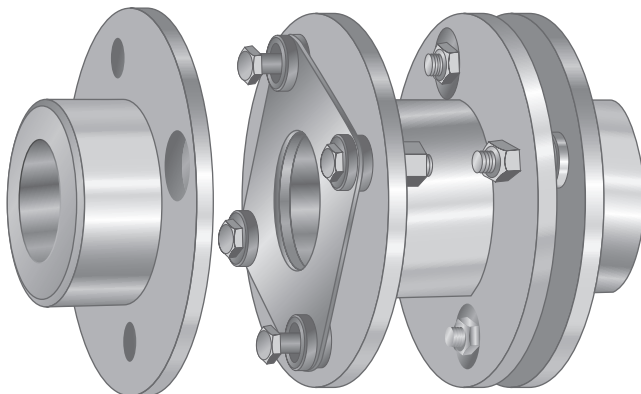


FIGURE 11-25 Flexible disk-type coupling. Torque is transmitted from hubs through laminated flexible elements to the spacer (T. B. Wood's Incorporated, Chambersberg, PA)

to 0.030 in for many types of couplings. (See References 1–4 and Internet sites 3–8, 17, 19, 22, and 24 for the manufacturers.)

The following is an example of the procedure used to select flexible couplings. We are focusing on the flexible disc type as illustrated in Figures 11–26 and 11–27.

Figure 11–26 shows a spacer-type flexible coupling, specifically used in pump drive applications. The spacer allows the coupling to be removed without moving the pump or motor, providing access to change the pump seal.

Figure 11–27 shows a floating shaft-type flexible coupling used for long-span applications. Each side of the flexible coupling needs to be rigidly supported by the connecting shafts.

The data given in Tables 11–8 and 11–9 can be used for problems in this book and they are only examples of the types of data available; different suppliers may show their data in different formats. See the Internet sites listed earlier.

Flexible Disc Coupling Selection Procedure

The procedure uses a pump drive as an example and assumes that the following data are known: (Using U.S. units)

- Power being transmitted by the coupling in horsepower, P
- The normal rotational speed of the coupling in rpm, n

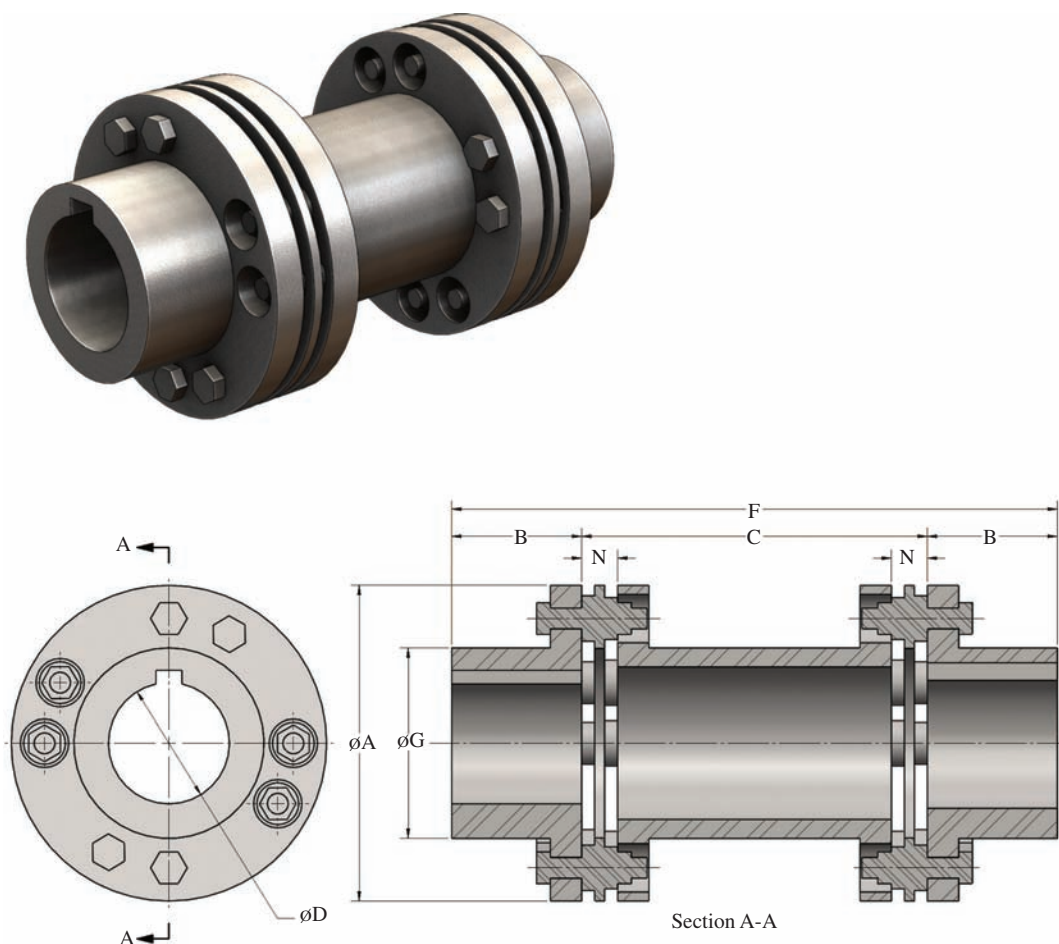


FIGURE 11–26 Flexible disc coupling—Spacer type

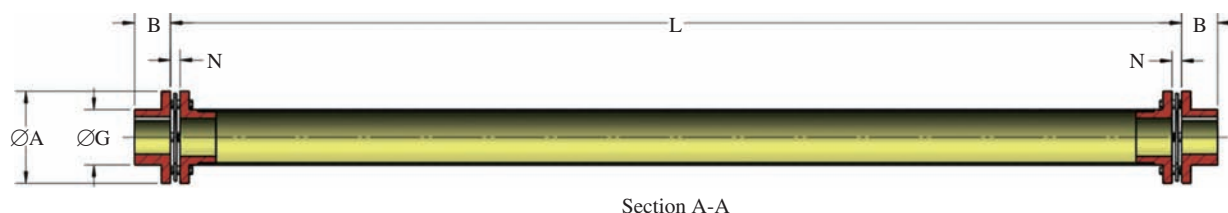


FIGURE 11–27 Flexible disc coupling—Floating shaft type

TABLE 11-8 Service Factor of Motor- and Turbine-Driven Equipment

Application	Service factor	Application	Service factor
Blowers		Centrifugal pump	
Centrifugal	1.0	General Duty	1.0
Lobe	1.5	Boiler Feed	1.0
Vane	1.5	Slurry	1.5
Compressors		Dredge	2.0
Centrifugal	1.0	Reciprocating pump	
Lobe	1.5	Double Acting	2.0
Vane	1.5	Single Acting 1 or 2 cylinders	2.5
Screw	1.5	Single Acting 3 or more cylinders	2.0
Fans		Pump	
Centrifugal	1.5	Rotary	1.5
Axial	1.5	Lobe	1.5
Light-Duty Blower	1.0	Vane	1.5
Conveyors		Printing presses	
Uniformly loaded	1.5	General Equipment	1.5

TABLE 11-9 Spacer-Type Flexible Coupling Engineering Data

Coupling size	Max bore	A	B	C	F	G	N	Max. horsepower per 100 rpm			Max		Max. continuous torque (lb · in)	Peak overload torque (lb · in)
								Service factor			rpm			
								1	1.5	2	Not balanced	balanced		
A	7/8	2 11/16	1 1/8	3 1/2	5 3/4	1 7/16	21/64	0.65	0.43	0.33	6000	8200	410	820
B	1 5/16	3 7/32	1 3/8	3 1/2	6 1/4	1 11/16	7/16	1.30	0.87	0.65	6000	7500	820	1640
C	1 3/8	3 27/32	1 5/8	3 1/2	6 3/4	2 1/16	1/2	2.20	1.50	1.10	5500	6800	1400	2800
D	1 7/8	4 9/16	1 7/8	3 1/2	7 1/4	2 3/4	1/2	3.10	2.10	1.60	5000	6300	1950	3900
E	2 1/4	5 11/32	2 1/8	3 1/2	7 3/4	3 9/32	19/32	5.60	3.70	2.80	4600	5700	3530	7060
F	2 5/8	6 1/16	2 5/8	5	10 1/4	3 25/32	31/32	10.00	6.70	5.00	4100	5500	6300	12 600
G	3	7	3	5	11	4 7/16	1 1/32	15.70	10.50	7.90	3700	5000	9900	19 800

- The maximum rotational speed that the coupling will see during any operating condition.
- The desired distance between the pump shaft and the drive motor shaft
- The diameters of the drive motor shaft and the pump shaft
- The nature of the application, as listed in Table 11-8, used to determine a service factor

Selection Procedure

1. Determine the power capacity required per 100 rpm:

$$\frac{\text{HP}}{100 \text{ rpm}} = \frac{\text{HP} \cdot 100}{n_{\text{coupling}}}$$

2. Determine the service factor, SF , for the given application from Table 11-8.
3. Select one or more candidate couplings from Table 11-9 using the computed power rating capacity per 100 rpm at the given service factor.
4. Verify that the maximum bore size of the coupling will fit the given shaft diameters. Bore diameters smaller than the maximum can be ordered.
5. Check that the maximum speed of the coupling does not exceed the maximum running speed of the equipment.
6. Verify that the system running torque and the maximum torque due to start up or shock loading does

not exceed the coupling's peak continuous torque and overload torque. Use the equation

$$\text{Torque} = \text{Power/rotational speed} = P/n$$

7. Summarize the pertinent data for sizes and allowable misalignment.

Example Problem 11-2

Figure 11-28 shows a 30-hp electric motor driving a centrifugal pump with a flexible coupling connecting the two shafts. The motor shaft diameter is 1.875 in and the pump shaft diameter is 1.75 in. The normal rotational speed of the drive motor is 1760 rpm. The desired spacing between the motor shaft and the pump shaft is 4.0 in minimum. Specify a suitable coupling using the sample data in Table 11-9.

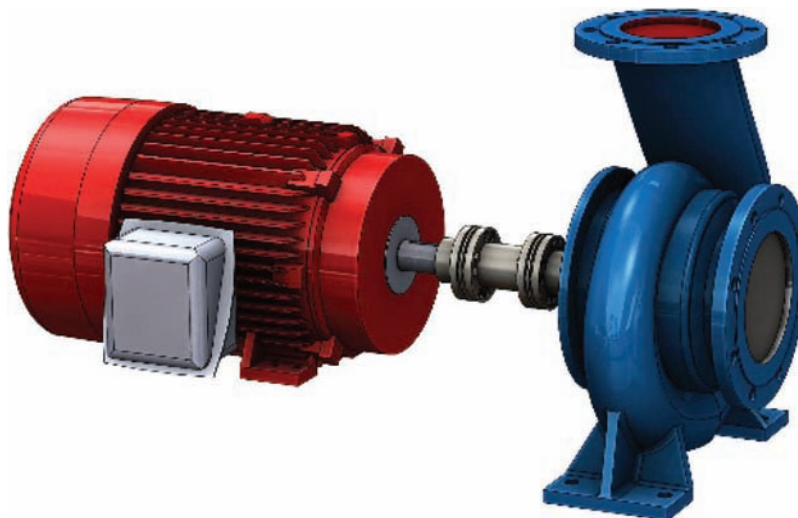


FIGURE 11-28 Electric motor driving a centrifugal pump through a flexible coupling

Solution

1. Determine the power capacity required per 100 rpm:

$$30 \text{ hp} (100)/n = 30 \text{ hp}(100)/1760 \text{ rpm} = 1.70 \text{ hp}/100 \text{ rpm}$$

2. Determine the service factor for the given application from Table 11-8:

For a centrifugal pump drive, $SF = 1.0$.

3. Select one or more candidate couplings from Table 11-9 using the 1.70 hp/100 rpm at 1.0 service factor.

We choose coupling sizes C and D as candidates:

1. Coupling C is rated at 2.2 hp/100 rpm.
2. Coupling D is rated at 3.3 hp/100 rpm.

4. Check the maximum allowable bore sizes for the two candidate couplings:

1. Coupling C has a maximum bore size of 1 3/8 in (1.375 in)—Too small for either shaft.
2. Coupling D has a maximum bore size of 1 7/8 in (1.875 in)—Fits both shaft sizes.

We continue the selection process using Coupling D.

5. Check that the maximum speed of the selected Coupling D:

Maximum speed is 5000 rpm > 1760 rpm of the drive motor—Satisfactory

6. Check the running torque for Coupling D:

$$\text{Actual running torque, } T = \frac{P}{n} = \frac{30 \text{ hp}}{1760 \text{ rev/min}} \cdot \frac{33\,000 \frac{\text{lb} \cdot \text{in}}{\text{min}}}{1.0 \text{ hp}} \cdot \frac{1 \text{ rev}}{2\pi \text{ rad}} \cdot \frac{12 \text{ in}}{\text{ft}} = 1074 \text{ lb} \cdot \text{in}$$

The listed maximum continuous torque for Coupling D is 1950 lb · in—Satisfactory

The allowable peak overload torque is 3900 lb · in and it is unlikely that this level will be reached.

7. Pertinent data for Coupling D:

- The spacer length is 3.50 in.
- The length between the flanges and the spacer ends is 0.50 in on each end.
- Assuming that the motor and pump shafts are inserted to the end of the two parts of the coupling, the total spacing is

$$\text{Spacing length} = 3.50 \text{ in} + 0.50 \text{ in} + 0.50 \text{ in} = 4.50 \text{ in}$$

This is greater than the specified minimum length of 4.00 in—Satisfactory.

d. Specify the bores of the two ends of Coupling D:

Left (motor) side: Bore = 1.875 in

Right (pump) side: Bore = 1.75 in

e. Total length of assembled Coupling D (see Table 11–9):

$$\text{Length} = C + 2(B) = 3.50 \text{ in} + 2(1.875 \text{ in}) = 7.25 \text{ in} \text{ (Dimension } F \text{ in the table)}$$

f. Diameters:

End pieces and middle tube: 2.75 in

OD of flanges: 4.5625 in

g. A key is required for both end pieces, specified by the motor and pump supplier.

h. The manufacturer's data must be consulted to determine the allowable misalignment. Typical values for this type and size of coupling are as follows:

$$\text{Maximum angular misalignment} = 0.50^\circ$$

$$\text{Allowable axial end float} = \pm 0.057 \text{ in}$$

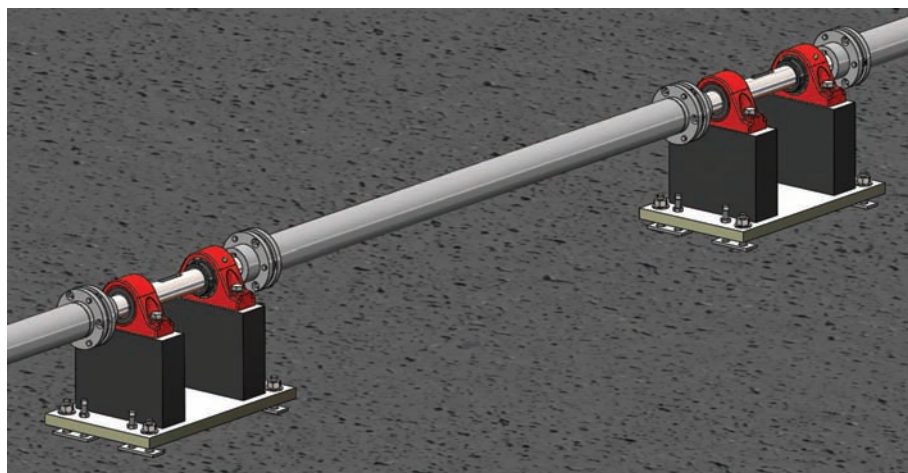
The procedure for selecting a floating shaft-type flexible coupling is virtually the same as that just demonstrated for the spacer-type coupling in Example Problem 11–2 with only small differences. Refer to Table 11–10

for the floating shaft-type coupling and compare the data with that in Table 11–9 for the spacer-type coupling.

Whereas the spacer-type coupling has a specified length value indicated by the *C* dimension in Table 11–9,

TABLE 11–10 Floating Shaft-Type Flexible Coupling Engineering Data

Coupling size	Max Bore	A	B	G	N	Max. Horsepower per 100 rpm			Max. Span <i>L</i> [in]			Max. Continuous torque (lb · in)	Peak overload torque (lb · in)
						Service Factor			For various angular speeds				
						1	1.5	2	500 rpm	1000 rpm	1500 rpm		
A2	7/8	2 11/16	1 1/8	1 7/16	21/64	0.65	0.43	0.33	94	94	88	410	820
B2	1 5/32	3 7/32	1 3/8	1 11/16	7/16	1.30	0.87	0.65	102	102	93	820	1640
C2	1 3/8	3 27/32	1 5/8	2 1/16	1/2	2.20	1.50	1.10	114	114	105	1400	2800
D2	1 7/8	4 11/32	1 7/8	2 3/4	17/32	3.90	2.60	2.00	133	133	122	2460	4920
E2	2 1/4	5 7/16	2 1/8	3 9/32	9/16	6.20	4.10	3.10	161	147	135	3900	7800
F2	2 5/8	5 13/16	2 5/8	3 25/32	19/32	11.20	7.50	5.60	172	157	144	7100	14 200
G2	3 1/8	6 5/8	3	4 1/2	15/32	19.40	12.90	9.70	191	173	160	12 200	24 400



(a) A floating shaft-type coupling mounted between two rigid supports



(b) Side view of the coupling in (a)

FIGURE 11-29 Example of an installation for a floating shaft-type coupling

the floating shaft-type coupling does not and that column has been deleted in Table 11-10. However, Table 11-10 shows new data for “Max Span L ” toward the right side of the table. The maximum values listed are dependent on the rotational speeds because of the dynamics of long rotating cylinders. Therefore, an additional Step 8 can be added to the procedure given earlier:

8. Specify the length L for the coupling, ensuring that it is smaller than the Max Span L value.

Figure 11-29 shows the floating shaft-type coupling installed in a typical machinery arrangement.

11-8 UNIVERSAL JOINTS

When an application calls for accommodating misalignment between mating shafts that is greater than 3° typically provided by flexible couplings, a *universal joint* is often employed. Figures 11-30 to 11-33 show some of the styles that are available. See Internet sites 5, 6, 9, 10, and 22 for commercially available universal joints. Angular misalignments of up to 45° are possible at low rotational speeds with single universal joints like that shown in Figure 11-30(a), consisting of two yokes, a center bearing block, and two pins that pass through the block at right angles. Approximately 20° to 30° is more reasonable for speeds above 10 rpm. Single universal joints have the disadvantage that the rotational

speed of the output shaft is nonuniform in relation to the input shaft.

A double universal joint, shown in Figure 11-30(b), allows the connected shafts to be parallel and offset by large amounts. Furthermore, the second joint cancels the nonuniform oscillation of the first joint so the input and output shafts rotate at the same uniform speed.

Figure 11-31 shows a vehicular universal joint connecting an engine or transmission to the drive wheels used in some rear-wheel-drive cars, light and heavy duty trucks, agricultural equipment, and construction vehicles. The spider assembly contains needle-bearing rollers on each arm. The right end shows a ball stud yoke, a flange yoke, and a center coupling yoke that make up a *double Cardan universal joint*. Another style, called a *constant velocity joint*, or simply a *CV joint*, is often used as a key component of front-wheel-drive and all-wheel-drive vehicle drivelines.

Figure 11-32 shows a heavy-duty industrial type double universal joint. Some of this type will have a two-part connecting tube that is splined to allow for sizable changes in axial position as well as accommodating the angular or parallel misalignment.

Figure 11-33 shows a novel design called the CornayTM universal joint that produces true constant velocity of the output shaft through all drive angles up to 90° . Compared with standard universal joint designs, the CornayTM joint can operate at higher speeds, carry



(a) Single universal joint components



(b) Double universal joint

FIGURE 11-30 Single and double universal joints (Curtis Universal Joint Co., Inc., Springfield, MA)

higher torque levels, and produce less vibration. See Internet site 10.

Manufacturers' literature should be consulted for specifying a suitable size of universal joint for a given application. The primary variables involved are the

torque to be transmitted, the rotational speed, and the angle at which the joint will be operating. A *speed/angle factor* is computed as the product of the rotational speed and the operating angle. From this value, an *operating use factor* is determined that is applied to the basic torque load to compute the required rated torque for the joint. All manufacturers provide such data in their catalogs.

Some drive shafts incorporating universal joints with relatively long, tubular sections between the end joints are often called *torque tubes*. This terminology is used in the following section giving an example of the selection of such a torque tube for an industrial machine.

Procedure to select an industrial universal joint or torque tube

As stated above, the selection of a universal joint or torque tube should be guided by the catalog data of the specific brand and model to be specified for the joint. The lists of Internet sites at the end of this chapter include information about several such manufacturers.

Here we show an example of a typical selection procedure for a torque tube of the type shown in Figure 11-34. We use data from Tables 11-11 to 11-13 that are representative of industry data. The uses of the tables are described within the following procedure.

TABLE 11-11 Service Factor for Industrial Universal Joint Applications

Application	Service factor
Blower	1.5
Conveyor	1.5
Fans	2.5
Food Industry	1.5
Centrifugal Pump	1
Reciprocating Pump	2
Paper Mills Press Roll Drives	2.5
Steel Mills: Cold Rolling Mills	5

TABLE 11-12 Industrial Universal Joint

Series	Peak torque rating (lb · in)	Bearing life factor	Min L (in)	Length compensation ΔL (in)	A (in)	B (in)	C (in)	D (in)	E (in)	F (in)	α (deg)	Maximum operating speed (rpm)	Speed to angle factor
A	800	230	13.66	3.12	1.38	2.375	3.88	2.50	3/8	3.125	20	5000	16 250
B	1240	366	14.96	3.62	1.56	2.750	4.56	3.00	7/16	3.750	20	5000	16 250
C	1500	443	15.81	3.47	1.69	2.750	4.56	3.50	7/16	3.750	22	5000	16 250
D	2000	591	15.5	2.5	2.00	3.750	5.88	3.50	1/2	4.750	22	5000	16 250
E	2400	756	15.75	2.5	2.00	3.750	5.88	3.50	1/2	4.750	22	5000	16 250

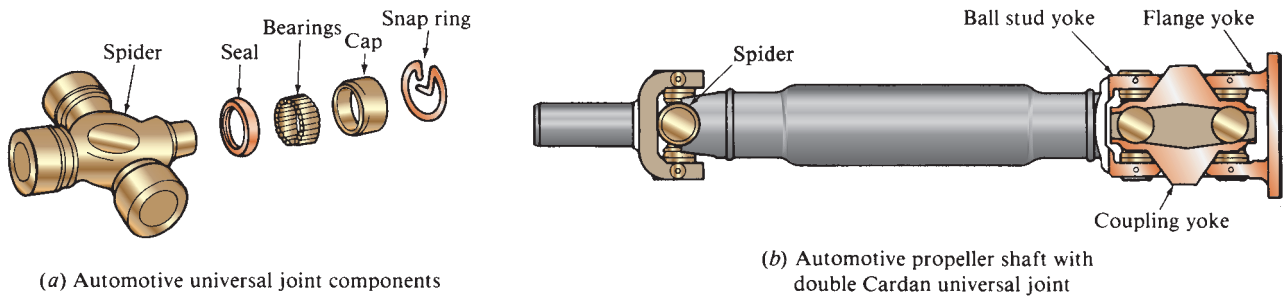


FIGURE 11-31 Automotive universal joints

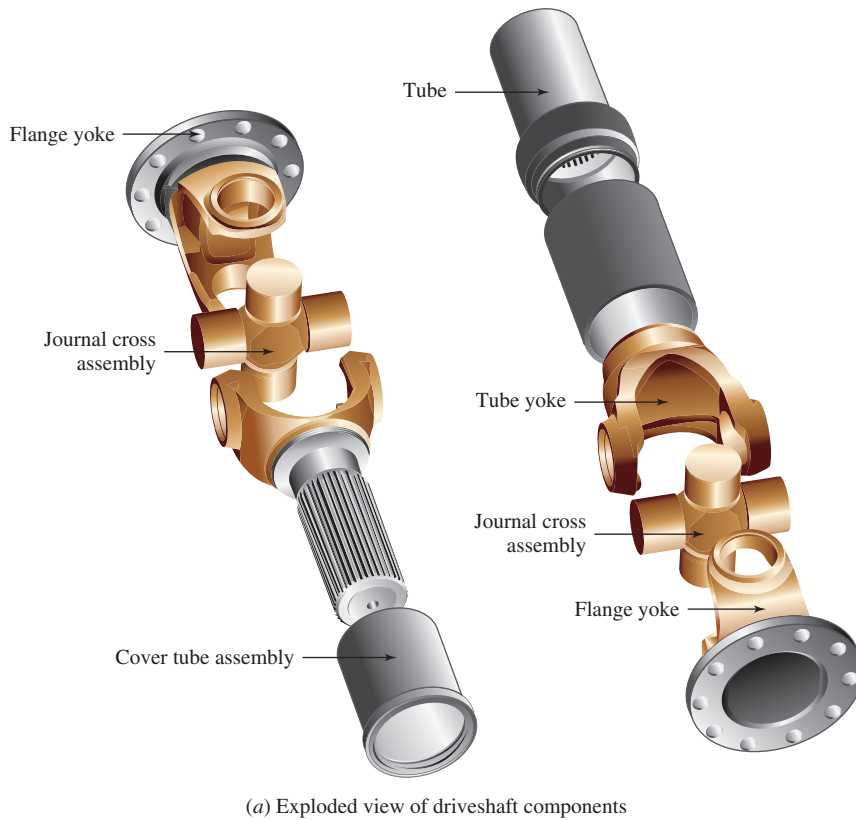


FIGURE 11-32 Industrial drive shaft incorporating a pair of universal joints (GWB - A Dana Brand)



FIGURE 11-33 Cornay™ universal joint
(Drive Technologies, Inc., Longmont, CO)

TABLE 11-13 Adjusted Normal Torque

Prime mover	Service factor
Electric Motor	1
Gas Engine	1.75
Diesel Engine	2

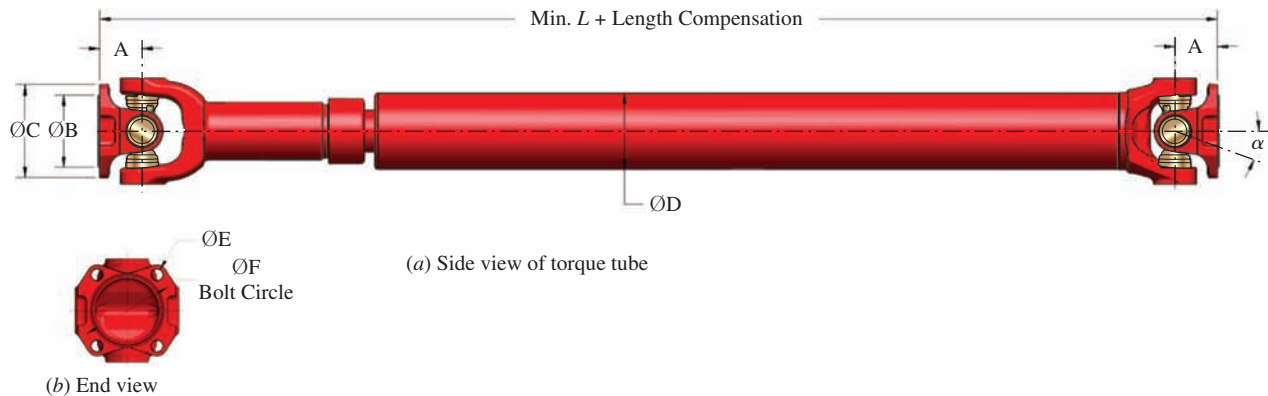


FIGURE 11-34 Type of torque tube used in the selection procedure

Typical data required to select a torque tube include the following:

1. Power to be transmitted by the torque tube.
2. Rotational speed of the torque tube, n .
3. The approximate length available between the two ends of the torque tube.
4. The expected angle of inclination of the torque tube.
5. The required life of the torque tube assembly.

With these data available, the following procedure can be used to specify a suitable torque tube.

1. Calculate the *equivalent torque* in the unit of lb·ft, using the following equations and a service factor from Table 11-11 for the particular application.

$$\text{Calculated Torque} = \frac{\text{Power}}{n}$$

$$\text{Equivalent Torque} = \text{Calculated Torque} \cdot \text{Service Factor}$$

2. Select a universal joint from Table 11-12, ensuring that the Equivalent Torque is less than the *Peak Torque Rating* of the joint.

3. Calculate the expected L_{10} life of the bearings using the following equation and the data for the selected joint.

$$L_{10} = \left(\frac{1.5 \cdot 10^7}{n \cdot \beta} \right) \cdot \left(\frac{\text{Bearing Life Factor}}{\text{Adjusted Normal Torque}} \right)^{10/3}$$

where

n : rotational speed of the universal joint (rpm)

β : Universal joint operating angle (degrees)

Bearing Life Factor: from Table 11-12

Adjusted normal torque in lb·ft from the following equation, using data from Table 11-13

Adjusted Normal Torque =

Equivalent Torque · Prime Mover Service Factor

4. Ensure that the rotational speed of the universal joint does not exceed the *maximum safe operating speed* (MSOS) from Table 11-12.
5. Verify that the universal joint operating speed times the operating angle is less than the *speed to angle factor* from Table 11-12.

$$n \cdot \beta < \text{Speed to angle factor}$$

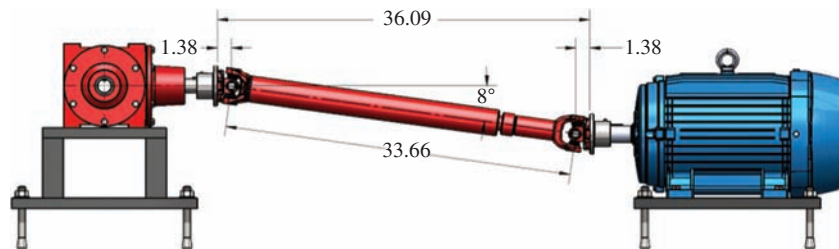
6. Specify pertinent dimensions and other data for the specified joint from Table 11-12.

Example Problem 11-3

Figure 11-35 shows part of a piece of machinery for the food industry in which an electric motor is driving a right-angle gearbox through a torque tube. The motor supplies 20 hp at a rotational speed of 1000 rpm. The nominal length between the motor and the gearbox is 30 in and the angle of inclination is 8°. Specify a suitable torque tube.



(a) Pictorial view of the drive system



(b) Side view

FIGURE 11-35 Electric motor driving a gearbox through a torque tube**Solution**

Given: Power = 20 hp; Rotational speed = 1000 rpm; Drive shaft joint angle = $\beta = 8^\circ$
 Use the **Procedure to select an industrial universal joint or torque tube**.

Results: 1. Calculate the *equivalent torque* in the unit of lb · ft, using the following equations and a service factor from Table 11-11 for the particular application.

From Table 11-11, the application service factor = 1.5

$$\text{Calculated Torque} = \frac{\text{Power}}{n} = \frac{20 \text{ hp}}{1000 \text{ rpm}} \cdot \frac{33\,000 \text{ lb} \cdot \text{ft}/\text{min}}{1 \text{ hp}} \cdot \frac{1 \text{ rev}}{2\pi \text{ rad}} = 105 \text{ ft} \cdot \text{lb}$$

$$\text{Equivalent Torque} = \text{Calculated Torque} \cdot \text{Service Factor} = (105 \text{ lb} \cdot \text{ft})(1.5) = 157.6 \text{ lb} \cdot \text{ft}$$

2. Select a universal joint from Table 11-12. The Series A joint has a peak torque rating of 800 lb · ft. Use this as a preliminary selection.
3. Calculate the expected L_{10} life of the bearings. The bearing life factor for the Series A joint is 230. The prime mover service factor from Table 11-13 for an electric motor is 1.0. Therefore, the Adjusted Normal Torque is the same as the equivalent torque, 157.6 lb · ft.

$$L_{10} = \left(\frac{1.5 \cdot 10^7}{n \cdot \beta} \right) \cdot \left(\frac{\text{Bearing Life Factor}}{\text{Adjusted Normal Torque}} \right)^{10/3}$$

$$L_{10} = \left(\frac{1.5 \cdot 10^7}{1000 \text{ rpm} \cdot 8^\circ} \right) \cdot \left(\frac{230}{157.6 \text{ lb} \cdot \text{ft}} \right)^{10/3} = 25\,594 \text{ hours}$$

This is deemed to be acceptable for the application.

4. Ensure that the rotational speed of the universal joint does not exceed the *maximum safe operating speed* from Table 11-12. For a Series A joint, MSOS = 5000 rpm which is far greater than the 1000 rpm speed of this joint.
5. Verify that the universal joint operating speed times the operating angle is less than the *speed to angle factor* from Table 11-12.

$$n \cdot \beta < \text{Speed to angle factor}$$

The Speed to angle factor = 16 250 from Table 11–12.

Then $n \cdot \beta = (1000 \text{ rpm}) \cdot (8^\circ) = 8000 < 16\,250$, which is satisfactory.

6. Pertinent data: The OD of the torque tube is 2.50 in. Other data are available from manufacturers' catalogs.

Summary: The Series A joint from Table 11–12 has been specified for this application to transmit 20 hp at a rotational speed of 1000 rpm while operating at an angle of 8° .

11-9 OTHER MEANS OF AXIAL LOCATION

The preceding sections of this chapter have focused on means of connecting machine elements to shafts for the purpose of transmitting power. Therefore, they emphasized the ability of the elements to withstand a given torque at a given speed of rotation. It must be recognized that the axial location of the machine elements must also be ensured by the designer.

The choice of the means for axial location depends heavily on whether or not axial thrust is transmitted by the element. Spur gears, V-belt sheaves, and chain sprockets produce no significant thrust loads. Therefore, the need for axial location affects only incidental forces due to vibration, handling, and shipping. Although not severe, these forces should not be taken lightly. Movement of an element in relation to its mating element in an axial direction can cause noise, excessive wear, vibration, or complete disconnection of the drive. Any bicycle rider who has experienced the loss of a chain can appreciate the consequences of misalignment. Recall that for spur gears, the strength of the gear teeth and the wear resistance are both directly proportional to the face width of the gear. Axial misalignment decreases the effective face width.

Some of the methods discussed in Section 11–6 for fastening elements to shafts for the purpose of transmitting power also provide some degree of axial location. Refer to the discussions of pinning, keyless hub to shaft connections, split taper bushings, set screws, the taper and screw, the taper and nut, the press fit, molding, and the several methods of mechanically locking the elements to the shaft.

Among the wide variety of other means available for axial location, we will discuss the following:

- Retaining rings
- Collars
- Shoulders
- Spacers
- Locknuts

Some of these items are shown in Figure 11–36. A machine shaft is supported on two tapered roller bearings with a gear on the left end and the middle portion

is a roller used as part of a paper handling system. The outer races of the bearings are supported by bearing caps and the housing of the machine. Because of the different widths of the inner and outer races, spacers are used to ensure axial positioning. The inner races of the bearings are press-fitted on the shaft, as discussed in Chapter 14. In addition, the outside faces of the inner race are held by locknuts.

The gear on the left end of the shaft is seated against a shoulder on its right side. Then, after the gear is in place, a retaining ring is inserted into a groove on the shaft to ensure that the gear does not move axially out of position or fall off the shaft. An alternative means of maintaining the axial position of the helical gear is to use a *retaining cap* that fastens to the end of the shaft and clamps the gear against the shaft shoulder.

More information on these other means of location follows.

Retaining Rings

Retaining rings are placed on a shaft, in grooves cut into the shaft, or in internal recesses to prevent the axial movement of a machine element or to hold internal components in place. Figure 11–37 shows sketches of common generic forms of retaining rings. The basic external design [Figure 11–37(a)] is most often used for retaining gears and other power transmission elements in axial position on shafts. This is discussed further in Chapter 12 on shaft design. There are numerous styles available from several vendors, including those listed in Internet sites 7, 11, and 22, from which catalog data may be found along with design guidance for preparing a shaft or housing to accept the rings. In general, retaining rings are classified as either *external* or *internal*. Some install in carefully sized grooves, while others hold their position with gripping force between the ring and the mating element. The amount of axial thrust capacity and the shoulder height provided by the different ring styles vary and catalog data should be consulted. Rings are made to either U.S. Customary dimensions or SI metric sizes. See the left end of the shaft in Figure 11–36 for an example of the use of retaining rings.

Collars

A *collar* is a ring slid over the shaft and positioned adjacent to a machine element for the purpose of axial location. It is held in position, typically, by set screws. Its advantage is that axial location can be set virtually anywhere along the shaft to allow adjustment of the position at the time of assembly. The disadvantages are chiefly those related to the use of set screws themselves (Section 11–6).

Shoulders

A *shoulder* is the vertical surface produced when a diameter change occurs on a shaft. Such a design is an excellent method for providing for the axial location of a machine element, at least on one side. Several of the shafts illustrated in Chapter 12 incorporate shoulders. The main design considerations are providing (1) a sufficiently large shoulder to locate the element effectively and (2) a fillet at the base of the shoulder that produces an acceptable stress concentration factor and that is compatible with the geometry of the bore of the mating element (see Figures 12–2 and 12–7). Note the use of shoulders in the shaft shown in Figure 11–36 to locate one side of the gear and the two bearings.

Spacers

A *spacer* is similar to a collar in that it is slid over the shaft against the machine element that is to be located. The primary difference is that set screws and the like are not necessary because the spacer is positioned *between* two elements and thus controls only the relative position between them. Typically, one of the elements is positively

located by some other means, such as a shoulder or a retaining ring.

Locknuts

When an element is located at the end of a shaft, a *locknut* can be used to retain it on one side. Figure 11–38 shows a bearing retainer type of locknut. These are available as stock items from bearing suppliers.

The shaft is threaded to match the threads on the locknut which can be seen in part (c) of this figure. Note that the lockwasher with the peripheral teeth has a tang on the ID that is positioned in a groove machined into the shaft. After inserting the lockwasher on the shaft against the element being held, the nut is screwed on so that it seats against the lockwasher, thus securing the bearing in place. Furthermore, four of the external tangs on the lockwasher are bent into the four grooves in the outer periphery of the nut to ensure that the nut is not rotated off the shaft.

Caution against Overconstraint

One practical consideration in the matter of axial location of machine elements is exercising care that elements are not *overconstrained*. Under certain conditions of differential thermal expansion or with an unfavorable tolerance stack, the elements may be forced together so tightly as to cause dangerous axial stresses. At times it may be desirable to locate only one bearing positively on a shaft and to permit the other to float slightly in the axial direction. The floating element may be held lightly with an axial spring force accommodating the thermal expansion without creating dangerous forces.

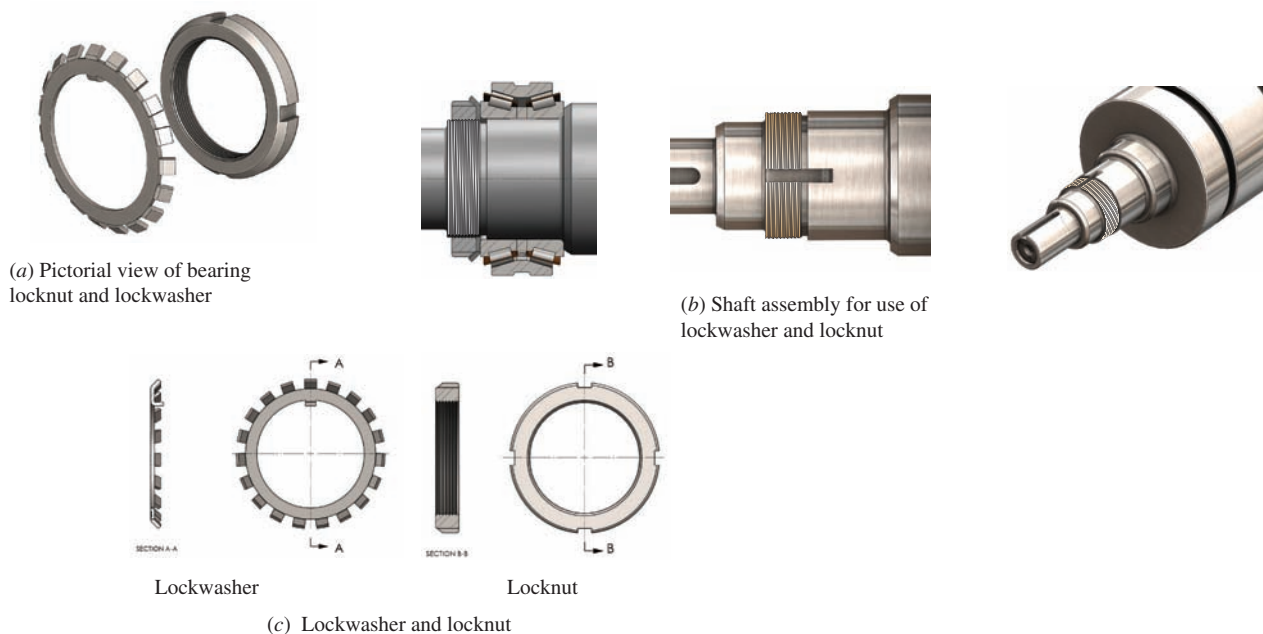


FIGURE 11–38 Locknut and lockwasher for retaining a bearing on a shaft.

In Figure 11–36, the left bearing is fixed into its housing, while the right bearing is permitted to slide (float) axially a small amount to accommodate thermal expansion and varying tolerances on shaft dimensions.

11-10 TYPES OF SEALS

Seals are important parts of machine design in situations where the following conditions apply:

1. Contaminants must be excluded from critical areas of a machine.
2. Lubricants must be contained within a space.
3. Pressurized fluids must be contained within a component such as a valve or a hydraulic cylinder.

Some of the parameters affecting the choice of the type of sealing system, the materials used, and the details of its design are as follows:

1. The nature of the fluids to be contained or excluded
2. Pressures on both sides of the seal
3. The nature of any relative motion between the seal and the mating components
4. Temperatures on all parts of the sealing system
5. The degree of sealing required: Is some small amount of leakage permissible?
6. The life expectancy of the system

7. The nature of the solid materials against which the seal must act: corrosion potential, smoothness, hardness, wear resistance
8. Ease of service for replacement of worn sealing elements

The number of designs for sealing systems is virtually limitless, and only a brief overview will be presented here. More comprehensive coverage can be found in References 12–15. Often, designers rely on technical information provided by manufacturers of complete sealing systems or specific sealing elements. Also, in critical or unusual situations, testing of a proposed design is advised. See Internet sites 12–15, 22, and 23.

The choice of a type of sealing system depends on the function that it must perform. Common conditions in which seals must operate are listed here, along with some of the types of seals used.

1. Static conditions such as sealing a closure on a pressurized container: elastomeric O-rings; T-rings; hollow metal O-rings; and sealants such as epoxies, silicones, and butyl caulking (Figure 11–39).
2. Sealing a closed container while allowing relative movement of some part, such as diaphragms, bellows, and boots (Figure 11–40).
3. Sealing around a continuously reciprocating rod or piston, such as in a hydraulic cylinder or a spool valve in a hydraulic system: lip seal; U-cup seal;

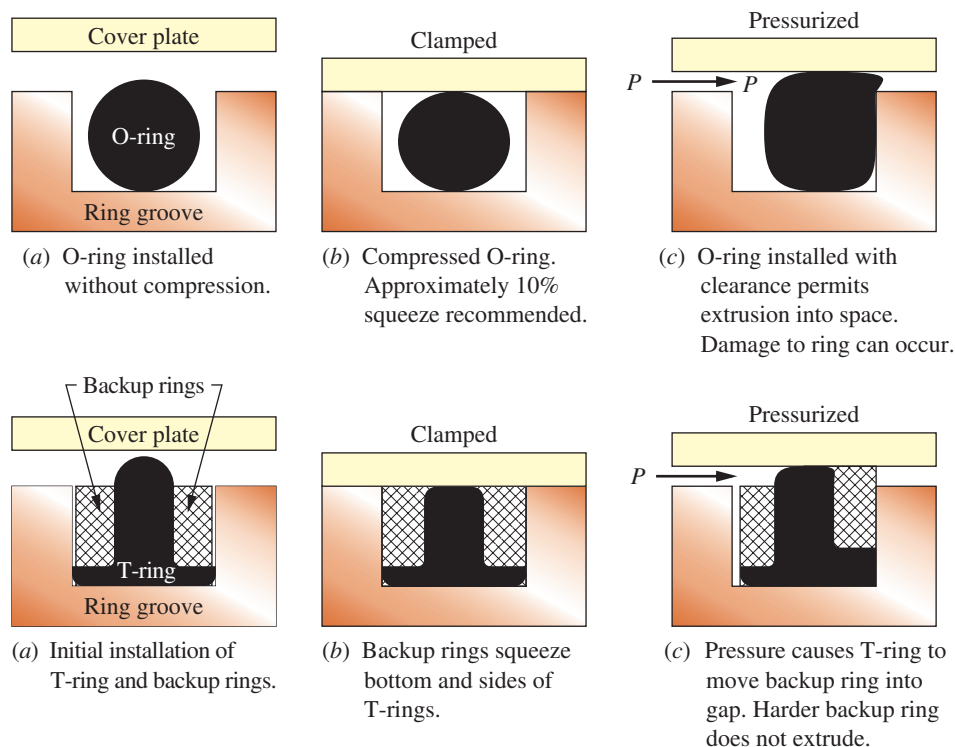


FIGURE 11–39 O-rings and T-rings used as static seals

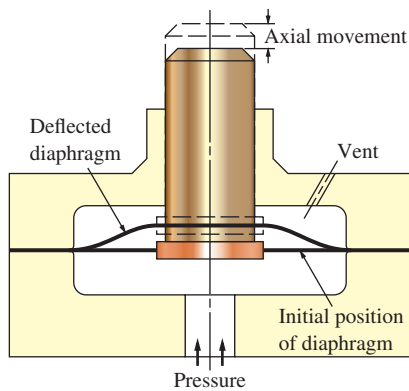


FIGURE 11-40 Application of a diaphragm seal

V-packing; and split ring seals, sometimes called *piston rings* (Figure 11-41).

4. Sealing around a rotating shaft such as the input or output shafts of a speed reducer, transmission, or engine: lip seal, wipers and scrapers, and face seals (Figure 11-42).
5. Protection of rolling-element bearings supporting shafts to keep contaminants from the balls and rollers (Figure 11-43).

6. Sealing the active elements of a pump or compressor to retain the pumped fluid: face seals and V-packing.
7. Sealing infrequently moved elements such as a valve stem of a fluid-flow control valve: compression packings and V-packings.
8. Sealing between hard, rigid surfaces such as between a cylinder head and the block of an engine: resilient gaskets. See Internet site 15.
9. Circumferential seals, such as at the tips of turbine blades, and on large, high-speed rotating elements: labyrinth seals, abradable seals, and hydrostatic seals.

11-11 SEAL MATERIALS

Most seal materials are resilient to permit the sealing points to follow minor variations in the geometry of mating surfaces. Flexing of parts of the seal cross section also occurs in some designs, calling for resiliency in the materials. Alternatively, as in the case of hollow metal O-rings, the shape of the seal allows the flexing of hard materials to occur. Face seals require rigid, hard materials that can withstand constant sliding motion and that can be produced with fine accuracy, flatness, and smoothness.

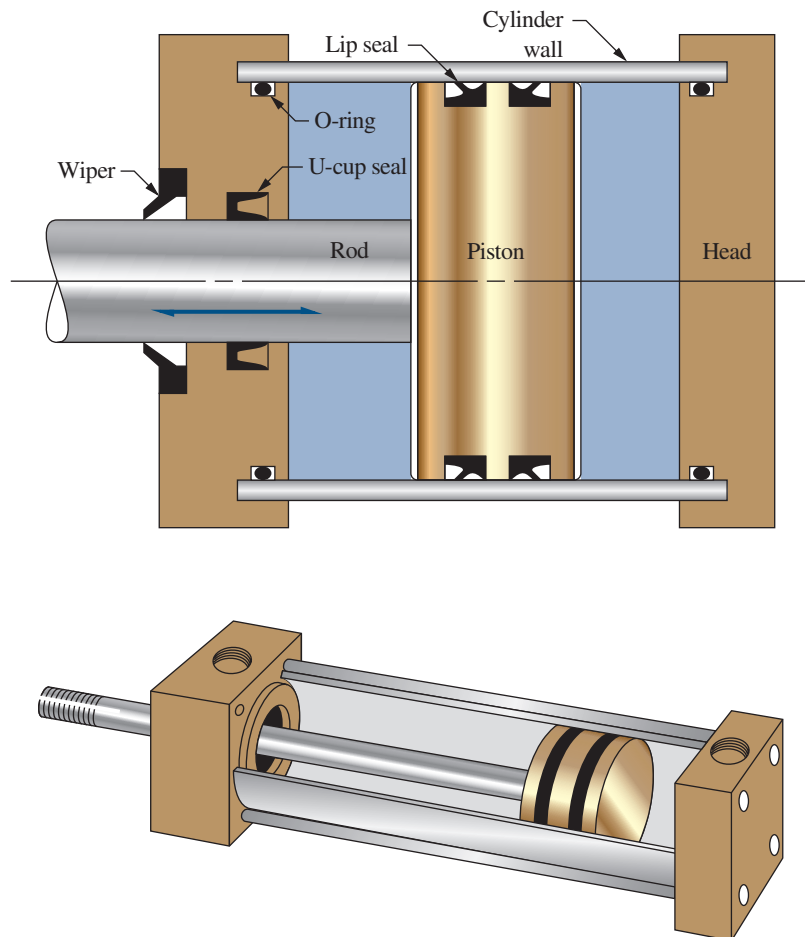


FIGURE 11-41 Lip seals, U-cup seal, wiper, and O-rings applied to a hydraulic actuator

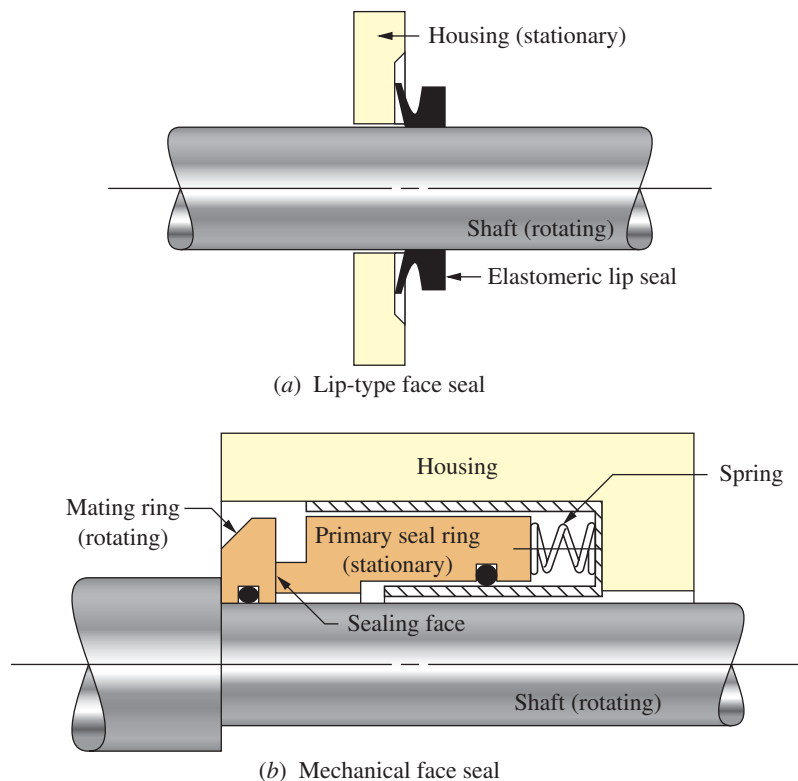


FIGURE 11-42 Face seals

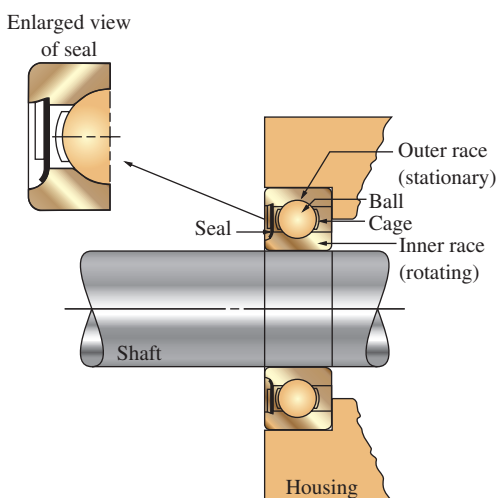


FIGURE 11-43 Seal for ball bearing

Elastomers

Resilient seals such as O-rings, T-rings, and lip seals are often made from synthetic elastomers such as the following:

Neoprene	Butyl	Nitrile (Buna-N)
Fluorocarbon	Silicone	Fluorosilicone
Butadiene	Polyester	Ethylene propylene
Polysulfide	Polyurethane	Epichlorohydrin
Polyacrylate	PNF (Phosphonitrilic fluoroelastomer)	

Many proprietary formulations within these general classifications are available under trade names from seal producers and plastics manufacturers.

Properties needed in a given installation will limit the selection of candidate materials. The following list gives some of the more prevalent requirements for seals and some of the materials that meet those requirements:

Weather resistance: Silicone, fluorosilicone, fluorocarbon, ethylene propylene, polyurethane, polysulfide, polyester, neoprene, epichlorohydrin, and PNF.

Petroleum fluid resistance: Polyacrylate, polyester, PNF, nitrile, polysulfide, polyurethane, fluorocarbon, and epichlorohydrin.

Acid resistance: Fluorocarbon.

High-temperature operation: Ethylene propylene, fluorocarbon, polyacrylate, silicone, and PNF.

Cold-temperature operation: Silicone, fluorosilicone, ethylene propylene, and PNF.

Tensile strength: Butadiene, polyester, and polyurethane.

Abrasion resistance: Butadiene, polyester, and polyurethane.

Impermeability: Butyl, polyacrylate, polysulfide, and polyurethane.

Rigid Materials

Face seals, and the parts of other types of sealing systems against which elastomers seal, require rigid materials that can withstand the sliding action and that are compatible with the environment around the seal. Some typical rigid materials used in sealing systems are described in the following list:

Metals: Carbon steel, stainless steel, cast iron, nickel alloys, bronze, and tool steels.

Plastics: Nylon, filled polytetrafluoroethylene (PTFE), and polyimide.

Carbon, ceramics, and tungsten-carbide.

Plating: Chromium, cadmium, tin, nickel, and silver.

Flame-sprayed compounds.

Packings

Packings for sealing shafts, rods, valve stems, and similar applications are made from a variety of materials, including leather, cotton, flax, several types of plastics, braided or twisted wire made from copper or aluminum, laminated cloth and elastomeric materials, and flexible graphite.

Gaskets

Common gasket materials are cork, cork and rubber compounds, filled rubber, paper, resilient plastics, and foams. See Internet site 15.

Shafts

When radial lip seals are required around shafts, the shafts are typically steel. They should be hardened to HRC 30 to resist scoring of the surface. Tolerance on the diameter of the shaft on which the seal bears should conform to the following recommendations to ensure that the seal lip can follow the variations:

Shaft Diameter (in)	Tolerance (in)
$D \leq 4.000$	± 0.003
$4.000 < D \leq 6.000$	± 0.004
$D > 6.000$	± 0.005

The surface of the shaft and any areas over which the seal must pass during installation should be free of burrs to protect against tearing the seal. A surface finish of 10 to 20 μin is recommended, with adequate lubrication to ensure full contact and to reduce friction between the seal and the shaft surface.

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INTERNET SITES FOR KEYS, COUPLINGS, AND SEALS

- Fastenal Company.** Provider of a wide variety of mechanical products including keystock in low-, medium-, and high-carbon steel, 18-8 and 316 stainless steel. Both U.S. and metric sizes. Square, rectangular, and woodruff types.
- Ringfeder Power Transmission GMBH.** Manufacturer of Ringfeder and ECOLOC keyless hub to shaft locking devices, along with Tschan and Gerwah couplings.
- Baldor/Dodge.** Manufacturer of a wide variety of power transmission components including flexible couplings, gearing, belt drives, clutches and brakes, and bearings.
- Regal Beloit Americas, Inc.** Manufacturer of a wide variety of power transmission equipment including flexible couplings and universal joints under the Kop-Flex, Browning, and Morse brands. Other brands include McGill, Rollway, and Sealmaster for gearing, belt drives, chain drives, bearings, conveyor components, drive shafts, and clutches.
- Dana Limited—Spicer Drivetrain Components.** Manufacturer of universal joints, and drive shafts for vehicular and industrial applications using the Dana and Spicer brand names.

6. **GWB—A Dana Brand.** Manufacturer of heavy-duty drive shafts for industrial equipment, locomotives, and similar applications.
7. **Stock Drive Products/Sterling Instrument.** Manufacturer and distributor of precision machine components and assemblies, including flexible couplings, gearing, clutches and brakes, and fasteners.
8. **T. B. Wood's Sons Company.** Manufacturer of mechanical, industrial power transmission products including flexible couplings, synchronous belt drives, V-belt drives, gearmotors, and gearing.
9. **Curtis Universal Joint Company.** Manufacturer of universal joints for the industrial and aerospace markets in U.S. and metric sizes.
10. **Cooper Tools/Apex Universal Joints.** Manufacturer of universal joints for the military, aerospace, performance racing, and industrial power transmission markets. Product line includes the Cornay™ universal joints.
11. **Rotor Clip Company, Inc.** Manufacturer of a wide variety of retaining rings and other products in U.S. and metric sizes for industrial, commercial, military, and consumer products.
12. **Federal Mogul Powertrain.** Manufacturer of seals for engines, transmissions, wheels, differentials, and industrial applications. Select Products, then Engine or Sealing.
13. **American Seal & Packing Company.** Manufacturer of O-rings, gaskets, mechanical seals, and related products.
14. **Mechanical Seals.net.** Part of the site for American Seal and Packing Company providing technical information about the design of various types of seals, how they work, how they are maintained, and the materials used.
15. **IGS Industries.** Manufacturer of gaskets, shims, and custom fabricated seals, and industrial sealants.
16. **American High Performance Seals, Inc.** Manufacturer of a wide variety of wipers and seals for rods, pistons, and rotary applications. Site includes numerous graphic representations of seal cross sections and tables of materials and their properties.
17. **Lord Corporation—Vibration & Motion Control Products.** Manufacturer of flexible couplings, vibration mounts, and shock isolation mounts using elastomeric materials bonded to metals. Select Products & Solutions, then Vibration & Motion Control.
18. **General Polygon Systems, Inc.** Providers of mechanical shaft to hub connections using the General Polygon System.
19. **Belden Universal.** Manufacturer of universal joints, drive shaft assemblies, and couplings for industrial, nuclear, aerospace, and other applications.
20. **G. L. Huyett Co.** Supplier of numerous machine components, including square and rectangular keys and keystock, bars, shafts, shim and shimstock in both U.S. and SI sizes and in several grades of material. From the home page, select *Power Transmission*.
21. **LOCTITE® Industrial Adhesives & Sealants/Henkel Corporation.** Manufacturer of numerous types of adhesives and sealants, including LOCTITE® Threadlockers® that inhibit loosening of screws and other threaded products.
22. **Grainger.** Distributor of a huge array of products for machinery components, hardware, motors, power transmission drive products, bearings, flexible couplings, universal joints, fasteners, retaining rings, keys and key stock, raw materials, welding products, adhesives, seals and sealants, threadlocking materials, linear motion devices, and many more.
23. **Federal Mogul Corporation/Powertrain.** Manufacturer of automotive components and products for industrial uses, energy, and transport, including seals crankshafts, camshafts, and oil control rings on pistons.
24. **Rexnord – Thomas Flexible Disc Couplings.** Manufacturer of a broad line of flexible disc couplings for pump drives, fans, compressors, and general industrial machinery.

PROBLEMS

For Problems 1–4 and 7, determine the required key geometry: length, width, and height. Use SAE 1018 steel for the keys if a satisfactory design can be achieved. If not, use a higher-strength material from Table 11–4. Unless otherwise stated, assume that the key material is weakest when compared with the shaft material or the mating elements.

1. Specify a key for a gear to be mounted on a shaft with a 2.00 in diameter. The gear transmits 21 000 lb·in of torque and has a hub length of 4.00 in. See Figure P11–1.
2. Specify a key for a gear carrying 21 000 lb·in of torque if it is mounted on a 3.60-in-diameter shaft. The hub length of the gear is 4.00 in.

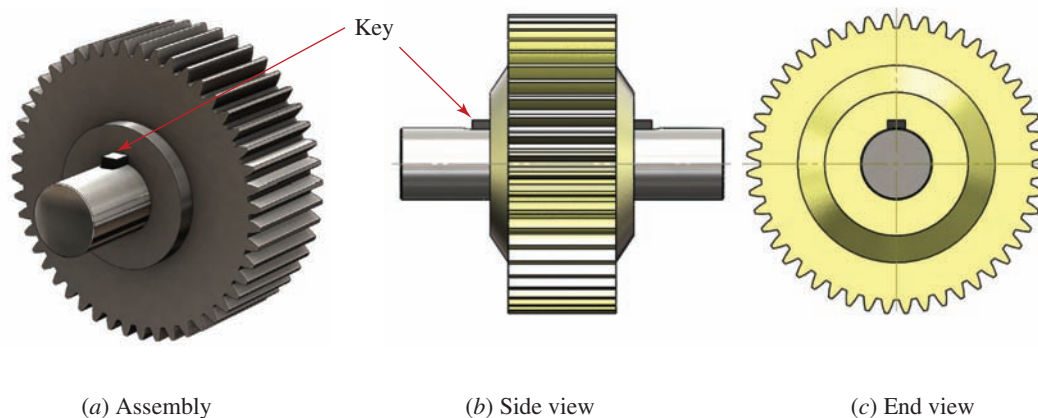


FIGURE P11–1 Gear, shaft, and key for Problem 11–1

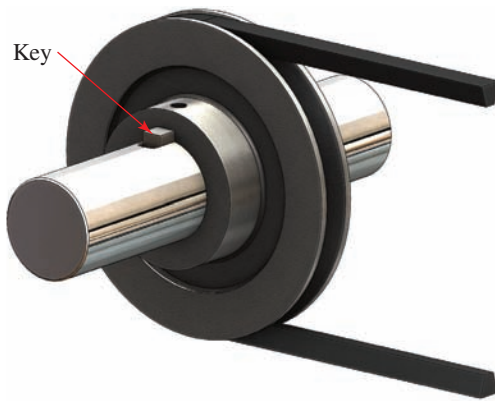


FIGURE P11-3 Belt sheave, shaft, and key for Problem 11-3

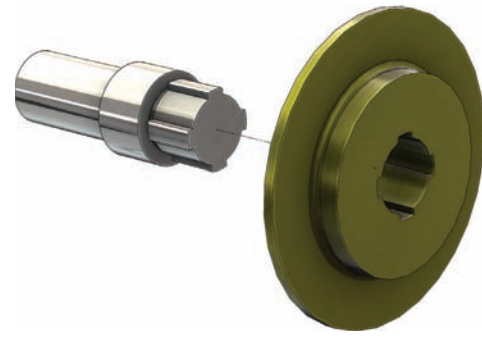
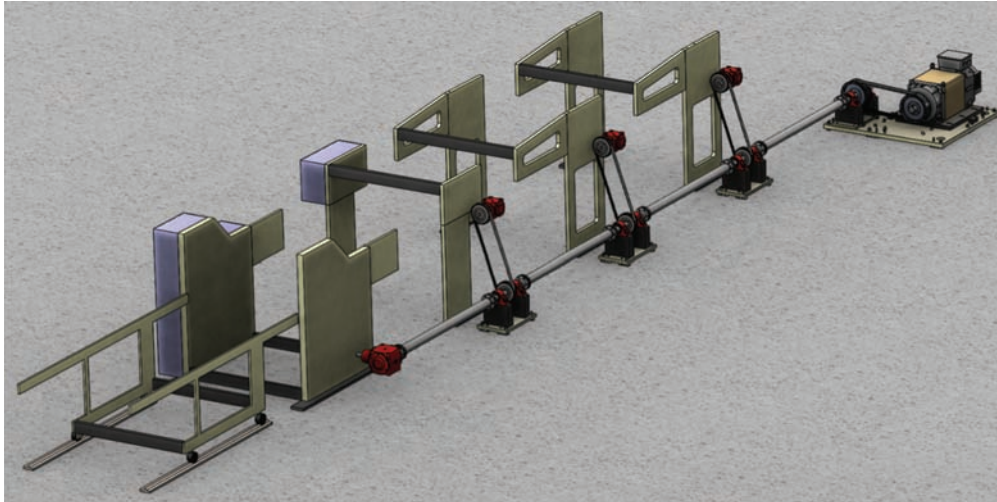


FIGURE P11-16 Pictorial view of splined shaft connection

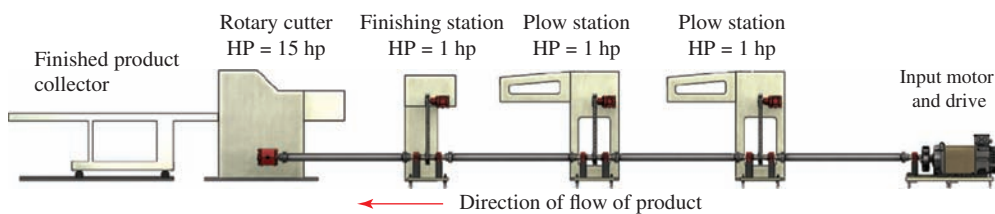
3. A V-belt sheave transmits 1112 lb·in of torque to a 1.75-in-diameter shaft. The sheave is made from ASTM class 20 cast iron and has a hub length of 1.75 in. See Figure P11-3.
4. A chain sprocket delivers 110 hp to a shaft at a rotational speed of 1700 rpm. The sprocket has a bore diameter of 2.50 in. The hub length is 3.25 in.
5. Specify a suitable spline having a *B* fit for each of the applications in Problems 1-4.
6. Design a cylindrical pin to transmit the power, as in Problem 4. But design it so that it will fail in shear if the power exceeds 220 hp.
7. Specify a key for both the sprocket and the wormgear from Example Problem 12-3. Note the specifications for the final shaft diameters at the end of the problem.
8. Describe a Woodruff key no. 204.
9. Describe a Woodruff key no. 1628.
10. Make a detailed drawing of a Woodruff key connection between a shaft and the hub of a gear. The shaft has a diameter of 1.500 in. Use a no. 1210 Woodruff key. Dimension the keyseat in the shaft and the hub.
11. Repeat Problem 10, using a no. 406 Woodruff key in a shaft having a 0.500 in diameter.
12. Repeat Problem 10, using a no. 2428 Woodruff key in a shaft having a 3.250 in diameter.
13. Compute the torque that could be transmitted by the key of Problem 10 on the basis of shear and bearing if the key is made from SAE 1018 cold-drawn steel with a design factor of $N = 3$.
14. Repeat Problem 13 for the key of Problem 11.
15. Repeat Problem 13 for the key of Problem 12.
16. Make a drawing of a four-spline connection having a major diameter of 1.500 in and an *A* fit. Show critical dimensions. See Figure P11-16 for the general layout.
17. Make a drawing of a 10-spline connection having a major diameter of 3.500 in and a *B* fit. Show critical dimensions.
18. Make a drawing of a 16-spline connection having a major diameter of 2.500 in and a *C* fit. Show critical dimensions.
19. Determine the torque capacity of the splines in Problems 16-18.
20. Describe the manner in which a set screw transmits torque if it is used in place of a key. Discuss the disadvantages of such an arrangement.
21. Describe a press fit as it would be used to secure a power transmission element to a shaft.
22. Describe the main differences between rigid and flexible couplings as they affect the stresses in the shafts that they connect.
23. Discuss a major disadvantage of using a single universal joint to connect two shafts with angular misalignment.
24. Describe five ways to locate power transmission elements positively on a shaft axially.
25. Describe three situations in which seals are applied in machine design.
26. List eight parameters that should be considered when selecting a type of seal and specifying a particular design.
27. Name three means of sealing a pressurized container under static conditions.
28. Name three methods of sealing a closed container while allowing relative movement of some part.
29. Name four types of seals used on reciprocating rods or pistons.
30. Name three types of seals applied to rotating shafts.
31. Describe the method of sealing a ball bearing from contaminants.
32. Describe an O-ring seal, and sketch its installation.
33. Describe a T-ring seal, and sketch its installation.
34. Describe some advantages of T-ring seals over O-rings.
35. Describe a diaphragm seal and the type of situation in which it is used.
36. Describe suitable methods of sealing the sides of a piston against the inner walls of the cylinder of a hydraulic actuator.
37. Describe the function of a scraper or wiper on a cylinder rod.
38. Describe the essential elements of a mechanical face seal.
39. Name at least six kinds of elastomers commonly used for seals.
40. Name at least three kinds of elastomers that are recommended for use when exposed to weather.
41. Name at least three kinds of elastomers that are recommended for use when exposed to petroleum-based fluids.
42. Name at least three kinds of elastomers that are recommended for use when exposed to cold-temperature operation.
43. Name at least three kinds of elastomers that are recommended for use when exposed to high-temperature operation.
44. A sealing application involves the following conditions: exposure to high-temperature petroleum fluids and impermeability. Specify a suitable elastomer for the seal.
45. A sealing application involves the following conditions: exposure to high temperature and weather, impermeability, and high strength and abrasion resistance. Specify a suitable elastomer for the seal.
46. Describe suitable shaft design details where elastomeric seals contact the shaft.

47. A driveline is to be designed to drive a series of four pieces of equipment for the printing industry as shown in Figure P11-47(a). An electric motor at the right drives the driveline at a rotational speed of 500 rpm and the power demand for each piece of equipment is shown in Figure P11-47(b). Figure P11-47(c) of the figure shows an enlarged view of the drive motor and the first segment

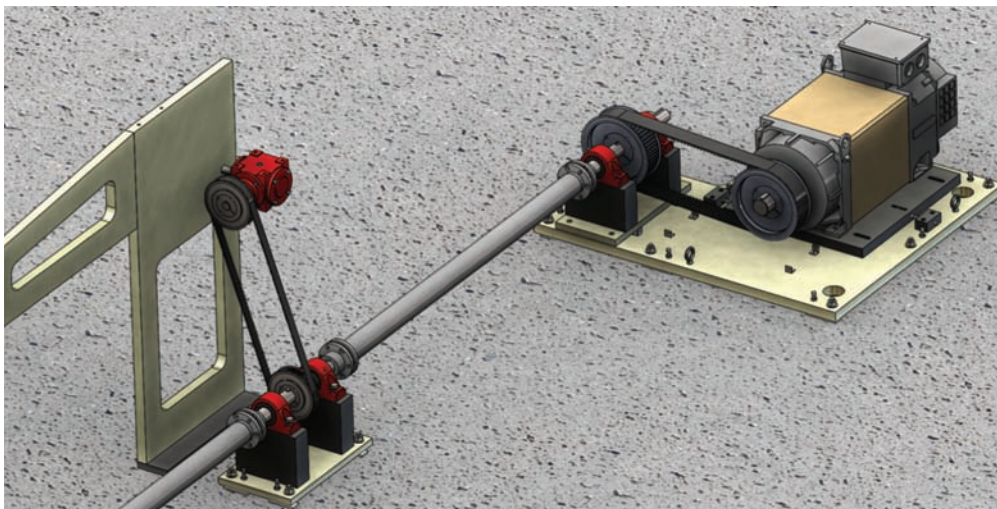
of the driveline. All four segments are to be identical. Figure P11-47(d) of the figure shows the general design of one driveline segment. The distance between the ends of each drive shaft segment is to be 63.00 in. Select a floating shaft-type coupling to connect each piece of equipment to a 2.00-in-diameter rigid jack shaft assembly. Refer Section 11-7 for design data.



(a) General layout of the machinery for a press line for the printing industry.



(b) Power demand for the four pieces of equipment



(c) Detail for the drive motor, synchronous belt reducer from the motor to the drive shaft, the floating shaft-type coupling and the belt drive to the first plow station



(d) General design of the coupling for each segment of the driveline

FIGURE P11-47

SHAFT DESIGN

The Big Picture

You Are the Designer

- 12-1 Objectives of This Chapter
- 12-2 Shaft Design Procedure
- 12-3 Forces Exerted on Shafts by Machine Elements
- 12-4 Stress Concentrations in Shafts
- 12-5 Design Stresses for Shafts
- 12-6 Shafts in Bending and Torsion Only
- 12-7 Shaft Design Examples—Bending and Torsion Only
- 12-8 Shaft Design Example—Bending and Torsion with Axial Forces
- 12-9 Spreadsheet Aid for Shaft Design
- 12-10 Shaft Rigidity and Dynamic Considerations
- 12-11 Flexible Shafts

THE BIG PICTURE

Shaft Design

Discussion Map

- A shaft is a rotating machine component that transmits power.
- Designers must create a practical shape for the shaft and specify suitable diameters that ensure that the shaft is safe under expected loads.

Discover

Identify examples of mechanical systems that incorporate power-transmitting shafts. Describe their geometry and the forces and torques that are exerted on them.

What kinds of stresses are produced in the shaft?

How are other elements mounted on the shaft? How does the shaft geometry accommodate them? How is the shaft supported? What kinds of bearings are used?

This chapter provides approaches that you can use to design shafts that are safe for their intended use. But you have the final responsibility for the design.

A *shaft* is the component of a mechanical device that transmits rotational motion and power. It is integral to any mechanical system in which power is transmitted from a prime mover, such as an electric motor or an engine, to other rotating parts of the system. Can you identify some kinds of mechanical systems incorporating rotating elements that transmit power?

Here are some examples: gear-type speed reducers, belt or chain drives, conveyors, pumps, fans, agitators, and many types of automation equipment. What others can you think of? Consider household appliances, lawn maintenance equipment, parts of a car, power tools, and machines around an office or in your workplace. Describe them and discuss how shafts are used. From what source is power delivered into the shaft? What kind of power-transmitting

element, if any, is on the shaft itself? Or is the shaft simply transmitting the rotational motion and torque to some other element? If so, how is the shaft connected to that element?

Visualize the forces, torques, and bending moments that are created in the shaft during operation. In the process of transmitting power at a given rotational speed, the shaft is inherently subjected to a torsional moment, or *torque*. Thus, torsional shear stress is developed in the shaft. Also, a shaft usually carries power-transmitting components, such as gears, belt sheaves, or chain sprockets, which exert forces on the shaft in the transverse direction (perpendicular to its axis). These transverse forces cause bending moments to be developed in the shaft, requiring analysis of the stress due to bending. In fact, most

shafts must be analyzed for combined stress. Often, the forces applied to shafts act in virtually any direction, not just in a single plane.

Describe the specific geometry of shafts from some types of equipment that you can examine. Make a sketch of any variations in geometry that may occur, such as changes in diameter, to produce shoulders, grooves, keyseats, or holes. How are any power-transmitting elements held in position along the length of the shaft? How are the shafts supported? Typically, bearings are used to support the shaft while permitting rotation relative to the housing of the machine. What kinds of bearings are used? Do they have rolling elements such as ball bearings? Or,

are they smooth-surfaced bearings? What materials are used?

It is likely that you will find much diversity in the design of the shafts in different kinds of equipment. You should see that the functions of a shaft have a large influence on its design. Shaft geometry is greatly affected by the mating elements such as bearings, couplings, gears, chain sprockets, or other kinds of power-transmitting elements.

This chapter provides approaches that you can use to design shafts that are safe for their intended use. But the final responsibility for the design is yours because it is impractical to predict in a book all conditions to which a given shaft will be subjected.

YOU ARE THE DESIGNER

Consider the gear-type speed reducer shown in Figure 1–12. Note that there are three shafts that must be designed. The input shaft carries the first gear in the gear train and rotates at the speed of the prime mover, typically an electric motor or an engine. The middle shaft carries two gears and rotates more slowly than the input shaft because of the first stage of speed reduction. The final gear in the train is carried by the third shaft that rotates slower than shaft 2 and which also transmits the power to the driven machine. From what material should each shaft be made? What torque is being transmitted by each shaft, and over what part of the shaft is it

acting? How are the gears to be located on the shafts? How is the power to be transmitted from the gears to the shafts, or vice versa? What forces are exerted on the shaft by the gears, and what bending moments result? What forces must the bearings that support each shaft resist? What are the minimum acceptable diameters for the shafts at all sections to ensure safe operation? What should be the final dimensional specifications for the many features of the shafts, and what should be the tolerances on those dimensions? The material in this chapter will help you make these and other shaft design decisions.

12-1 OBJECTIVES OF THIS CHAPTER

After completing this chapter, you will be able to:

1. Propose reasonable geometries for shafts to carry a variety of types of power-transmitting elements, providing for the secure location of each element and the reliable transmission of power.
2. Compute the forces exerted on shafts by gears, belt sheaves, and chain sprockets.
3. Determine the torque distribution on shafts.
4. Prepare shearing force and bending moment diagrams for shafts in two planes.
5. Account for stress concentration factors commonly encountered in shaft design.
6. Specify appropriate design stresses for shafts.
7. Apply the shaft design procedure shown in this chapter to determine the required diameter of shafts at any section to resist the combination of torsional shear stress and bending stress.
8. Specify reasonable final dimensions for shafts that satisfy strength requirements and installation considerations and that are compatible with the elements mounted on the shafts.

9. Consider the influence of shaft rigidity on its dynamic performance.

12-2 SHAFT DESIGN PROCEDURE

Because of the simultaneous occurrence of torsional shear stresses and normal stresses due to bending, the stress analysis of a shaft virtually always involves the use of a combined stress approach. The recommended approach for shaft design and analysis is the *distortion energy theory of failure*. This theory was introduced in Chapter 5 and will be discussed more fully in Section 12–5. Vertical shear stresses and direct normal stresses due to axial loads may also occur. On very short shafts or on portions of shafts where no bending or torsion occurs, such stresses may be dominant. The discussions in Chapters 3–5 explain the appropriate analysis.

The specific tasks to be performed in the design and analysis of a shaft depend on the shaft's proposed design in addition to the manner of loading and support. With this in mind, the following is a recommended general approach for the design of a shaft.

This process will be demonstrated after the concepts of force and stress analysis are presented.

PROCEDURE FOR DESIGN OF A SHAFT ▼

1. Determine the rotational speed of the shaft.
2. Determine the power or the torque to be transmitted by the shaft.
3. Determine the design of the power-transmitting components or other devices that will be mounted on the shaft, and specify the required location of each device.
4. Specify the location of bearings to support the shaft. Normally two and only two bearings are used to support a shaft. The reactions on bearings supporting radial loads are assumed to act at the midpoint of the bearings. For example, if a single-row ball bearing is used, the load path is assumed to pass directly through the balls. If thrust (axial) loads exist in the shaft, you must specify which bearing is to be designed to react against the thrust load. Then the bearing that does not resist the thrust load should be permitted to move slightly in the axial direction to ensure that no unexpected and undesirable thrust load is exerted on that bearing. Bearings should be placed on either side of the power-transmitting elements if possible to provide stable support for the shaft and to produce reasonably well-balanced loading of the bearings. The bearings should be placed close to the power-transmitting elements to minimize bending moments. Also, the overall length of the shaft should be kept small to keep deflections at reasonable levels.
5. Propose the general form of the geometry for the shaft, considering how each element on the shaft will be held in position axially and how power transmission from each element to the shaft is to take place. For example, consider the shaft in Figure 12–1, which is to carry two gears

as the intermediate shaft in a double-reduction, spur gear-type speed reducer. Gear *A* accepts power from gear *P* by way of the input shaft. The power is transmitted from gear *A* to the shaft through the key at the interface between the gear hub and the shaft. The power is then transmitted down the shaft to point *C*, where it passes through another key into gear *C*. Gear *C* then transmits the power to gear *Q* and thus to the output shaft. The locations of the gears and bearings are dictated by the overall configuration of the reducer.

It is now decided that the bearings will be placed at points *B* and *D* on the shaft to be designed. But how will the bearings and the gears be located so as to ensure that they stay in position during operation, handling, shipping, and so forth? Of course, there are many ways to do this. One way is proposed in Figure 12–2. Shoulders are to be machined in the shaft to provide surfaces against which to seat the bearings and the gears on one side of each element. The gears are restrained on the other side by retaining rings snapped into grooves in the shaft. The bearings will be held in position by the housing acting on the outer races of the bearings. Keyseats will be machined in the shaft at the location of each gear. This proposed geometry provides for positive location of each element.

6. Determine the magnitude of torque that the shaft sees at all points. It is recommended that a torque diagram be prepared, as will be shown later.
7. Determine the forces that are exerted on the shaft, both radially and axially.
8. Resolve the radial forces into components in perpendicular directions, usually vertically and horizontally.
9. Solve for the reactions on all support bearings in each plane.

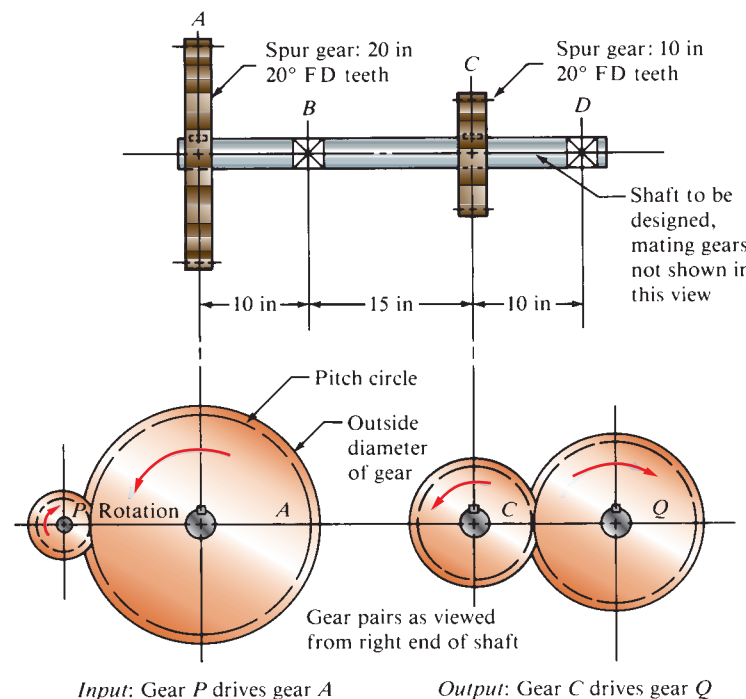


FIGURE 12–1 Intermediate shaft for a double-reduction, spur gear-type speed reducer

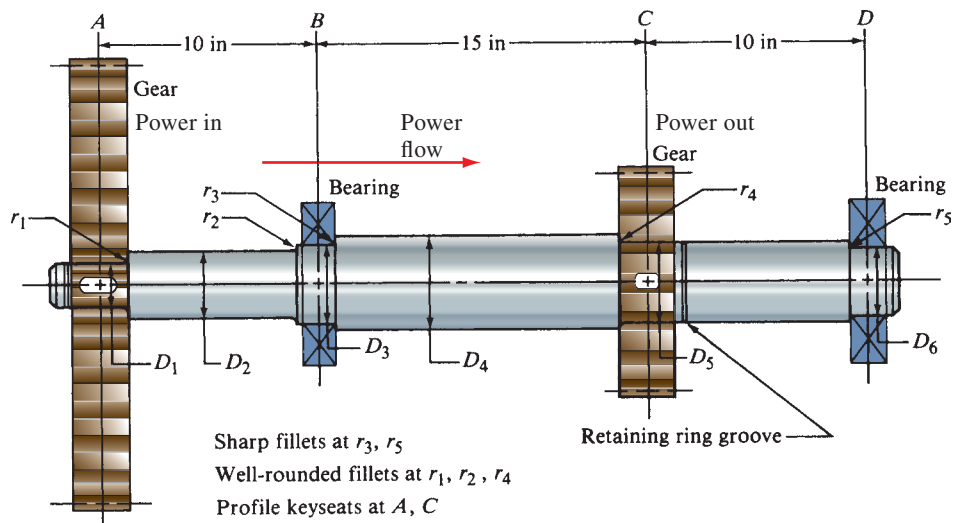


FIGURE 12-2 Proposed geometry for the shaft in Figure 12-1. Sharp fillets at r_3, r_5 ; well-rounded fillets at r_1, r_2, r_4 ; profile keyseats at A, C

10. Produce the complete shearing force and bending moment diagrams to determine the distribution of bending moments in the shaft.
11. Select the material from which the shaft will be made, and specify its condition: cold-drawn, heat-treated, and so on. As indicated in Table 2-9, suggested steel materials for shafts are plain carbon or alloy steels with medium carbon content, such as SAE 1040, 4140, 4340, 4640, 5150, 6150, and 8650. Good ductility with percent elongation above about 12% is recommended. Determine the ultimate strength, yield strength, and percent elongation of the selected material.
12. Determine an appropriate design stress, considering the manner of loading (smooth, shock, repeated and reversed, or other).
13. Analyze each critical point of the shaft to determine the minimum acceptable diameter of the shaft at that point in order to ensure safety under the loading at that point. In general, the critical points are several and include those where a change of diameter takes place, where the higher values of torque and bending moment occur, and where stress concentrations occur.
14. Specify the final dimensions, surface finishes, tolerances, geometric dimensioning details, fillet radii, shoulder heights, keyseat dimensions, retaining ring groove geometry, and other details for each part of the shaft, ensuring that the minimum diameter dimensions from Step 13 are satisfied. Some considerations are listed here while additional information and guidelines are presented in Chapters 13-15.
 - a. Power-transmission elements such as gears, belt sheaves, and chain sprockets are typically placed on shafts with small clearances to facilitate assembly while retaining reliable operation. Chapter 13 describes fits between elements that should be applied to final tolerancing decisions. For high-precision devices and those operating at high speeds, the elements are often applied with a light interference fit to minimize vibration and to maintain precise

alignment of mating members. The chosen fit determines the final limit dimensions for the shaft. See References 2 and 12.

- b. The inner races of rolling contact bearings are typically mounted on shafts with a light interference fit as specified by the bearing manufacturer. It is essential that bearing manufacturer's specifications are shown on the shaft drawing. Chapters 13-15 give more information and some example data. Table 15-5 shows that for a ball bearing the specified total tolerance for the shaft diameter at a bearing seat is typically in the range of 0.0003 in to 0.0011 in [approximately 7.6 micrometers (μm) to 28.0 μm]. See examples of the bearing seat dimensioning in Figures 15-6 and 15-7, typically shown as *limit dimensions*. For the purposes of this chapter, it is sufficient to state that the specified bearing inner race diameter determines the final shaft dimension.
- c. Because of the close tolerances at bearing seats, these parts of the shaft are typically ground. Other parts may be produced by turning with fine-finishing passes to achieve the desired fit and satisfactory fatigue performance.
- d. Keyseats in the shaft should be dimensioned as shown in Figure 11-2(b), giving the S dimension for ease of manufacture. See also Figures 15-6 and 15-7 for examples.
- e. This chapter presents a method of accounting for stress concentrations due to shoulder fillets that is initially approximate. The designer is expected to specify the final fillet radii that result in safe stresses.
- f. Retaining ring manufacturers specify the dimensioning and tolerancing required for the grooves into which the rings are seated. Designers must show these details on the shaft drawings. See Figures 15-6 and 15-7 for examples.
- g. Axial dimensions for shaft features are typically shown relative to some critical datum feature such as a shoulder against which a gear or bearing is mounted.

12-3 FORCES EXERTED ON SHAFTS BY MACHINE ELEMENTS

Gears, belt sheaves and pulleys, chain sprockets, and other elements typically carried by shafts exert forces on the shaft that cause bending moments. The following is a discussion of the methods for computing these forces for some cases. In general, you will have to use the principles of statics and dynamics to determine the forces for any particular element.

Spur Gears

The force exerted on a gear tooth during power transmission acts normal (perpendicular) to the involute-tooth profile, as discussed in Chapter 9 and shown in Figure 12-3. It is convenient for the analysis of shafts to consider the rectangular components of this force acting in the radial and tangential directions. It is most convenient to compute the tangential force, W_t , directly from the known torque being transmitted by the gear. For U.S. Customary units,

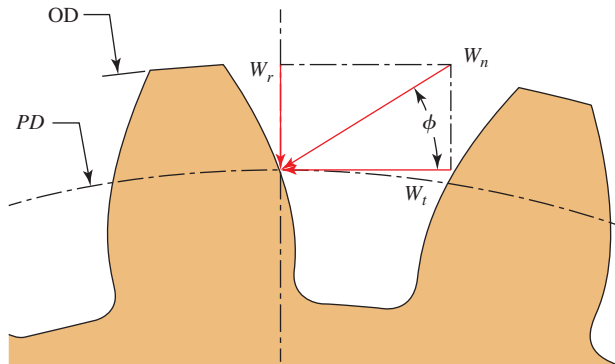


FIGURE 12-3 Forces on gear teeth: Tangential force, W_t ; Radial force, W_r ; Normal force, W_n

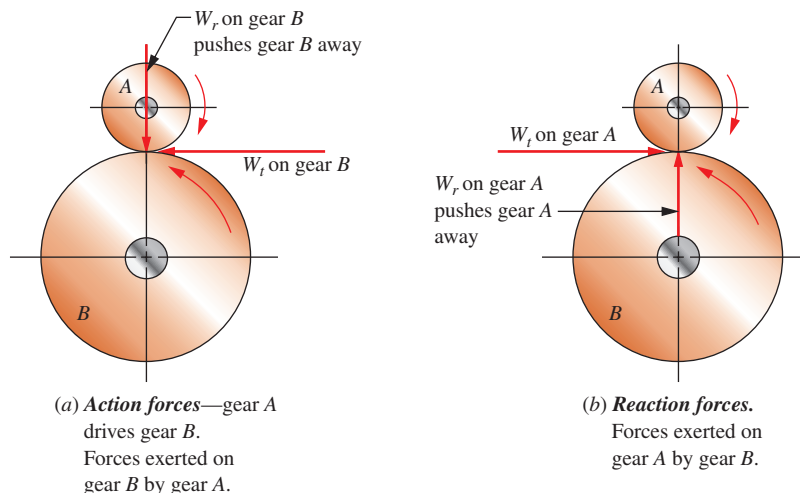


FIGURE 12-4 Directions for forces on mating spur gears

⊞ Torque

$$T = 63\,000 (P)/n \quad (12-1)$$

⊞ Tangential Force

$$W_t = T/(D/2) \quad (12-2)$$

where P = power being transmitted in hp

n = rotational speed in rpm

T = torque on the gear in lb·in

D = pitch diameter of the gear in in

The angle between the total force and the tangential component is equal to the pressure angle, ϕ , of the tooth form. Thus, if the tangential force is known, the radial force can be computed directly from

⊞ Radial Forces

$$W_r = W_t \tan \phi \quad (12-3)$$

and there is no need to compute the normal force at all. For gears, the pressure angle is typically $14\frac{1}{2}^\circ$, 20° , or 25° .

Directions for Forces on Mating Spur Gears

Representing the forces on gears in their correct directions is essential to an accurate analysis of forces and stresses in the shafts that carry the gears. The force system shown in Figure 12-4(a) represents the action of the driving gear A on the driven gear B. The tangential force, W_t , pushes perpendicular to the radial line causing the driven gear to rotate. The radial force, W_r , exerted by the driving gear A, acts along the radial line tending to push the driven gear B away.

An important principle of mechanics states that for each action force, there is an equal and opposite reaction force. Therefore, as shown in Figure 12-4, the driven

gear pushes back on the driving gear with a tangential force opposing that of the driving gear and a radial force that tends to push the driving gear away. For the orientation of the gears shown in Figure 12-4, note the following directions for forces:

Action: Driver pushes
on driven gear

W_t : Acts to the left
 W_r : Acts downward

Reaction: Driven gear
pushes back on driver

W_t : Acts to the right
 W_r : Acts upward

In summary, whenever you need to determine the direction of forces acting on a given gear, first determine whether it is a driver or a driven gear. Then visualize the action forces of the driver. If the gear of interest is the driven gear, these are the forces on it. If the gear of interest is the driver gear, the forces on it act in the opposite directions to the action forces.

Helical Gears

In addition to the tangential and radial forces encountered with spur gears, helical gears produce an axial force (as discussed in Chapter 10). First compute the tangential force from Equations (12-1) and (12-1). Then, if the helix angle of the gear is ψ , and if the normal pressure angle is ϕ_n , the radial force can be computed from

Radial Force

$$W_r = W_t \tan \phi_n / \cos \psi \quad (12-4)$$

The axial force is

Axial Force

$$W_x = W_t \tan \psi \quad (12-5)$$

Bevel Gears

Refer to Chapter 10 to review the formulas for the three components of the total force on bevel gear teeth in the tangential, radial, and axial directions. Example Problem 10-5 gives a comprehensive analysis of the forces, torques, and bending moments on shafts carrying bevel gears.

Worms and Wormgears

Chapter 10 also gives the formulas for computing the forces on worms and wormgears in the tangential, radial, and axial directions. See Example Problem 10-9.

Chain Sprockets

Figure 12-5 shows a pair of chain sprockets transmitting power. The upper part of the chain is in tension and produces the torque on either sprocket. The lower part of the chain, referred to as the *slack side*, exerts no force on either sprocket. Therefore, the total bending force on the shaft carrying the sprocket is equal to the tension in the tight side of the chain. If the torque on a certain sprocket is known,

Force in Chain

$$F_c = T/(D/2) \quad (12-6)$$

where D = pitch diameter of that sprocket

Notice that the force, F_c , acts along the direction of the tight side of the belt. Because of the size difference between the two sprockets, that direction is at some angle from the centerline between the shaft centers. A precise analysis would call for the force, F_c , to be resolved into components parallel to the centerline and perpendicular to it. That is,

$$F_{cx} = F_c \cos \theta \quad \text{and} \quad F_{cy} = F_c \sin \theta$$

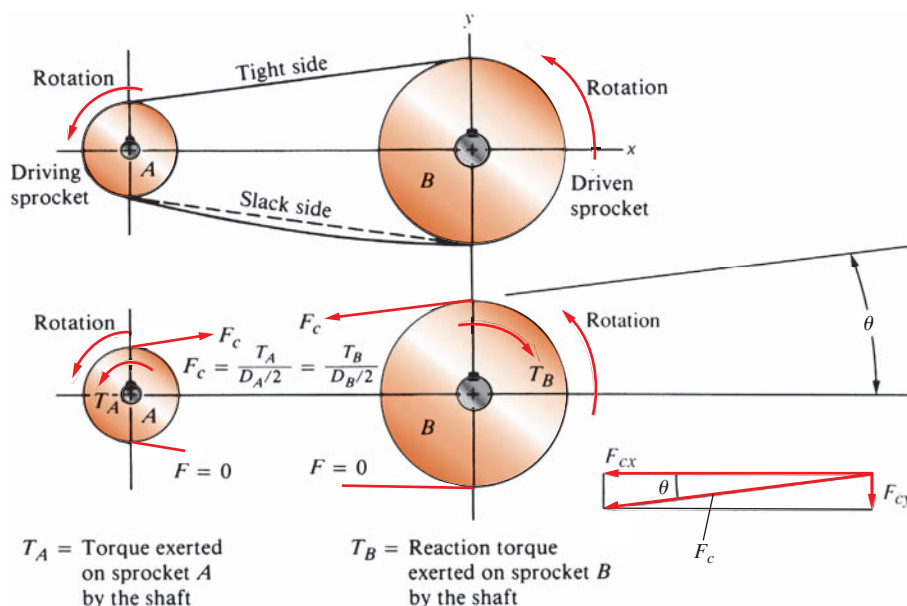


FIGURE 12-5 Forces on chain sprockets

where the x -direction is parallel to the centerline, the y -direction is perpendicular to it, and the angle θ is the angle of inclination of the tight side of the chain with respect to the x -direction.

These two components of the force would cause bending in both the x -direction and the y -direction. Alternatively, the analysis could be carried out in the direction of the force, F_c , in which single plane bending occurs.

If the angle θ is small, little error will result from the assumption that the entire force, F_c , acts along the x -direction. *Unless stated otherwise, this book will use this assumption.*

V-Belt Sheaves

The general appearance of the V-belt drive system looks similar to the chain drive system. But there is one important difference: Both sides of the V-belt are in tension, as indicated in Figure 12-6. The tight side tension, F_1 , is greater than the “slack side” tension, F_2 , and thus there is a net driving force on the sheaves equal to

Net Driving Force

$$F_N = F_1 - F_2 \quad (12-7)$$

The magnitude of the net driving force can be computed from the torque transmitted:

Net Driving Force

$$F_N = T/(D/2) \quad (12-8)$$

But notice that the bending force on the shaft carrying the sheave is dependent on the *sum*, $F_1 + F_2 = F_B$. To be more precise, the components of F_1 and F_2 parallel to the line of centers of the two sprockets should be used. But unless the two sprockets are radically different in diameter, little error will result from $F_B = F_1 + F_2$.

To determine the bending force, F_B , a second equation involving the two forces F_1 and F_2 is needed. This is provided by assuming a ratio of the tight side tension to the slack side tension. For V-belt drives, the ratio is normally taken to be

$$F_1/F_2 = 5 \quad (12-9)$$

It is convenient to derive a relationship between F_N and F_B of the form

$$F_B = CF_N \quad (12-10)$$

where C = constant to be determined

$$C = \frac{F_B}{F_N} = \frac{F_1 + F_2}{F_1 - F_2} \quad (12-11)$$

But from Equation (12-9), $F_1 = 5F_2$. Then

$$C = \frac{F_1 + F_2}{F_1 - F_2} = \frac{5F_2 + F_2}{5F_2 - F_2} = \frac{6F_2}{4F_2} = 1.5$$

Equation (12-10) then becomes, for V-belt drives,

Bending Force on Shaft for V-Belt Drive

$$F_B = 1.5 F_N = 1.5T/(D/2) \quad (12-12)$$

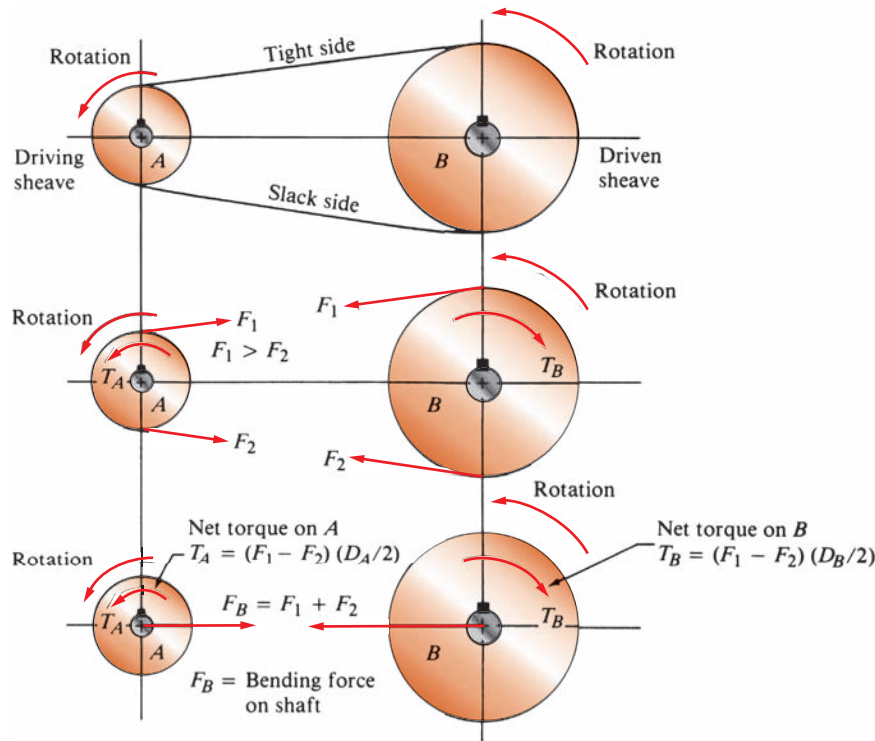


FIGURE 12-6 Forces on belt sheaves or pulleys

It is customary to consider the bending force, F_B , to act as a single force in the direction along the line of centers of the two sheaves as shown in Figure 12–6.

Flat-Belt Pulleys

The analysis of the bending force exerted on shafts by flat-belt pulleys is identical to that for V-belt sheaves except that the ratio of the tight side to the slack side tension is typically taken to be 3 instead of 5. Using the same logic as with V-belt sheaves, we can compute the constant C to be 2.0. Then, for flat-belt drives,

▢ Bending Force on Shaft for Flat-Belt Drive

$$F_B = 2.0 F_N = 2.0 T/(D/2) \quad (12-13)$$

Flexible Couplings

More detailed discussion of flexible couplings was presented in Chapter 11, but it is important to observe here how the use of a flexible coupling affects the design of a shaft.

A flexible coupling is used to transmit power between shafts while accommodating minor misalignments in the radial, angular, or axial directions. Thus, the shafts adjacent to the couplings are subjected to torsion, but the misalignments cause no axial or bending loads.

12-4 STRESS CONCENTRATIONS IN SHAFTS

In order to mount and locate the several types of machine elements on shafts properly, a final design typically contains several diameters, keyseats, ring grooves, and other geometric discontinuities that create stress concentrations. The shaft design proposed in Figure 12–2 is an example of this observation.

These stress concentrations must be taken into account during the design analysis. But a problem exists because the true design values of the stress concentration factors, K_t , are unknown at the start of the design process. Most of the values are dependent on the diameters of the shaft and on the fillet and groove geometries, and these are the objectives of the design.

You can overcome this dilemma by establishing a set of preliminary design values for commonly encountered stress concentration factors, which can be used to produce initial estimates for the minimum acceptable shaft diameters. Then, after the refined dimensions are selected, you can analyze the final geometry to determine the real values for stress concentration factors. Comparing the final values with the preliminary values will enable you to judge the acceptability of the design. The final values of K_t can be determined using Appendix 18 or Internet site 3 in Chapter 3.

Preliminary Design Values for K_t

Considered here are the types of geometric discontinuities most often found in power-transmitting shafts: keyseats, shoulder fillets, and retaining ring grooves. In each case, a suggested design value is relatively high in order to produce a conservative result for the first approximation to the design. Again it is emphasized that the final design should be checked for safety. That is, if the final value is less than the original design value, the design is still safe. Conversely, if the final value is higher, the stress analysis for the design must be rechecked.

Keyseats. A *keyseat* is a longitudinal groove cut into a shaft for the mounting of a key, permitting the transfer of torque from the shaft to a power-transmitting element, or vice versa. The detail design of keys was covered in Chapter 11.

Two types of keyseats are most frequently used: profile and sled runner (see Figure 12–7). The profile keyseat is milled into the shaft, using an end mill having a diameter equal to the width of the key. The resulting groove is flat-bottomed and has a sharp, square corner at its end. The sled runner keyseat is produced by a circular milling cutter having a width equal to the width of the key. As the cutter begins or ends the keyseat, it produces a smooth radius. For this reason, the stress concentration factor for the sled runner keyseat is lower than that for the profile keyseat. Normally used design values are

$$K_t = 2.0 \quad (\text{profile})$$

$$K_t = 1.6 \quad (\text{sled runner})$$

Each of these is to be applied to the bending stress calculation for the shaft, using the full diameter of the shaft. The factors take into account both the reduction in cross

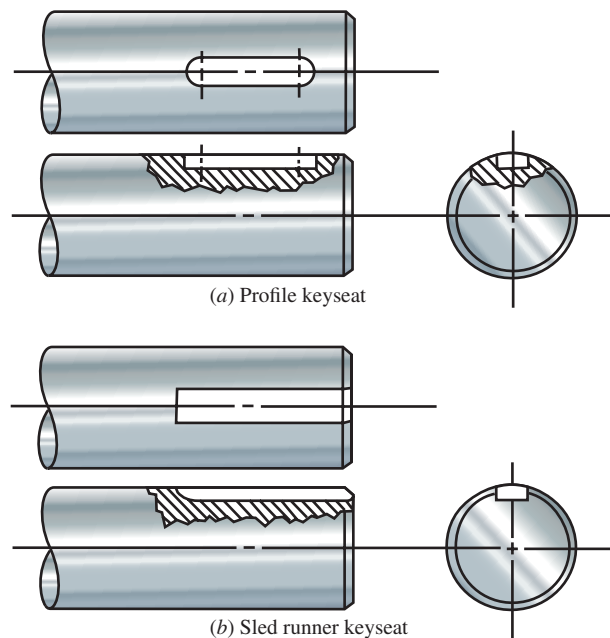
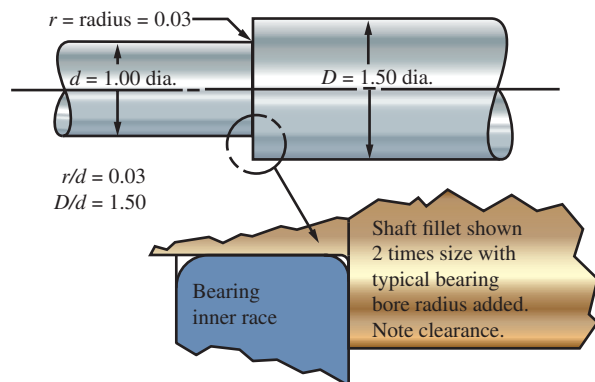


FIGURE 12–7 Keyseats

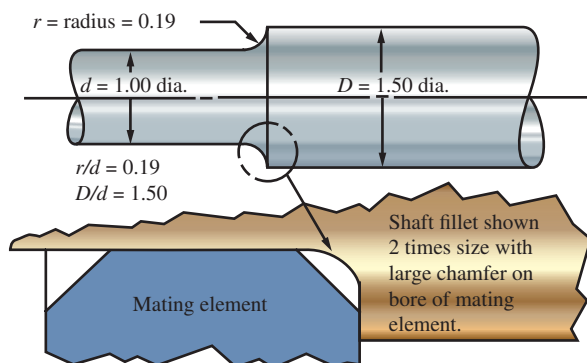
section and the effect of the discontinuity. Consult the references listed for more detail about stress concentration factors for keyseats. (See Reference 6.) If the torsional shear stress is fluctuating rather than steady, the stress concentration factor is also applied to that.

Shoulder Fillets. When a change in diameter occurs in a shaft to create a shoulder against which to locate a machine element, a stress concentration dependent on the ratio of the two diameters and on the radius in the fillet is produced (see Figure 12–8). It is recommended that the fillet radius be as large as possible to minimize the stress concentration, but at times the design of the gear, bearing, or other element affects the radius that can be used. For the purpose of design, we will classify fillets into two categories: sharp and well-rounded.

The term *sharp* here does not mean truly sharp, without any fillet radius at all. Such a shoulder configuration would have a very high stress concentration factor and should be avoided. Instead, *sharp* describes a shoulder with a relatively small fillet radius. One situation in which this is likely to occur is where a ball or roller bearing is to be located. The inner race of the bearing has a factory-produced radius, but it is small. The fillet radius on the shaft must be smaller yet in order for the bearing to be seated properly against the shoulder.



(a) Example of sharp fillet
($K_t = 2.5$ for bending)



(b) Example of well-rounded fillet
($K_t = 1.5$ for bending)

FIGURE 12–8 Fillets on shafts

When an element with a large chamfer on its bore is located against the shoulder, or when nothing at all seats against the shoulder, the fillet radius can be much larger (*well-rounded*), and the corresponding stress concentration factor is smaller. We will use the following values for design for bending:

$$K_t = 2.5 \quad (\text{sharp fillet})$$

$$K_t = 1.5 \quad (\text{well-rounded fillet})$$

Referring to Appendix 18 or Internet site 3 in Chapter 3, you can see that these values correspond to ratios of r/d of approximately 0.03 for the sharp fillet case and 0.19 for the well-rounded fillet for a D/d ratio of 1.50.

Retaining Ring Grooves. Retaining rings are used for many types of locating tasks in shaft applications. The rings are installed in grooves in the shaft after the element to be retained is in place. The geometry of the groove is dictated by the ring manufacturer. Its usual configuration is a shallow groove with straight side walls and bottom and a small fillet at the base of the groove. The behavior of the shaft in the vicinity of the groove can be approximated by considering two sharp-filleted shoulders positioned close together. Thus, the stress concentration factor for a groove is fairly high.

For preliminary design, we will apply $K_t = 3.0$ to the bending stress at a retaining ring groove to account for the rather sharp fillet radii. The stress concentration factor is not applied to the torsional shear stress if it is steady in one direction.

The computed estimate for the minimum required diameter at a ring groove is at the base of the groove. You should increase this value by approximately 6% to account for the typical groove depth to determine the nominal size for the shaft. Apply a ring groove factor of 1.06 to the computed required diameter.

12-5 DESIGN STRESSES FOR SHAFTS

In a given shaft, several different stress conditions can exist at the same time. For any part of the shaft that transmits power, there will be a torsional shear stress, while bending stress is usually present on the same parts. Only bending stresses may occur on other parts. Some points may not be subjected to either bending or torsion but will experience vertical shearing stress. Axial tensile or compressive stresses may be superimposed on other stresses. Then there may be some points where no significant stresses at all are created.

Thus, the decision of what design stress to use depends on the particular situation at the point of interest. In many shaft design and analysis projects, computations must be done at several points to account completely for the variety of loading and geometry conditions that exist.

Several cases discussed in Chapter 5 for computing design factors, N , are useful for determining design

stresses for shaft design. The bending stresses will be assumed to be completely reversed and repeated because of the rotation of the shaft. Because ductile materials perform better under such loads, it will be assumed that the material for the shaft is ductile. It will also be assumed that the torsional loading is relatively constant and acting in one direction. If other situations exist, consult the appropriate analysis from Chapter 5.

The symbol τ_d will be used for the design stress when a shear stress is the basis for the design. The symbol σ_d will be used when a normal stress is the basis.

Design Shear Stress—Steady Torque

It was stated in Chapter 5 that the best predictor of failure in ductile materials due to a steady shear stress was the distortion energy theory in which the design shear stress is computed from

$$\tau_d = s_y / (N\sqrt{3}) = (0.577s_y) / N \quad (12-14)$$

We will use this value for steady torsional shear stress, vertical shear stress, or direct shear stress in a shaft.

Design Shear Stress—Reversed Vertical Shear

Points on a shaft where no torque is applied and where the bending moments are zero or very low are often subjected to significant vertical shearing forces which then govern the design analysis. This typically occurs where a

bearing supports an end of a shaft and where no torque is transmitted in that part of the shaft.

Figure 12-9(a) shows the distribution of vertical shearing stresses on such a circular cross section. Note that the maximum shearing stress is at the neutral axis of the shaft, that is, at the diameter. The stress decreases in a roughly parabolic manner to zero at the outer surface of the shaft.

The diagram in Figure 12-9(b) can be explained by visualizing an element of the shaft, beginning at the top, with a shearing force acting downward on the cross section. Then follow that element as it rotates one revolution.

1. Look straight at the circular cross section from the end of the shaft [see Figure 12-9(c)].
2. An element at the top of the shaft sees zero shearing stress as indicated in Figure 12-9(a).
3. After rotating 90° counterclockwise, placing it on the left side of the horizontal diameter, that same element sees the maximum shearing stress, directed downward.
4. That same element sees zero shearing stress again when it rotates 90° more, placing it at the bottom of the shaft.
5. Now as the same element rotates 90° more to a place on the right side of the horizontal diameter, note that the element is upside down from its orientation when it was on the left side, while the shearing force still acts downward.

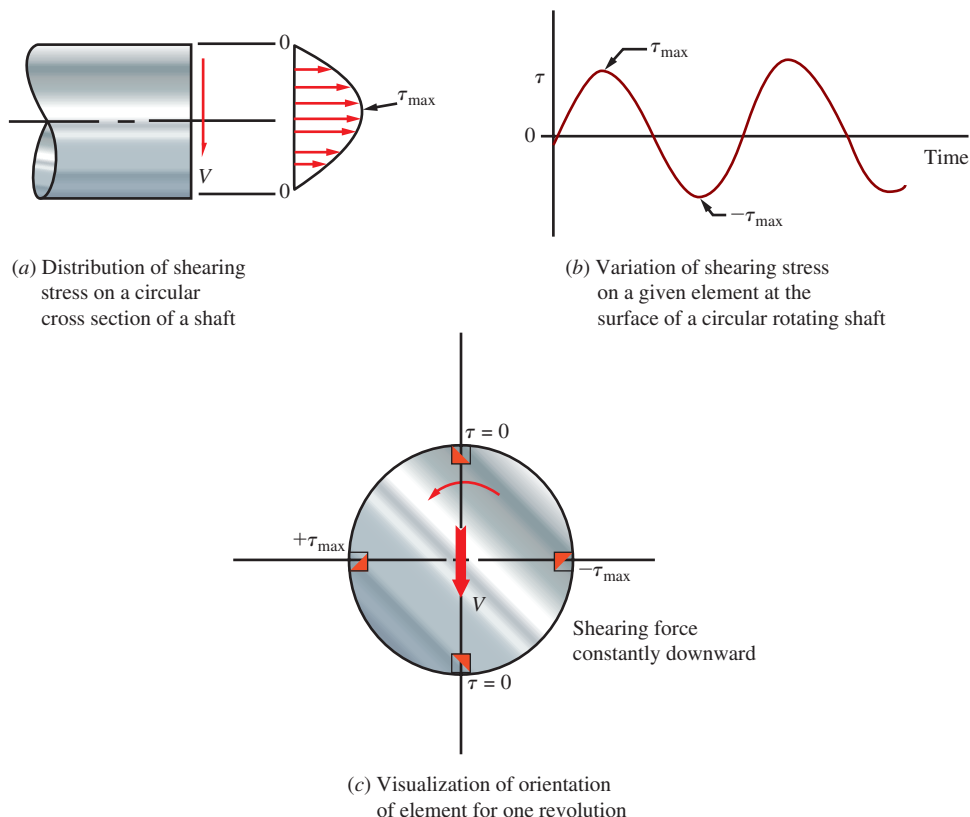


FIGURE 12-9 Shearing stress in a rotating shaft due to vertical shearing force, V

6. However, in relation to the element, the stress is acting in the *opposite direction* as compared with the direction in Step 3, and the stress on the element has reversed from the value seen in Step 3.
7. Rotating another 90° back to the top of the shaft, the element sees zero shearing stress again.

Therefore, we can conclude that any particle of the shaft on its outside surface sees a reversal of the direction of the shearing stress during each rotation, as shown in Figure 12–9(b).

Recall from strength of materials that the maximum vertical shearing stress for the special case of a solid circular cross section can be computed from

$$\tau_{\max} = 4V/3A$$

where V = vertical shearing force
 A = area of the cross section

Where stress concentration factors are to be considered,

$$\tau_{\max} = K_t(4V/3A)$$

Also note, as shown in Figure 12–9, that the rotation of the shaft causes any point at the outer part of the cross section to experience a reversing shearing stress that varies from $+\tau_{\max}$ to zero to $-\tau_{\max}$ to zero in each revolution. The design equations applied to this type of loading are developed next:

$$N = s'_{sn}/\tau_{\max}$$

We recommend using the distortion energy theory. Then the endurance strength in shear is

$$s'_{sn} = 0.577s'_n$$

Thus, the above equation can be written in the form

$$N = 0.577s'_n/\tau_{\max}$$

Expressed as a design stress, this is

$$\tau_d = 0.577s'_n/N$$

Now letting $\tau_{\max} = \tau_d = K_t(4V)/3A$ gives

$$\frac{K_t(4V)}{3A} = \frac{0.577s'_n}{N}$$

Solving for N gives

$$N = \frac{0.577s'_n(3A)}{K_t(4V)} = \frac{0.433s'_n(A)}{K_t(V)} \quad (12-15)$$

Equation (12–15) is useful if the goal is to evaluate the design factor for a given magnitude of loading, a given geometry for the shaft, and given material properties.

Now, solving for the required area to produce a specified design factor, N , gives

$$A = \frac{K_t(V)N}{0.433s'_n} = \frac{2.31K_t(V)N}{s'_n}$$

But our usual objective is the design of the shaft to determine the required diameter. By substituting

$$A = \pi D^2/4$$

we can solve for D :

⇒ Required Shaft Diameter

$$D = \sqrt{2.94 K_t(V)N/s'_n} \quad (12-16)$$

This equation should be used to compute the required diameter for a shaft where a vertical shearing force V is the only significant loading present. In most shafts, the resulting diameter will be much smaller than that required at other parts of the shaft where significant values of torque and bending moment occur. Also, practical considerations may require that the shaft be somewhat larger than the computed minimum to accommodate a reasonable bearing at the place where the shearing force is equal to the radial load on the bearing.

Implementation of Equations (12–15) and (12–16) has the complication that values for the stress concentration factor under conditions of vertical shearing stress are not well known. Published data such as that in Reference 6 in this chapter and Internet site 3 in Chapter 3 report values for stress concentration factors for axial normal stress, bending normal stress, and torsional shear stress. But values for vertical shearing stress are rarely reported. As an approximation, we will use the values for K_t for torsional shear stress when using these equations.

Design Normal Stress—Fatigue Loading

For the repeated and reversed bending in a shaft caused by transverse loads applied to the rotating shaft, the design stress is related to the endurance strength of the shaft material. The actual conditions under which the shaft is *manufactured* and *operated* should be considered when specifying the design stress.

Refer to the discussion in Section 5–6 for the method of computing the estimated actual endurance strength, s'_n , for use in shaft design. The process starts with using the graph in Figure 5–11 to determine the endurance strength as a function of the ultimate tensile strength of the material, adjusted for the surface finish. Equation (5–21) adjusts this value by applying four factors for the type of material, the type of stress, reliability, and the size of the cross section. When rotating steel shafts are being designed, the values for the material factor and the type of stress factor are both equal to 1.0. Use Table 5–3 for the reliability factor. Use Figure 5–12 or the equations in Table 5–4 to determine the size factor.

Stress concentration factors will be accounted for in the design equation to be developed later.

The design stress used here for parts of the shaft subjected to reversed bending stress due to the rotation of the shaft is

$$\sigma_d = s'_n/N \quad (12-17)$$

Note: Other factors not considered here could have adverse effect on the endurance strength of the shaft

material, and therefore on the design stress. Examples are as follows:

- Variation in peak stress levels above the nominal endurance strength for some periods of time, even quite short periods for high overstressing.
- Temperatures above approximately 400°F (204°C).
- Vibration that induces variations in stress not included in the analysis.
- Residual stresses: dangerous when tensile, possibly beneficial when compressive.
- Case hardening of the shaft material, resulting in non-uniform strength distribution near the surface and changing the net endurance strength.
- Interference fits of mating elements that introduce additional local stresses and stress concentrations.
- Corrosion that may roughen the surface and decrease the effective endurance strength.
- Thermal cycling over and above the applied stresses, changing the span of stress cycles.
- Axial tensile or compressive stresses. (A later section addresses this case.)

Testing of actual components under realistic operating conditions is recommended when any of these conditions exist.

Design Factor, N

Refer to Section 5–9 for discussion of factors that affect the choice of design factor. We will typically use $N = 2.5$ to 3.0 in this book, indicating a moderate level of uncertainty about actual material strengths, loading conditions, and long-term environmental factors. Higher values are sometimes justified for critical designs without clear knowledge of actual conditions. Conversely, lower values can be used when extensive, reliable data are available.

12-6 SHAFTS IN BENDING AND TORSION ONLY

Examples of shafts subjected to bending and torsion only are those carrying spur gears, V-belt sheaves, or chain sprockets. The power being transmitted causes the torsion, and the transverse forces on the elements cause bending. In the general case, the transverse forces do not all act in the same plane. In such cases, the bending moment diagrams for two perpendicular planes are prepared first. Then the resultant bending moment at each point of interest is determined. The process will be illustrated in Example Problem 12-1.

A design equation is now developed based on the assumption that the bending stress in the shaft is repeated and reversed as the shaft rotates, but that the torsional shear stress is nearly uniform. The design equation is based on the principle shown graphically in Figure 12-10 in which the vertical axis is the ratio of the

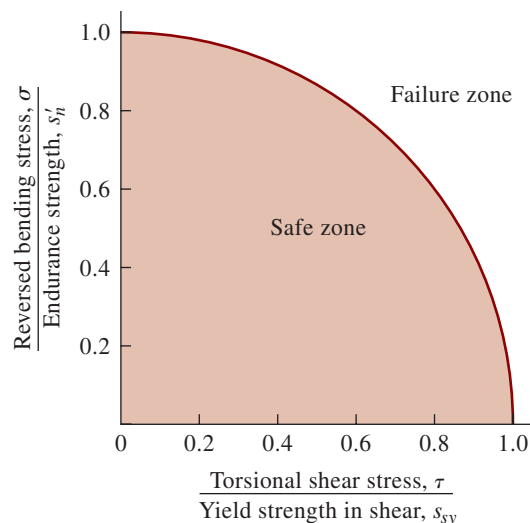


FIGURE 12-10 Basis for shaft design equation for repeated and reversed bending stress and steady torsional shear stress

reversed bending stress to the endurance strength of the material. (See Reference 8.) The horizontal axis is the ratio of the torsional shear stress to the yield strength of the material in shear. The points having the value of 1.0 on these axes indicate impending failure in pure bending or pure torsion, respectively. Experimental data show that failure under combinations of bending and torsion roughly follows the curve connecting these two points, which obeys the following equation:

$$(\sigma/s'_n)^2 + (\tau/s_{ys})^2 = 1 \quad (12-18)$$

We will use $s_{ys} = s_y/\sqrt{3}$ for the distortion energy theory. Also, a design factor can be introduced to each term on the left side of the equation to yield an expression based on design stresses:

$$(N\sigma/s'_n)^2 + (N\tau\sqrt{3}/s_y)^2 = 1$$

Now we can introduce a stress concentration factor for bending in the first term only, because this stress is repeated. No factor is needed for the torsional shear stress term because it is assumed to be steady, and stress concentrations have little or no effect on the failure potential. Then

$$(K_t N \sigma/s'_n)^2 + (N \tau \sqrt{3}/s_y)^2 = 1 \quad (12-19)$$

For rotating solid circular shafts, the bending stress due to a bending moment, M , is

$$\sigma = M/S \quad (12-20)$$

where $S = \pi D^3/32$ is the section modulus. The torsional shear stress is

$$\tau = T/Z_p \quad (12-21)$$

where $Z_p = \pi D^3/16$ is the polar section modulus.

Note that $Z_p = 2S$ and that, therefore,

$$\tau = T/(2S)$$

Substituting these relationships into Equation (12-19) gives

$$\left[\frac{K_t N M}{S s'_n} \right]^2 + \left[\frac{N T \sqrt{3}}{2 S s_y} \right]^2 = 1 \quad (12-22)$$

Now the terms N and S can be factored out, and the terms $\sqrt{3}$ and 2 can be brought outside the bracket in the torsion term:

$$\left[\frac{N}{S} \right]^2 \left[\left[\frac{K_t M}{s'_n} \right]^2 + \frac{3}{4} \left[\frac{T}{s_y} \right]^2 \right] = 1$$

We now take the square root of the entire equation:

$$\frac{N}{S} \sqrt{\left[\frac{K_t M}{s'_n} \right]^2 + \frac{3}{4} \left[\frac{T}{s_y} \right]^2} = 1$$

Let $S = \pi D^3/32$ for a solid circular shaft.

$$\frac{32 N}{\pi D^3} \sqrt{\left[\frac{K_t M}{s'_n} \right]^2 + \frac{3}{4} \left[\frac{T}{s_y} \right]^2} = 1 \quad (12-23)$$

Now we can solve for the diameter D :

Design Equation for Shaft Design

$$D = \left[\frac{32 N}{\pi} \sqrt{\left[\frac{K_t M}{s'_n} \right]^2 + \frac{3}{4} \left[\frac{T}{s_y} \right]^2} \right]^{1/3} \quad (12-24)$$

Equation (12-24) is used for shaft design in this book. It is compatible with the standard ANSI B106.1M-1985. (See Reference 1.) Note that Equation (12-24) can also be used for pure bending or pure torsion.

12-7 SHAFT DESIGN EXAMPLES—BENDING AND TORSION ONLY

Next we show two design examples for shaft design using the General Design Procedure given in Section 12-2. The general process must be adapted to the specific nature of each problem. Example Problems 12-1 and 12-2 carry some combinations of spur gears, belt sheaves, and chain sprockets that exert forces normal to the shaft to cause bending. The forces may act in any direction around the shaft depending on the placement of mating elements. Notice that none of these types of elements exerts forces parallel to the axis of the shaft tending to cause direct axial tension or compression.

Section 12-8 will modify this procedure for cases in which axial loads do exist, as with shafts carrying helical gears or wormgears.

Recommended Basic Sizes for Shafts

When mounting a commercially available element, of course, follow the manufacturer's recommendation for the basic size of the shaft and the tolerance.

In the U.S. Customary unit system, diameters are usually specified to be common fractions or their decimal equivalents. Appendix 2 lists the preferred basic sizes that you can use for dimensions over which you have control in decimal-inch, fractional-inch, and metric units. (See Reference 2.)

When commercially available unmounted bearings are to be used on a shaft, it is likely that their bores will be in metric dimensions. Typical sizes available and their decimal equivalents are listed in Table 14-3.

Design Example 12-1

Design the shaft shown in Figures 12-1 and 12-2. It is to be machined from SAE 1144 OQT 1000 steel. The shaft is part of the drive for a large blower system supplying air to a furnace. Gear A receives 200 hp from gear P . Gear C delivers the power to gear Q . The shaft rotates at 600 rpm. Use $D_A = 20.00$ in. and $D_C = 10.00$ in. Pressure angle $= \phi = 20^\circ$.

Solution

First determine the properties of the steel for the shaft. From Figure A4-2, $s_y = 83\,000$ psi, $s_u = 118\,000$ psi, and the percent elongation is 19%. Thus, the material has good ductility. Using Figure 5-11, we can estimate $s_n = 42\,000$ psi. For reversed bending, let $C_m = C_{st} = 1.0$.

A size factor should be applied to the endurance strength because the shaft will be quite large to be able to carry 200 hp. Although we do not know the actual size at this time, we might select $C_s = 0.75$ from Figure 5-12 as an estimate.

A reliability factor should also be specified. This is a design decision and we can use Table 5-3 for some common factors. For this problem, let's design for a reliability of 0.99 and use $C_R = 0.81$. Now we can compute the estimated actual endurance strength:

$$s'_n = s_n C_s C_R = (42\,000)(0.75)(0.81) = 25\,000 \text{ psi}$$

The design factor is taken to be $N = 2$. The blower is not expected to present any unusual shock or impact.

Now we can compute the torque in the shaft from Equation (12-1):

$$T = (63\,000)(P)/n = (63\,000)(200)/600 = 21\,000 \text{ lb} \cdot \text{in}$$

Note that only that part of the shaft from A to C is subjected to this torque. There is zero torque from the right of gear C over to bearing D . See Figure 12-11(d).

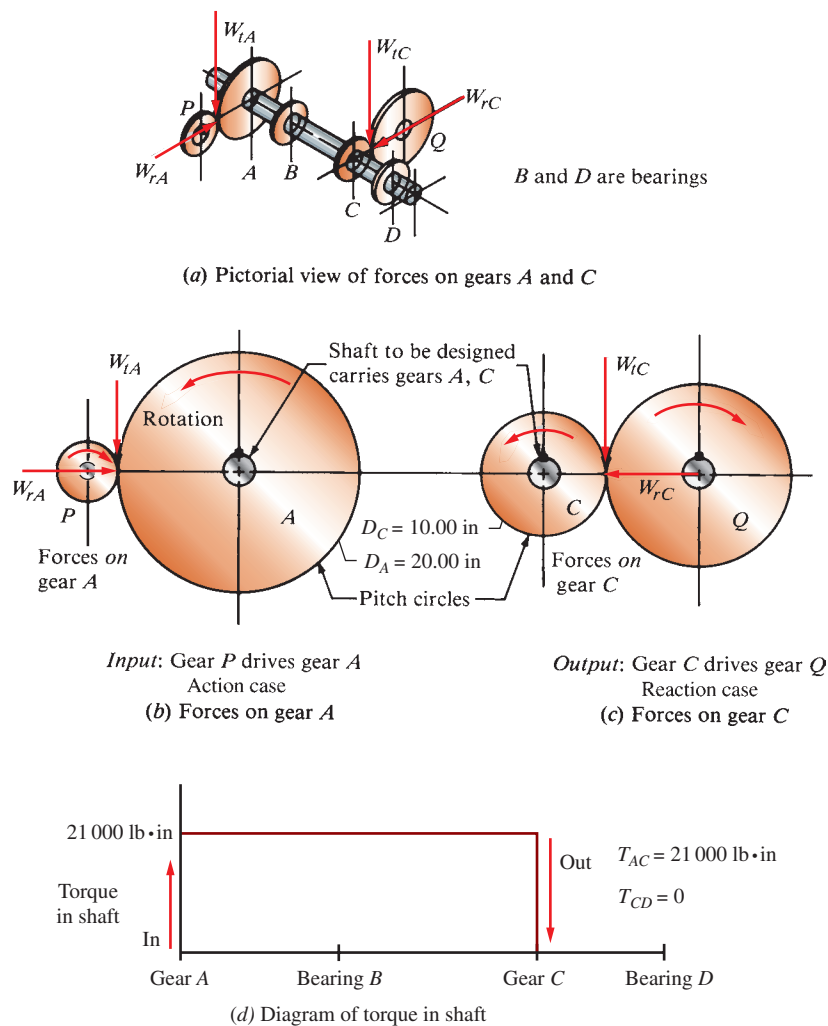


FIGURE 12-11 Forces on gears A and C and the torque in the shaft

Forces on the Gears: Figure 12-11 shows the two pairs of gears with the forces acting on gears A and C as shown. Observe that gear A is driven by gear P, and gear C drives gear Q. It is very important for the directions of these forces to be correct. The values of the forces are found from Equations (12-2) and (12-3).

$$W_{tA} = T_A / (D_A/2) = 21\,000 \text{ lb} \cdot \text{in} / (20 \text{ in}/2) = 2100 \text{ lb} \downarrow$$

$$W_{rA} = W_{tA} \tan(\phi) = 2100 \text{ lb} [\tan(20^\circ)] = 764 \text{ lb} \rightarrow$$

$$W_{tC} = T_C / (D_C/2) = 21\,000 \text{ lb} \cdot \text{in} / (10 \text{ in}/2) = 4200 \text{ lb} \downarrow$$

$$W_{rC} = W_{tC} \tan(\phi) = 4200 \text{ lb} [\tan(20^\circ)] = 1529 \text{ lb} \leftarrow$$

Forces on the Shaft: The next step is to show these forces on the shaft in their proper planes of action and in the proper direction. The reactions at the bearings are computed, and the shearing force and bending moment diagrams are prepared. The results are shown in Figure 12-12.

We continue the design by computing the minimum acceptable diameter of the shaft at several points along the shaft. We use the proposed design from Figure 12-2. At each point, we will observe the magnitude of torque and the bending moment that exist at the point, and we will estimate the value of any stress concentration factors. If more than one stress concentration exist in the vicinity of the point of interest, the larger value is used for design. This assumes that the geometric discontinuities themselves do not interact, which is good practice. For example, at point A, the keyseat should end well before the shoulder fillet begins.

1. **Point A:** Gear A produces torsion in the shaft from A and to the right. To the left of A, where there is a retaining ring, there are no forces, moments, or torques.

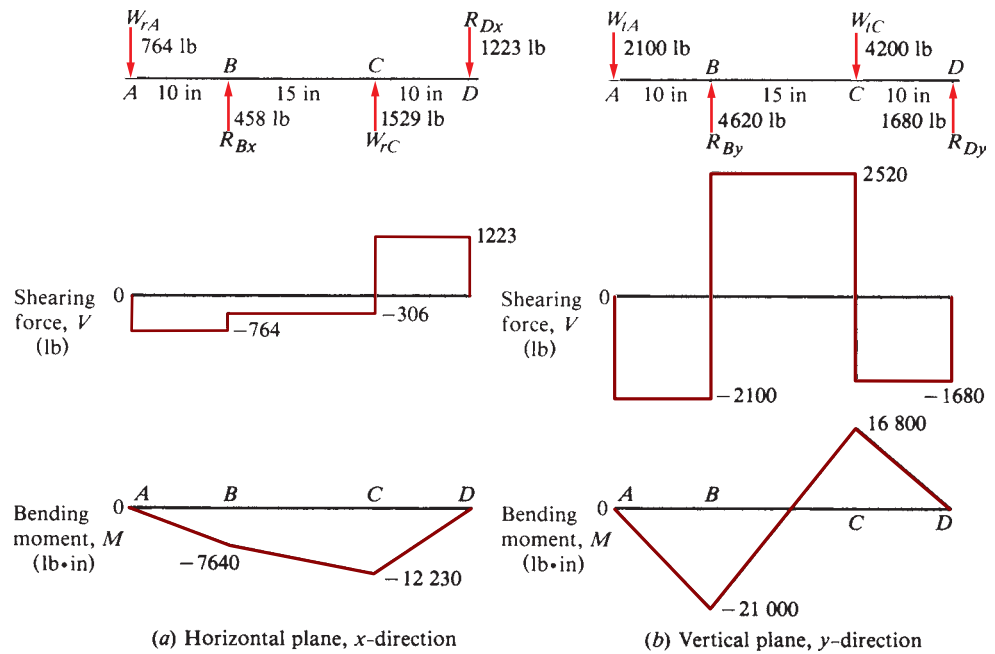


FIGURE 12-12 Load, shear, and moment diagrams for the shaft in Figure 12-10

The moment at A is zero because it is a free end of the shaft. Now we can use Equation (12-24) to compute the required diameter for the shaft at A, using only the torsion term.

$$D_1 = \left[\frac{32 N}{\pi} \sqrt{\frac{3}{4} \left(\frac{T}{s_y} \right)^2} \right]^{1/3}$$

$$D_1 = \left[\frac{32(2)}{\pi} \sqrt{\frac{3}{4} \left(\frac{21\,000}{83\,000} \right)^2} \right]^{1/3} = 1.65 \text{ in}$$

2. **Point B:** Point B is the location of a bearing with a sharp fillet to the right of B and a well-rounded fillet to the left. It is desirable to make D_2 at least slightly smaller than D_3 at the bearing seat to permit the bearing to be slid easily onto the shaft up to the place where it is pressed to its final position. There is usually a light press fit between the bearing bore and the shaft seat. The torque in the shaft at B is

$$T_B = 21\,000 \text{ lb} \cdot \text{in}$$

The bending moment at B is the resultant of the moment in the x- and y-planes from Figure 12-12:

$$M_B = \sqrt{M_{Bx}^2 + M_{By}^2} = \sqrt{(7640)^2 + (21\,000)^2} = 22\,350 \text{ lb} \cdot \text{in}$$

To the left of B (diameter D_2),

$$K_t = 1.5 \text{ (well-rounded fillet)}$$

Using Equation (12-24) because of the combined stress condition,

$$D_2 = \left[\left(\frac{32N}{\pi} \right) \sqrt{\left(\frac{K_t M}{s'_n} \right)^2 + \frac{3}{4} \left(\frac{T}{s_y} \right)^2} \right]^{1/3}$$

$$D_2 = \left[\frac{32(2)}{\pi} \sqrt{\left[\frac{1.5(22\,350)}{25\,500} \right]^2 + \frac{3}{4} \left[\frac{21\,000}{83\,000} \right]^2} \right]^{1/3} = 3.30 \text{ in} \quad (12-24a)$$

At B and to the right of B (diameter D_3), everything is the same, except the value of $K_t = 2.5$ for the sharp fillet. Then

$$D_3 = \left[\frac{32(2)}{\pi} \sqrt{\left[\frac{2.5(22\,350)}{25\,500} \right]^2 + \frac{3}{4} \left[\frac{21\,000}{83\,000} \right]^2} \right]^{1/3} = 3.55 \text{ in}$$